

Caracterización de jets — ALL VAL

15 salidas/modelos disponibles · 67.551 jets · score $\geq 0,8$

TP: b recuperados FN: b perdidos

FP: fondo aceptado TN: fondo rechazado

32 variables en unidades originales, media $\pm$ desviación poblacional (ddof=0).

Barras: altura=pT medio (GeV), color=conteo por celda, x/y=intervalos de impactos (mm).

Bins, cámara, escala de color y techo de pT comunes.

Incluye todas las semillas, ensembles y controles disponibles.

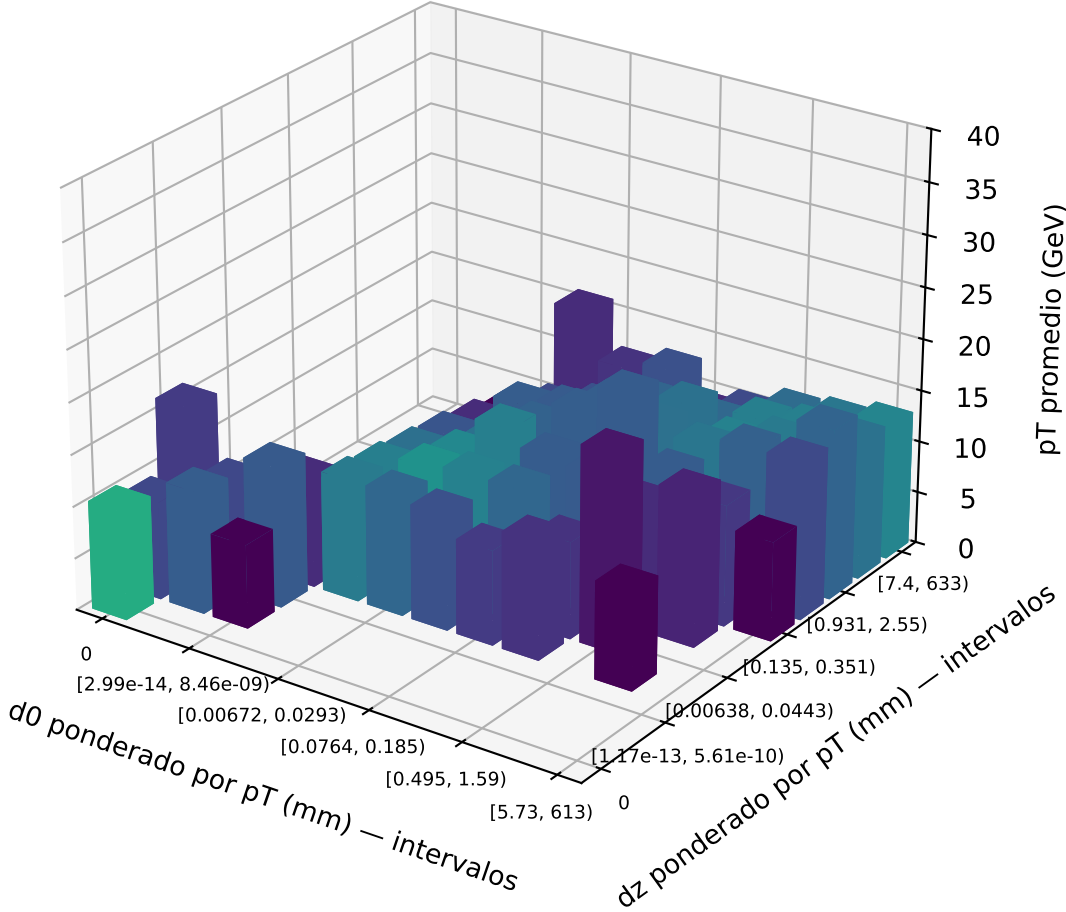
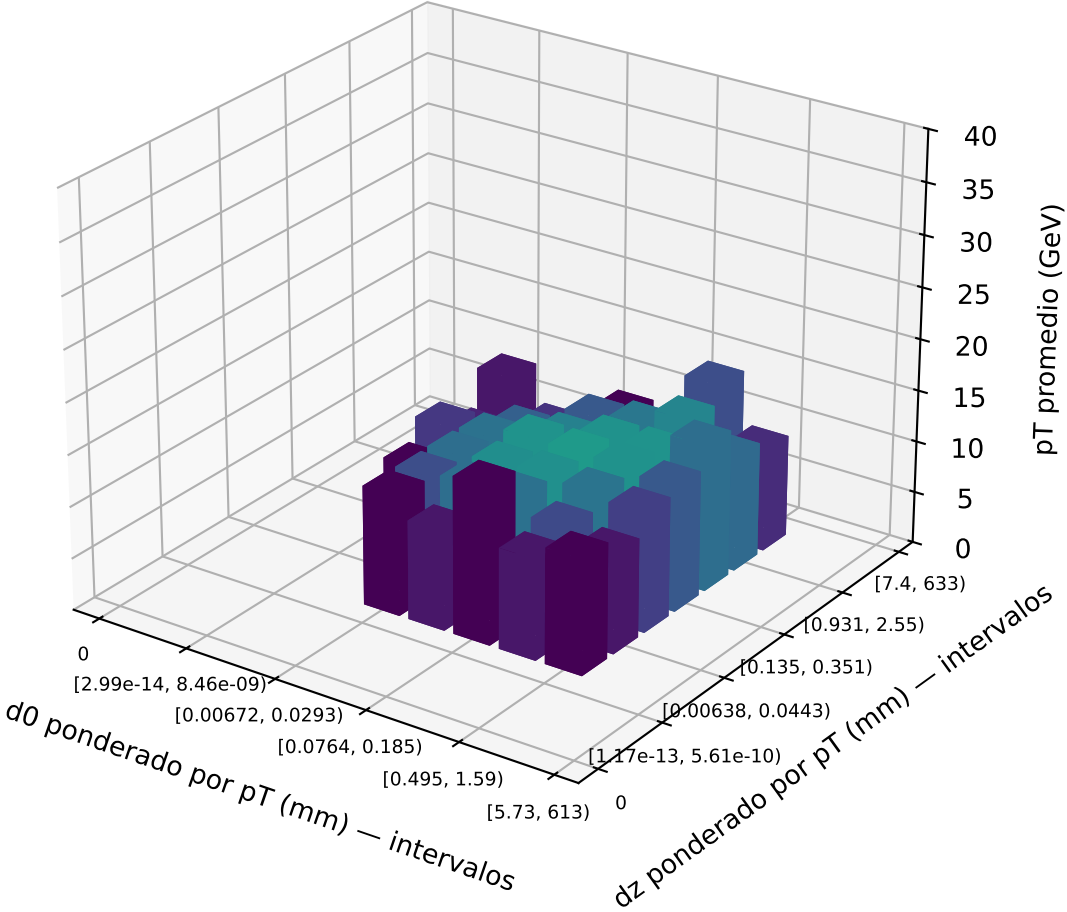
RF32 es una referencia local existente.

Notebook12 pendiente de finalización verificada.

Perfil de pT medio en intervalos de d0/dz — RF8

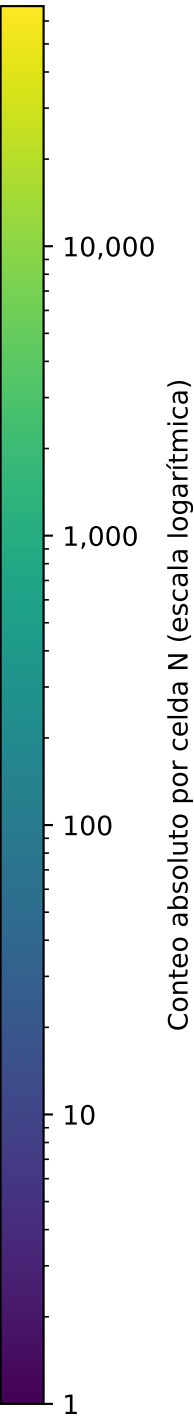
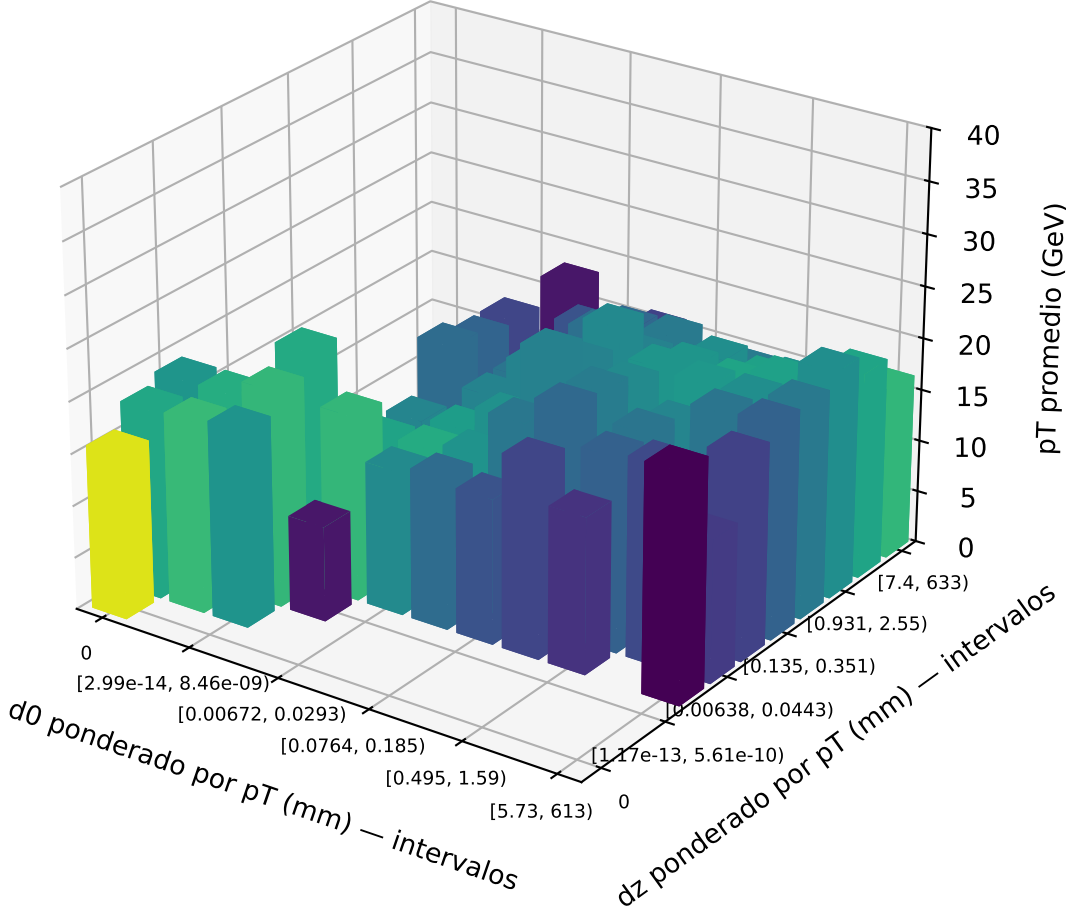
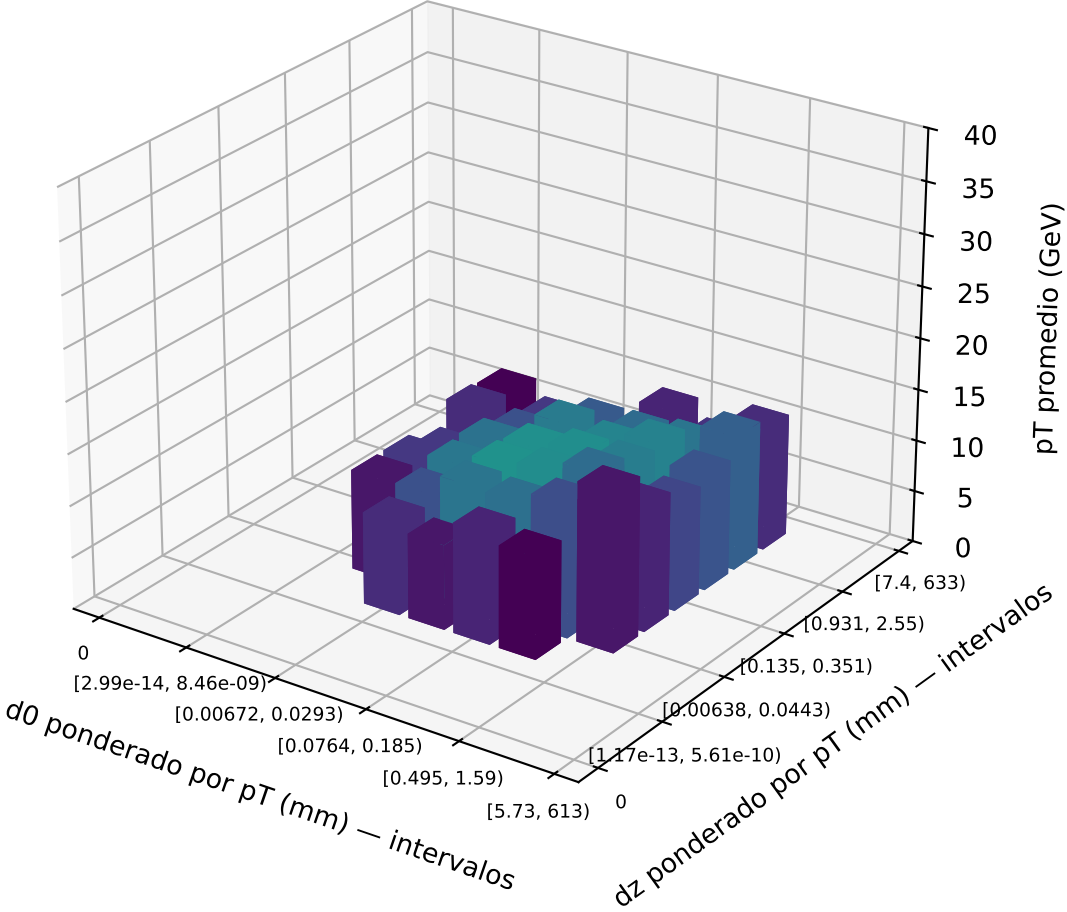
TP — N=2,987

FN — N=3,688



FP — N=2,112

TN — N=58,764



Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — RF8

Variable	Unidad	TP N=2,987	FN N=3,688	FP N=2,112	TN N=58,764
jet_pt	GeV	12.24 ± 6.241	12.28 ± 6.94	11.9 ± 5.522	16.58 ± 11.78
jet_eta	1	0.01025 ± 1.403	-0.003789 ± 1.424	0.01912 ± 1.483	0.002103 ± 1.496
jet_abs_eta	1	1.211 ± 0.7098	1.219 ± 0.736	1.284 ± 0.742	1.275 ± 0.7827
jet_mass	GeV	2.509 ± 1.35	2.547 ± 1.672	2.286 ± 1.112	2.872 ± 1.885
jet_ncharged	count	2.776 ± 1.538	2.676 ± 1.954	2.295 ± 1.248	2.598 ± 1.949
jet_nneutral	count	1.845 ± 1.586	2.03 ± 1.643	1.835 ± 1.387	2.388 ± 1.796
n_tracks_cone	count	2.798 ± 1.523	2.722 ± 1.957	2.313 ± 1.234	2.627 ± 1.949
track_pt_sum	GeV	5.862 ± 4.583	5.062 ± 5.084	4.955 ± 3.686	6.819 ± 8.163
track_pt_sum_over_jet_pt	1	0.4794 ± 0.2243	0.3978 ± 0.2638	0.419 ± 0.2287	0.3729 ± 0.2677
avg_track_pt	GeV	2.269 ± 1.43	1.681 ± 1.267	2.175 ± 1.231	2.165 ± 2.078
std_track_pt	GeV	0.788 ± 0.8746	0.695 ± 0.8594	0.6446 ± 0.761	1.017 ± 1.617
n_tracks_below_avg_pt	count	1.431 ± 1.14	1.444 ± 1.347	1.085 ± 0.9667	1.389 ± 1.363
n_tracks_above_avg_pt	count	1.367 ± 0.6474	1.278 ± 0.8374	1.229 ± 0.4907	1.239 ± 0.8107
max_pt_ratio	1	0.285 ± 0.1601	0.2228 ± 0.1631	0.2607 ± 0.135	0.2104 ± 0.1556
min_pt_ratio	1	0.1374 ± 0.1475	0.09213 ± 0.114	0.141 ± 0.1156	0.08456 ± 0.09788
max_dr_ratio	1	0.1623 ± 0.1473	0.1134 ± 0.1183	0.1622 ± 0.1178	0.1034 ± 0.1071
min_dr_ratio	1	0.2464 ± 0.1765	0.1877 ± 0.1683	0.2294 ± 0.1435	0.1795 ± 0.1548
dr_max_pt	1	0.1403 ± 0.08417	0.135 ± 0.09833	0.1489 ± 0.08477	0.1263 ± 0.09905
dr_min_pt	1	0.2011 ± 0.1002	0.1932 ± 0.1173	0.2012 ± 0.09407	0.1904 ± 0.1174
dr_max_dr	1	0.2388 ± 0.09644	0.2308 ± 0.1199	0.2267 ± 0.08966	0.2214 ± 0.1208
dr_min_dr	1	0.1113 ± 0.07327	0.1057 ± 0.08384	0.1287 ± 0.08117	0.1034 ± 0.08808
pt_difference	GeV	1.852 ± 2.15	1.686 ± 2.215	1.471 ± 1.76	2.496 ± 4.084
r50	1	0.147 ± 0.08168	0.1423 ± 0.09597	0.154 ± 0.08171	0.133 ± 0.09742
r95	1	0.2362 ± 0.09595	0.2282 ± 0.1191	0.2247 ± 0.0898	0.2151 ± 0.1194
mean_abs_d0 *	mm	0.1638 ± 0.1542	2.145 ± 10.62	0.1285 ± 0.1401	1.237 ± 7.854
max_abs_d0 *	mm	0.2771 ± 0.2871	5.839 ± 20.44	0.2083 ± 0.2123	3.277 ± 16.53
std_abs_d0 *	mm	0.0867 ± 0.1148	2.251 ± 7.915	0.06464 ± 0.08748	1.261 ± 6.641
mean_abs_dz *	mm	0.4011 ± 0.4659	3.194 ± 15.89	0.3367 ± 0.4563	1.65 ± 9.634
max_abs_dz *	mm	0.666 ± 0.7531	8.305 ± 26.2	0.5212 ± 0.6278	4.321 ± 22.93
std_abs_dz *	mm	0.2031 ± 0.2772	3.156 ± 9.411	0.1546 ± 0.2395	1.675 ± 9.459
pt_weighted_abs_d0 *	mm	0.1693 ± 0.1516	2.018 ± 10.52	0.133 ± 0.1361	1.179 ± 7.866
pt_weighted_abs_dz *	mm	0.4164 ± 0.4755	2.745 ± 15.66	0.3486 ± 0.4483	1.355 ± 8.341

* Variable utilizada como entrada del modelo.

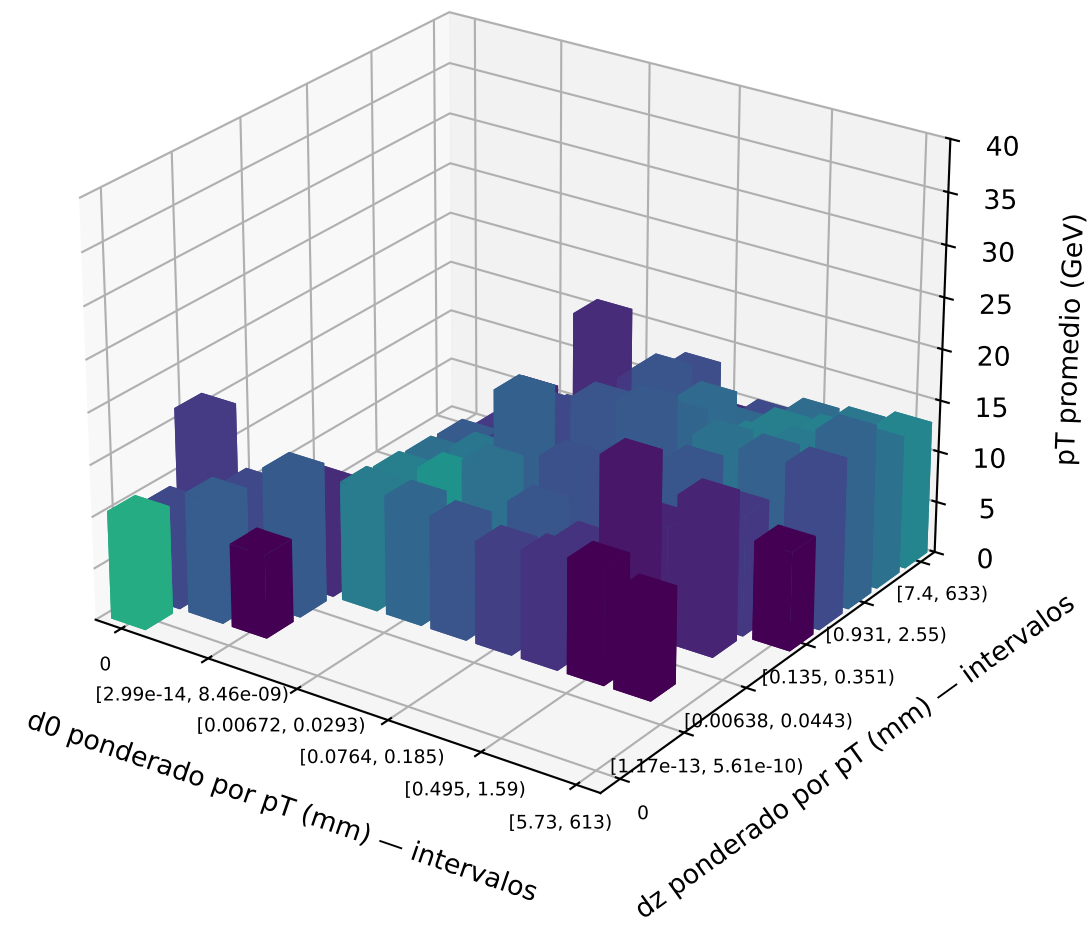
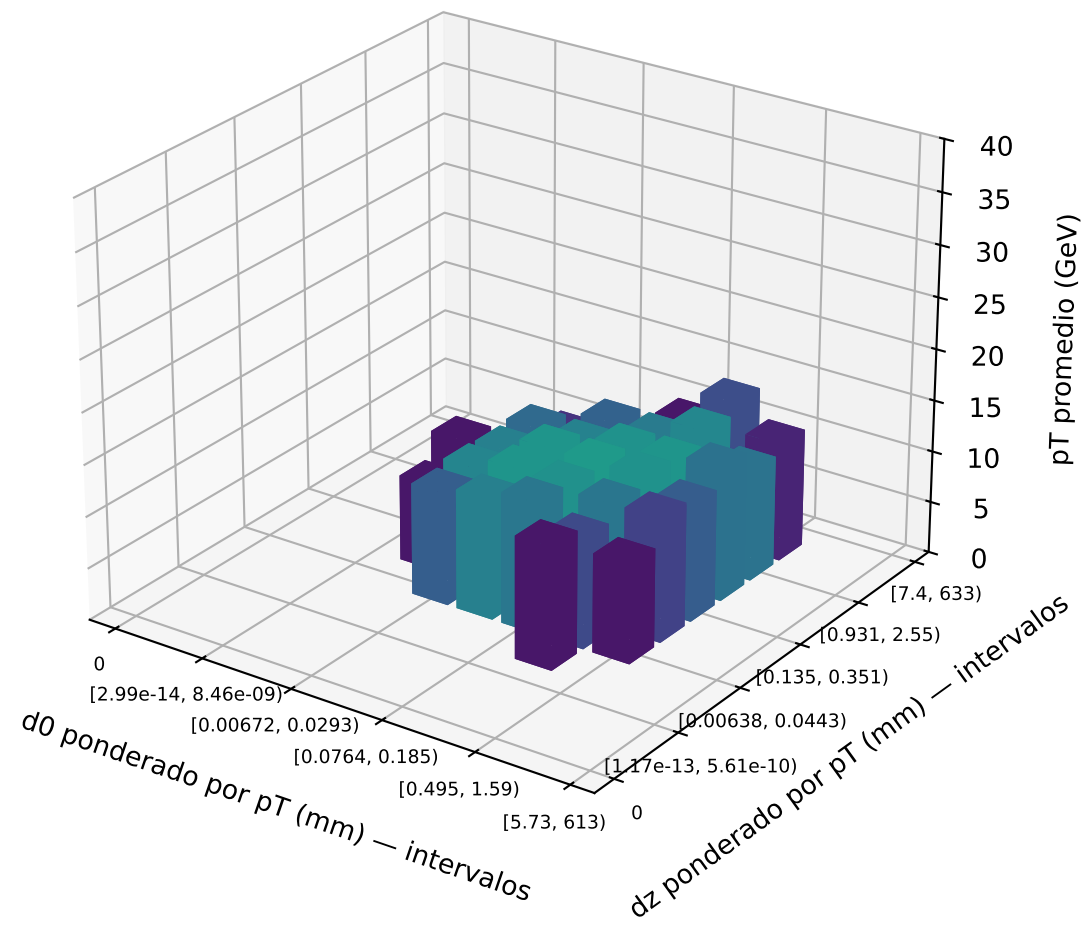
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC8 · ensemble

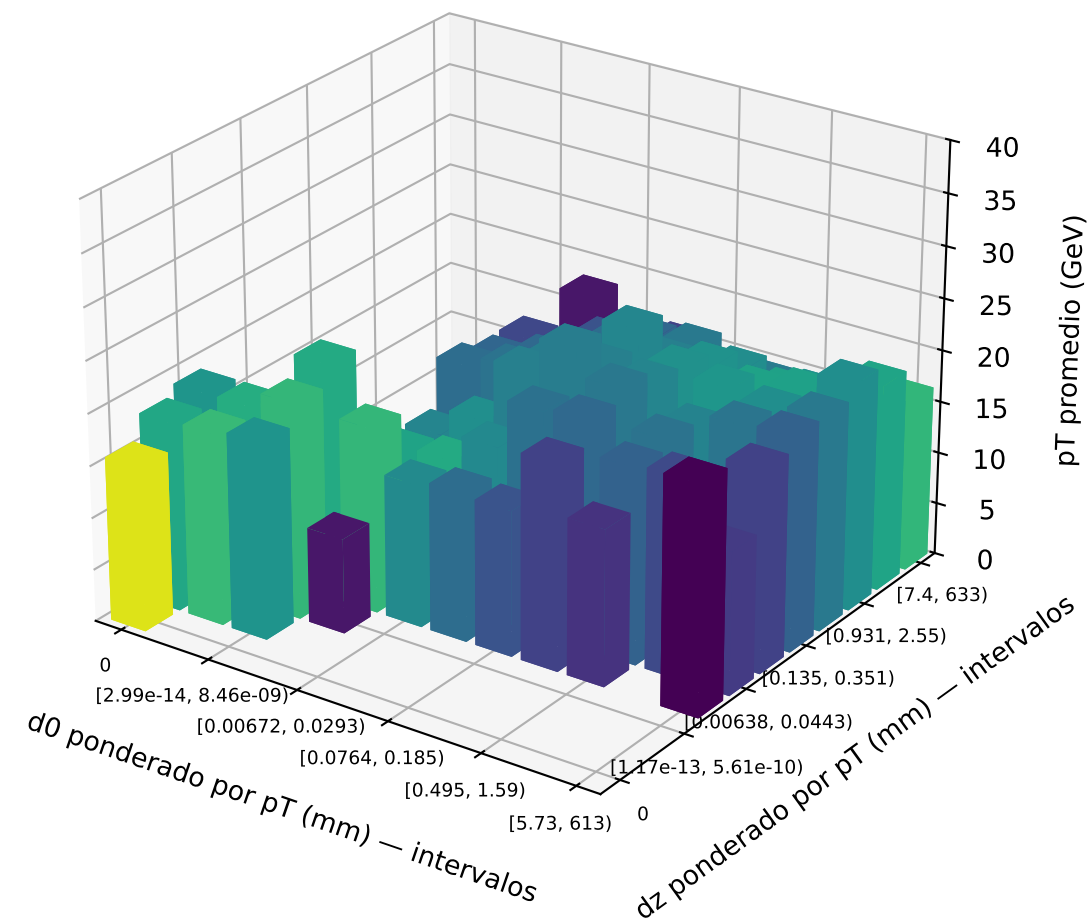
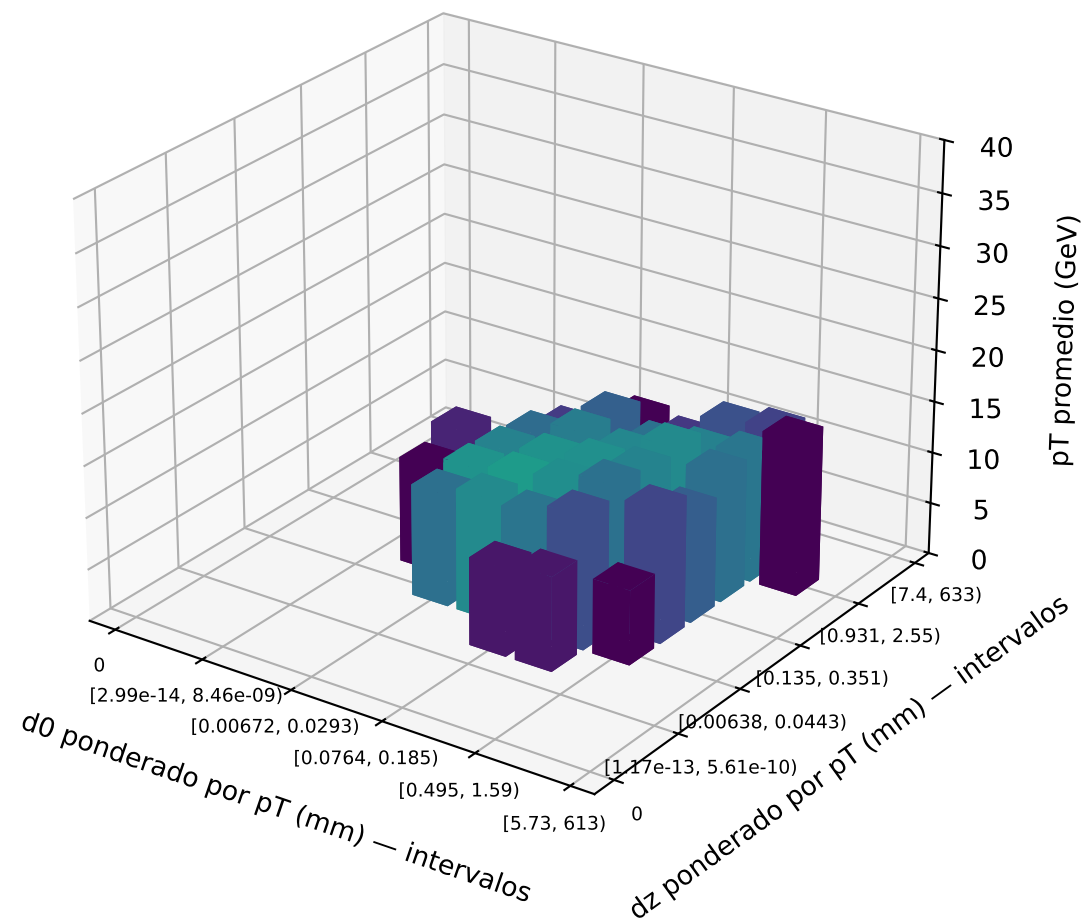
TP — N=3,513

FN — N=3,162



FP — N=3,175

TN — N=57,701



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC8 · ensemble

Variable	Unidad	TP N=3,513	FN N=3,162	FP N=3,175	TN N=57,701
jet_pt	GeV	12.23 ± 6.325	12.3 ± 6.966	12.1 ± 5.807	16.65 ± 11.85
jet_eta	1	0.001792 ± 1.398	0.003268 ± 1.434	-0.007309 ± 1.457	0.003244 ± 1.498
jet_abs_eta	1	1.205 ± 0.7081	1.227 ± 0.742	1.258 ± 0.7354	1.276 ± 0.7838
jet_mass	GeV	2.513 ± 1.374	2.549 ± 1.698	2.392 ± 1.196	2.877 ± 1.894
jet_ncharged	count	2.757 ± 1.537	2.681 ± 2.017	2.423 ± 1.304	2.597 ± 1.958
jet_nneutral	count	1.857 ± 1.6	2.046 ± 1.636	1.858 ± 1.428	2.397 ± 1.799
n_tracks_cone	count	2.785 ± 1.521	2.724 ± 2.022	2.447 ± 1.288	2.626 ± 1.958
track_pt_sum	GeV	5.783 ± 4.472	5.017 ± 5.273	5.126 ± 3.83	6.844 ± 8.217
track_pt_sum_over_jet_pt	1	0.4747 ± 0.2237	0.3895 ± 0.2697	0.4256 ± 0.225	0.3717 ± 0.2684
avg_track_pt	GeV	2.247 ± 1.413	1.607 ± 1.245	2.135 ± 1.222	2.167 ± 2.091
std_track_pt	GeV	0.7893 ± 0.8669	0.678 ± 0.8643	0.6658 ± 0.7858	1.023 ± 1.627
n_tracks_below_avg_pt	count	1.426 ± 1.141	1.452 ± 1.378	1.177 ± 0.9873	1.389 ± 1.369
n_tracks_above_avg_pt	count	1.359 ± 0.6416	1.272 ± 0.8699	1.27 ± 0.5317	1.237 ± 0.8139
max_pt_ratio	1	0.2845 ± 0.161	0.2129 ± 0.1605	0.2565 ± 0.1347	0.2097 ± 0.1559
min_pt_ratio	1	0.1361 ± 0.1466	0.08613 ± 0.1078	0.1344 ± 0.1145	0.08389 ± 0.09749
max_dr_ratio	1	0.1604 ± 0.1462	0.1074 ± 0.1133	0.157 ± 0.1171	0.1026 ± 0.1068
min_dr_ratio	1	0.2465 ± 0.1775	0.1779 ± 0.1637	0.2241 ± 0.1442	0.1789 ± 0.1549
dr_max_pt	1	0.1414 ± 0.08528	0.133 ± 0.09934	0.1498 ± 0.08548	0.1259 ± 0.0992
dr_min_pt	1	0.2023 ± 0.1004	0.1906 ± 0.1195	0.2037 ± 0.09471	0.19 ± 0.1178
dr_max_dr	1	0.2397 ± 0.09657	0.2285 ± 0.1232	0.2328 ± 0.08953	0.221 ± 0.1213
dr_min_dr	1	0.1123 ± 0.07407	0.1037 ± 0.08457	0.1262 ± 0.07955	0.1031 ± 0.08826
pt_difference	GeV	1.851 ± 2.116	1.66 ± 2.261	1.532 ± 1.829	2.512 ± 4.111
r50	1	0.1476 ± 0.08264	0.1409 ± 0.09718	0.1562 ± 0.08261	0.1325 ± 0.09758
r95	1	0.2371 ± 0.09604	0.2259 ± 0.1223	0.2307 ± 0.08954	0.2146 ± 0.1199
mean_abs_d0 *	mm	0.1552 ± 0.158	2.484 ± 11.44	0.135 ± 0.176	1.257 ± 7.925
max_abs_d0 *	mm	0.2678 ± 0.2677	6.774 ± 21.94	0.2453 ± 0.3015	3.332 ± 16.67
std_abs_d0 *	mm	0.08614 ± 0.1083	2.612 ± 8.494	0.08619 ± 0.1326	1.282 ± 6.7
mean_abs_dz *	mm	0.3828 ± 0.445	3.679 ± 17.11	0.3392 ± 0.4748	1.674 ± 9.721
max_abs_dz *	mm	0.6529 ± 0.7371	9.59 ± 28.09	0.5968 ± 0.7767	4.387 ± 23.14
std_abs_dz *	mm	0.2079 ± 0.2962	3.642 ± 10.08	0.2052 ± 0.3356	1.7 ± 9.543
pt_weighted_abs_d0 *	mm	0.1637 ± 0.1653	2.332 ± 11.33	0.146 ± 0.1879	1.197 ± 7.937
pt_weighted_abs_dz *	mm	0.4048 ± 0.4725	3.146 ± 16.88	0.3682 ± 0.5189	1.373 ± 8.416

* Variable utilizada como entrada del modelo.

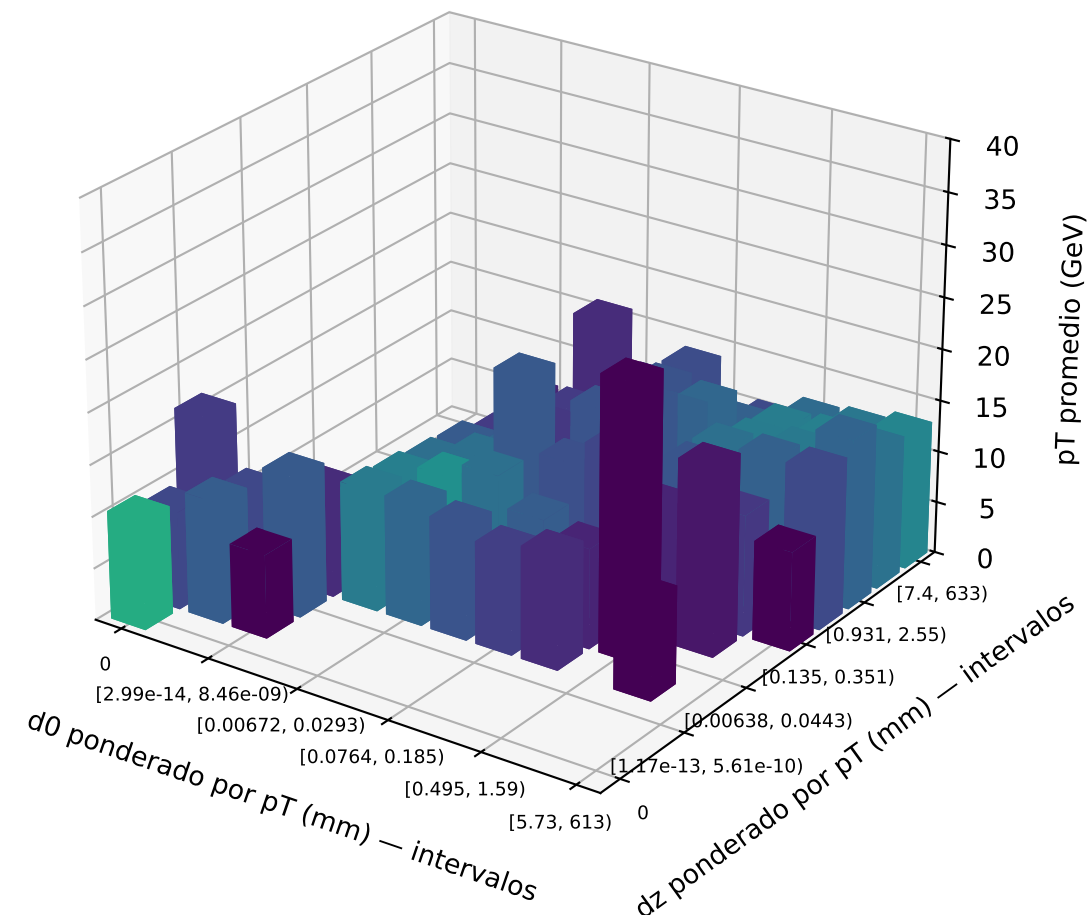
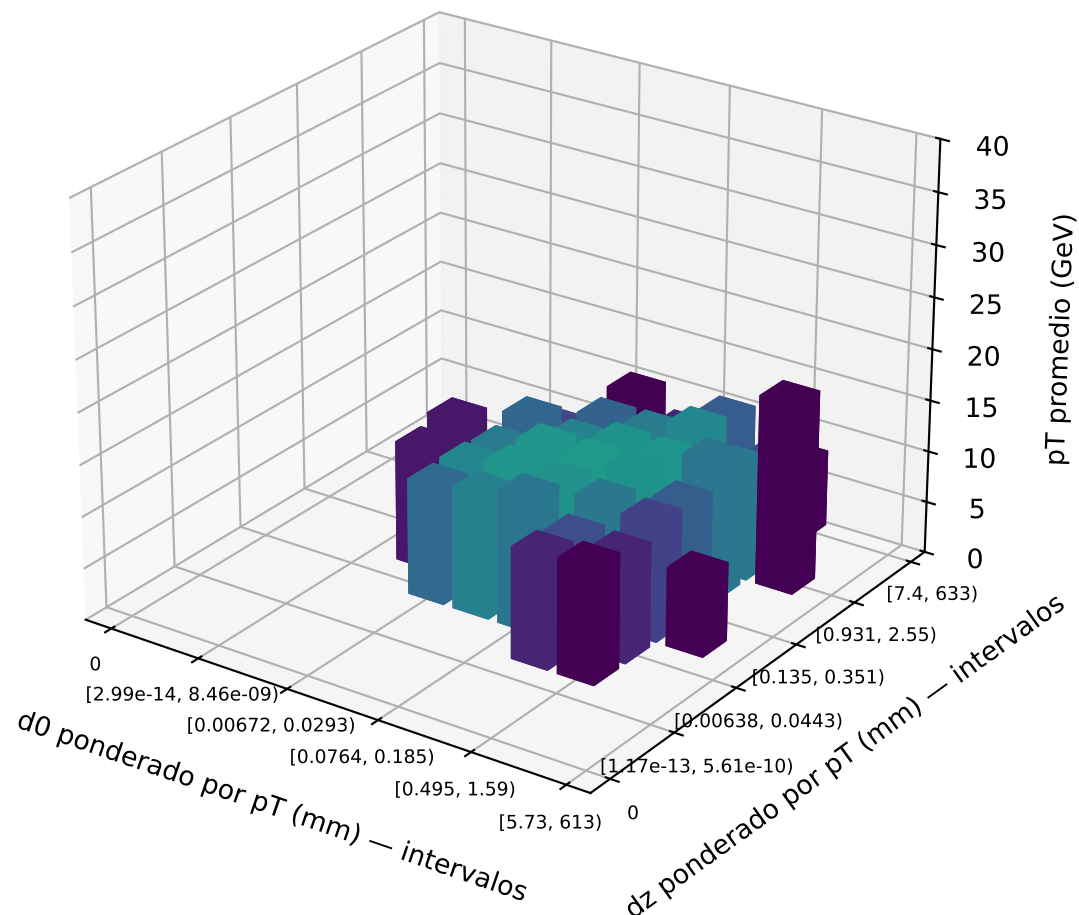
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC8 · semilla 101

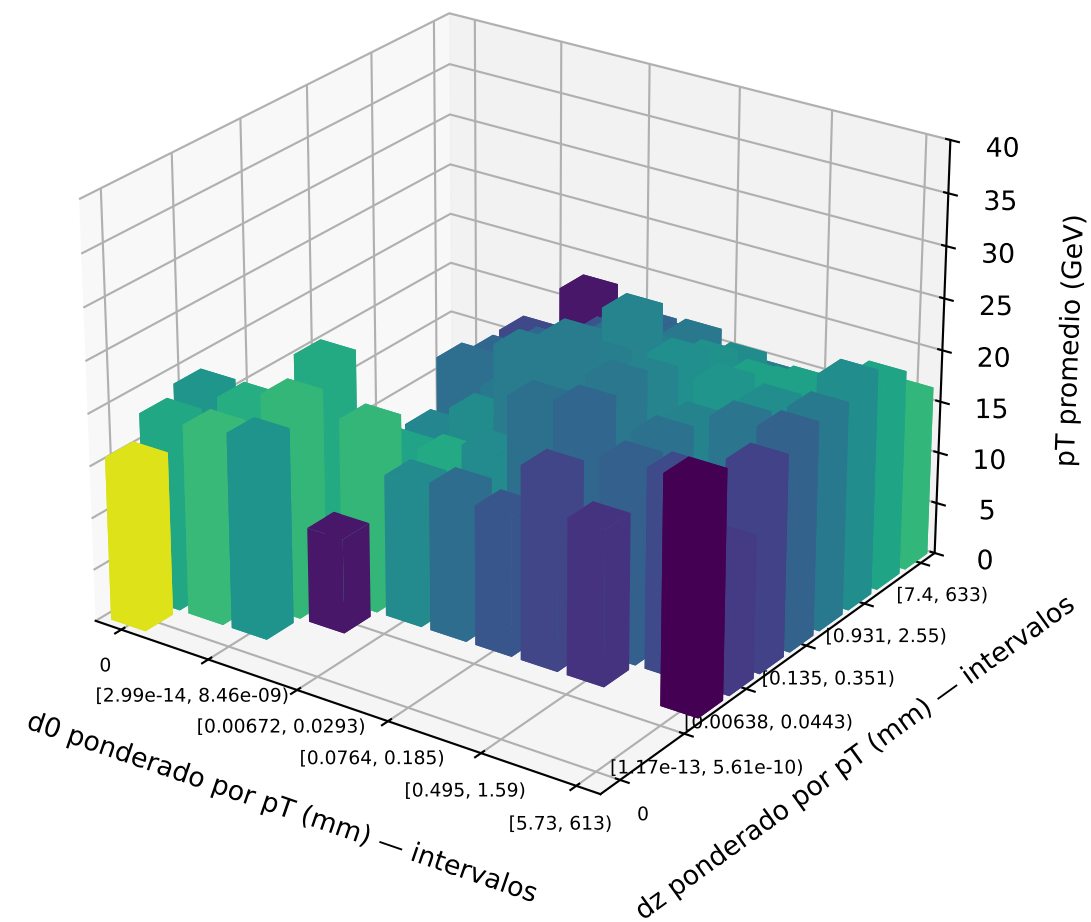
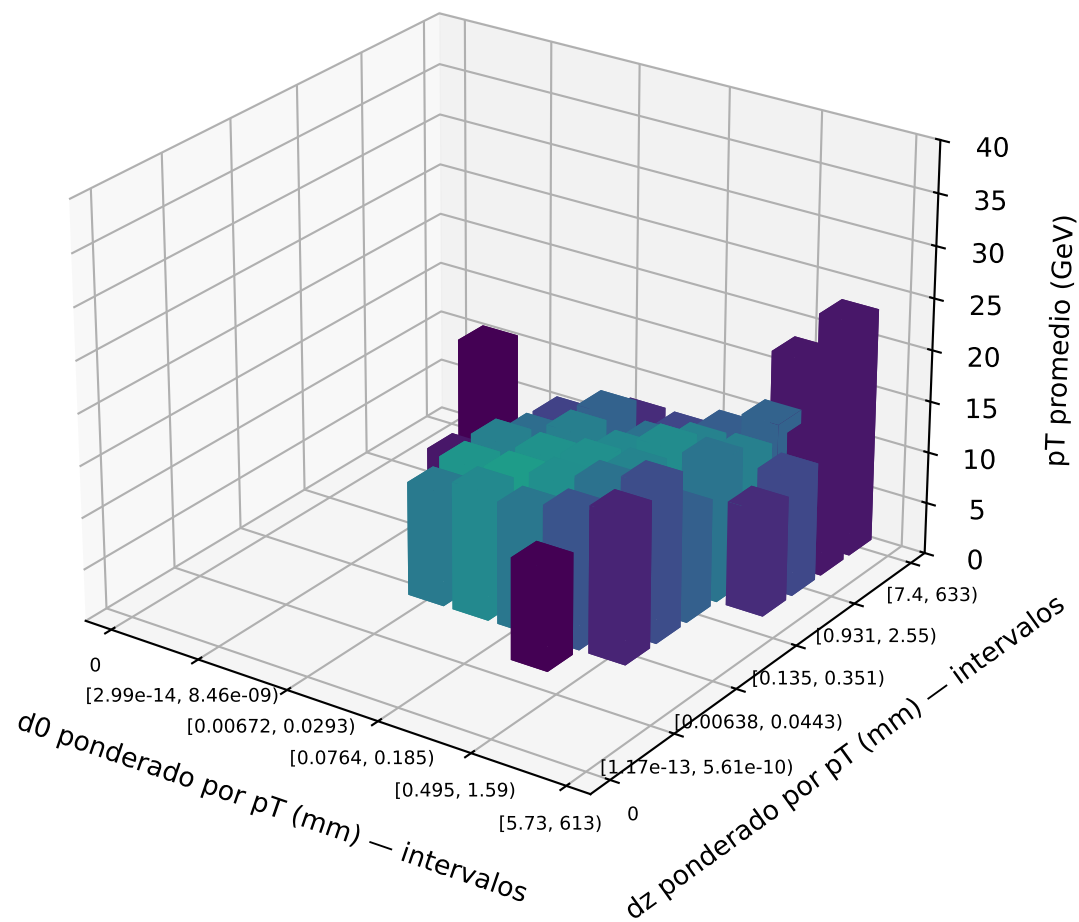
TP — N=3,614

FN — N=3,061



FP — N=3,443

TN — N=57,433



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC8 · semilla 101

Variable	Unidad	TP N=3,614	FN N=3,061	FP N=3,443	TN N=57,433
jet_pt	GeV	12.28 ± 6.338	12.24 ± 6.972	12.24 ± 5.929	16.67 ± 11.86
jet_eta	1	-0.001635 ± 1.399	0.007363 ± 1.433	0.01688 ± 1.448	0.001843 ± 1.499
jet_abs_eta	1	1.208 ± 0.7063	1.224 ± 0.7451	1.248 ± 0.7339	1.277 ± 0.7841
jet_mass	GeV	2.547 ± 1.38	2.51 ± 1.702	2.454 ± 1.23	2.876 ± 1.896
jet_ncharged	count	2.812 ± 1.545	2.613 ± 2.019	2.502 ± 1.322	2.593 ± 1.96
jet_nneutral	count	1.856 ± 1.596	2.054 ± 1.642	1.846 ± 1.431	2.401 ± 1.8
n_tracks_cone	count	2.842 ± 1.528	2.654 ± 2.025	2.528 ± 1.3	2.622 ± 1.961
track_pt_sum	GeV	5.858 ± 4.53	4.903 ± 5.222	5.249 ± 3.942	6.844 ± 8.23
track_pt_sum_over_jet_pt	1	0.4782 ± 0.2232	0.3825 ± 0.2697	0.4311 ± 0.2264	0.3711 ± 0.2685
avg_track_pt	GeV	2.215 ± 1.372	1.624 ± 1.305	2.111 ± 1.276	2.169 ± 2.092
std_track_pt	GeV	0.8078 ± 0.8757	0.6526 ± 0.85	0.6878 ± 0.7941	1.023 ± 1.63
n_tracks_below_avg_pt	count	1.472 ± 1.147	1.398 ± 1.377	1.241 ± 0.9986	1.386 ± 1.37
n_tracks_above_avg_pt	count	1.37 ± 0.6486	1.256 ± 0.8686	1.288 ± 0.543	1.235 ± 0.8145
max_pt_ratio	1	0.2828 ± 0.1589	0.2126 ± 0.1633	0.2543 ± 0.1335	0.2096 ± 0.156
min_pt_ratio	1	0.131 ± 0.1413	0.09042 ± 0.1164	0.1282 ± 0.1094	0.08402 ± 0.09787
max_dr_ratio	1	0.156 ± 0.1415	0.1108 ± 0.1207	0.1516 ± 0.1131	0.1027 ± 0.1071
min_dr_ratio	1	0.2435 ± 0.1756	0.1792 ± 0.1666	0.2203 ± 0.143	0.1789 ± 0.155
dr_max_pt	1	0.1419 ± 0.08509	0.132 ± 0.09988	0.1508 ± 0.08567	0.1257 ± 0.09922
dr_min_pt	1	0.2041 ± 0.09996	0.188 ± 0.1202	0.2069 ± 0.09483	0.1898 ± 0.1178
dr_max_dr	1	0.2432 ± 0.09514	0.2241 ± 0.1247	0.2377 ± 0.08841	0.2206 ± 0.1214
dr_min_dr	1	0.1123 ± 0.07351	0.1033 ± 0.08545	0.126 ± 0.07948	0.103 ± 0.08829
pt_difference	GeV	1.9 ± 2.145	1.596 ± 2.226	1.593 ± 1.867	2.512 ± 4.118
r50	1	0.1484 ± 0.08229	0.1397 ± 0.09788	0.1579 ± 0.08267	0.1323 ± 0.0976
r95	1	0.2405 ± 0.09467	0.2215 ± 0.1238	0.2353 ± 0.08855	0.2143 ± 0.12
mean_abs_d0 *	mm	0.1641 ± 0.1798	2.551 ± 11.62	0.1634 ± 0.25	1.261 ± 7.943
max_abs_d0 *	mm	0.2846 ± 0.2984	6.969 ± 22.27	0.294 ± 0.385	3.343 ± 16.71
std_abs_d0 *	mm	0.09134 ± 0.1181	2.689 ± 8.622	0.1009 ± 0.157	1.287 ± 6.715
mean_abs_dz *	mm	0.4125 ± 0.5419	3.753 ± 17.39	0.4421 ± 1.916	1.674 ± 9.733
max_abs_dz *	mm	0.7044 ± 0.8433	9.824 ± 28.51	0.7565 ± 2.081	4.395 ± 23.19
std_abs_dz *	mm	0.2228 ± 0.3208	3.737 ± 10.23	0.2469 ± 0.4058	1.705 ± 9.565
pt_weighted_abs_d0 *	mm	0.1716 ± 0.1847	2.394 ± 11.51	0.1749 ± 0.2622	1.2 ± 7.955
pt_weighted_abs_dz *	mm	0.4327 ± 0.5614	3.203 ± 17.15	0.4691 ± 1.913	1.371 ± 8.424

* Variable utilizada como entrada del modelo.

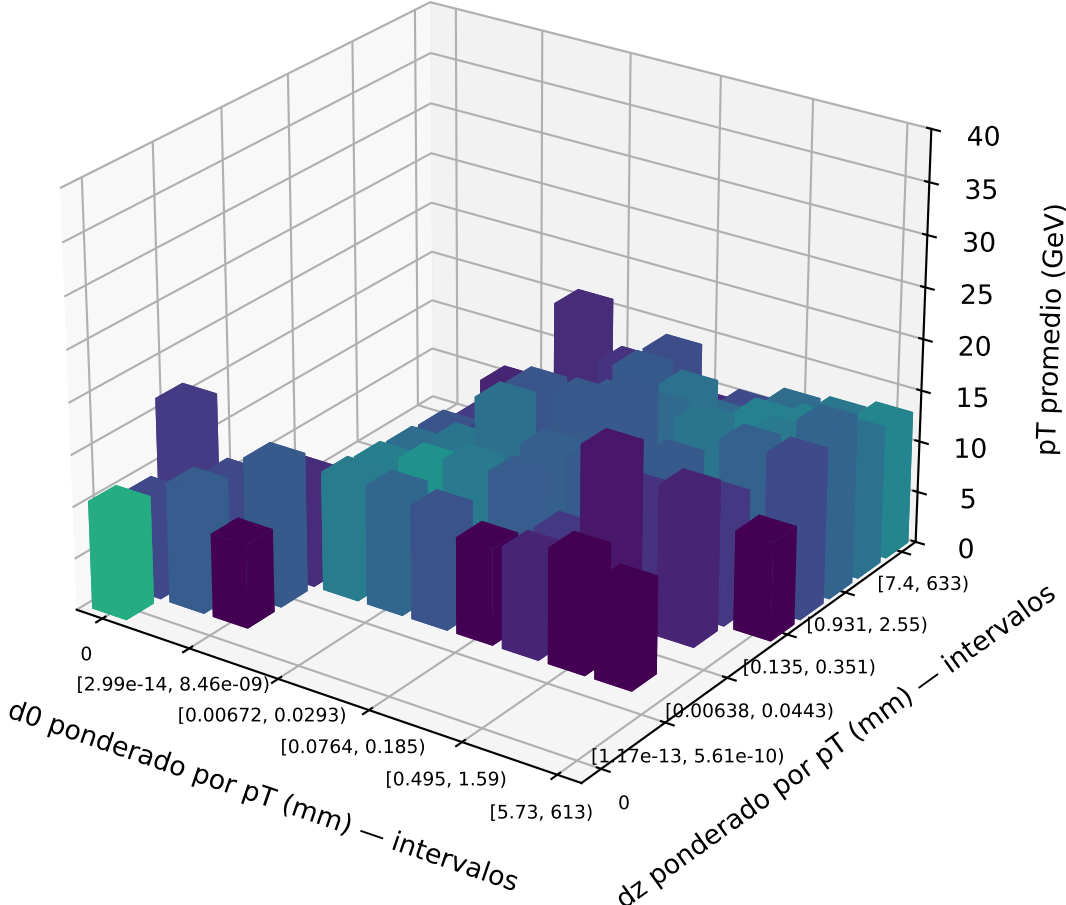
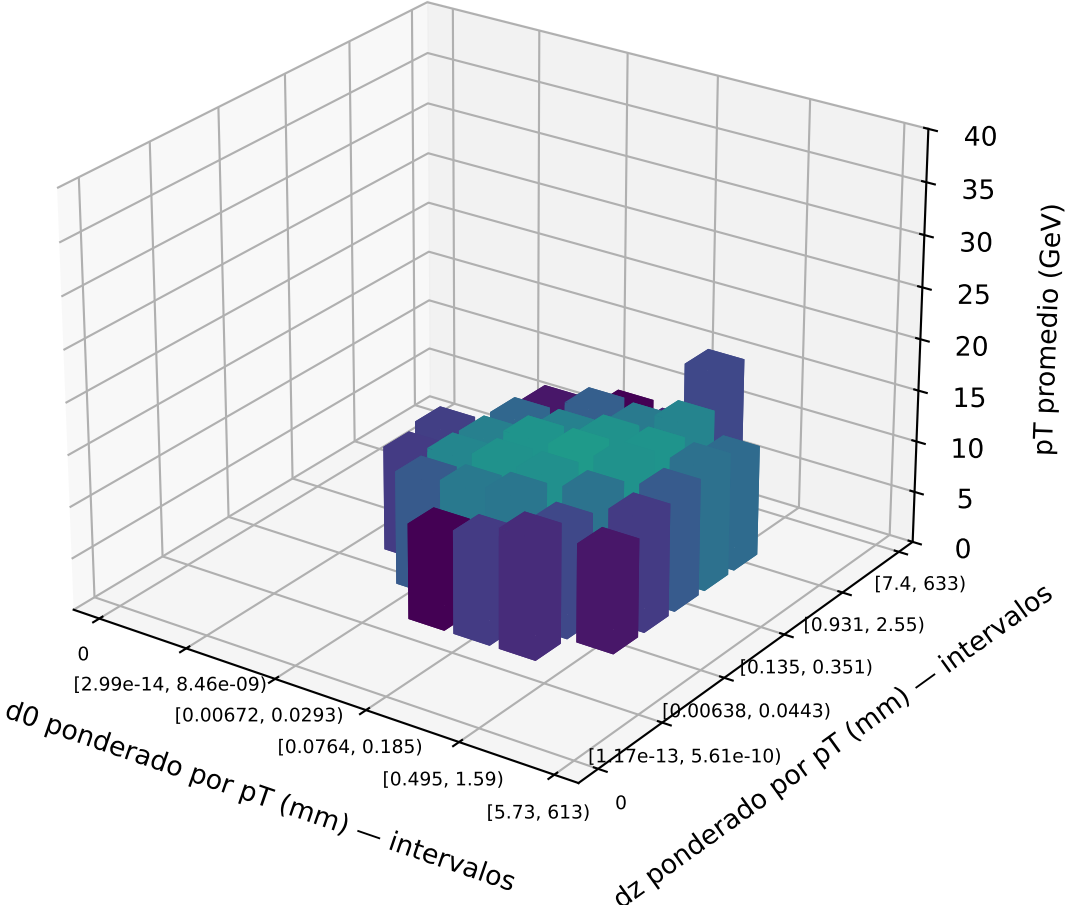
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC8 · semilla 42

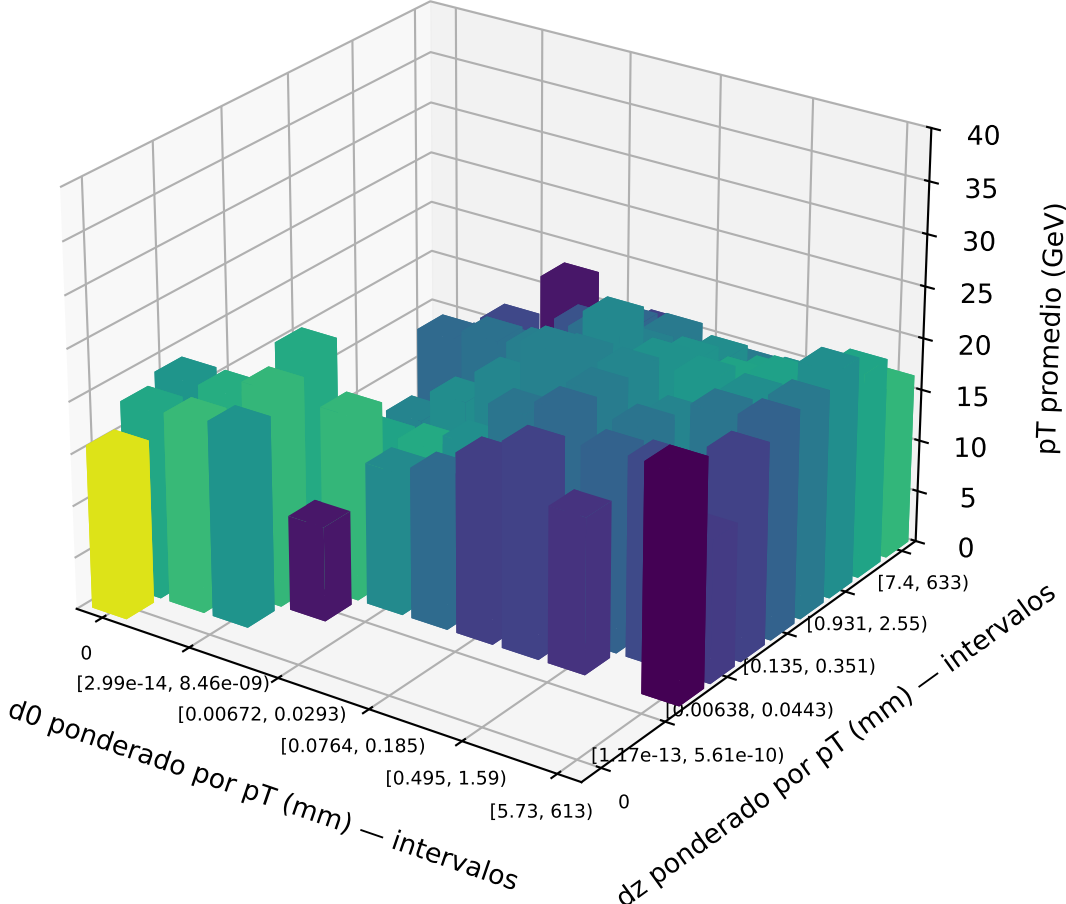
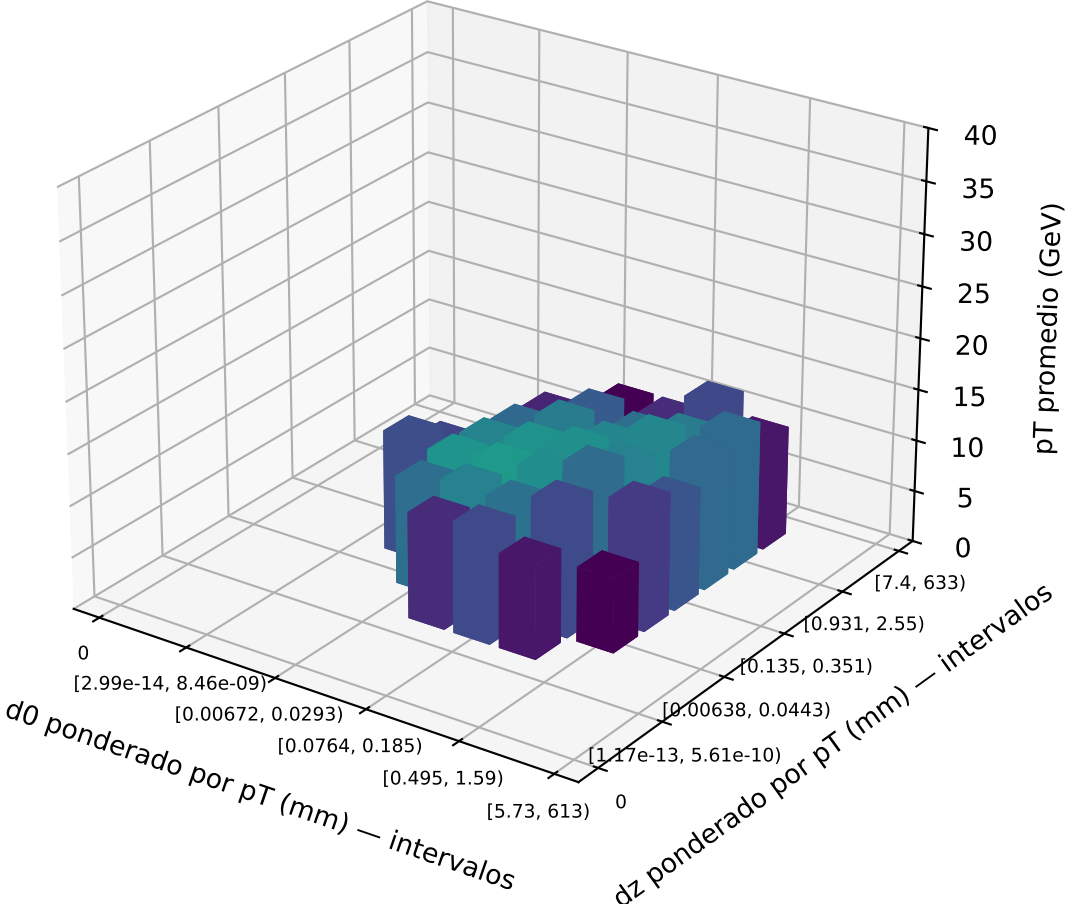
TP — N=3,339

FN — N=3,336



FP — N=2,886

TN — N=57,990



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC8 · semilla 42

Variable	Unidad	TP N=3,339	FN N=3,336	FP N=2,886	TN N=57,990
jet_pt	GeV	12.11 ± 6.142	12.42 ± 7.093	11.85 ± 5.49	16.64 ± 11.83
jet_eta	1	0.01025 ± 1.399	-0.005271 ± 1.43	-0.006923 ± 1.486	0.003172 ± 1.496
jet_abs_eta	1	1.207 ± 0.7077	1.223 ± 0.7407	1.29 ± 0.7362	1.275 ± 0.7835
jet_mass	GeV	2.475 ± 1.344	2.585 ± 1.706	2.305 ± 1.118	2.879 ± 1.892
jet_ncharged	count	2.697 ± 1.504	2.744 ± 2.02	2.292 ± 1.227	2.602 ± 1.957
jet_nneutral	count	1.844 ± 1.588	2.05 ± 1.645	1.859 ± 1.409	2.395 ± 1.799
n_tracks_cone	count	2.721 ± 1.487	2.792 ± 2.024	2.316 ± 1.209	2.631 ± 1.957
track_pt_sum	GeV	5.694 ± 4.352	5.146 ± 5.348	4.914 ± 3.658	6.846 ± 8.204
track_pt_sum_over_jet_pt	1	0.472 ± 0.2239	0.3966 ± 0.2688	0.418 ± 0.2255	0.3724 ± 0.2683
avg_track_pt	GeV	2.272 ± 1.437	1.616 ± 1.222	2.17 ± 1.251	2.166 ± 2.087
std_track_pt	GeV	0.7711 ± 0.8611	0.7021 ± 0.8724	0.6347 ± 0.7759	1.023 ± 1.624
n_tracks_below_avg_pt	count	1.378 ± 1.121	1.499 ± 1.38	1.084 ± 0.9368	1.393 ± 1.368
n_tracks_above_avg_pt	count	1.343 ± 0.6202	1.293 ± 0.8764	1.232 ± 0.4953	1.239 ± 0.8139
max_pt_ratio	1	0.2855 ± 0.1612	0.2157 ± 0.1607	0.2598 ± 0.1376	0.2098 ± 0.1556
min_pt_ratio	1	0.1403 ± 0.1491	0.08449 ± 0.1052	0.1423 ± 0.1205	0.08375 ± 0.09707
max_dr_ratio	1	0.1646 ± 0.1485	0.106 ± 0.1108	0.1644 ± 0.1225	0.1025 ± 0.1065
min_dr_ratio	1	0.2485 ± 0.1774	0.1795 ± 0.1644	0.2286 ± 0.1466	0.1789 ± 0.1547
dr_max_pt	1	0.1411 ± 0.08529	0.1337 ± 0.09868	0.1503 ± 0.08499	0.126 ± 0.09917
dr_min_pt	1	0.2008 ± 0.1005	0.1926 ± 0.1186	0.2006 ± 0.09458	0.1902 ± 0.1177
dr_max_dr	1	0.2367 ± 0.09697	0.2321 ± 0.1218	0.2276 ± 0.09001	0.2213 ± 0.1212
dr_min_dr	1	0.1129 ± 0.07475	0.1035 ± 0.08341	0.1283 ± 0.08031	0.1031 ± 0.08817
pt_difference	GeV	1.8 ± 2.096	1.721 ± 2.275	1.445 ± 1.793	2.511 ± 4.103
r50	1	0.1472 ± 0.08284	0.1415 ± 0.09634	0.1556 ± 0.08248	0.1327 ± 0.09754
r95	1	0.2342 ± 0.0964	0.2294 ± 0.121	0.2256 ± 0.09016	0.215 ± 0.1197
mean_abs_d0 *	mm	0.1497 ± 0.146	2.368 ± 11.14	0.122 ± 0.1586	1.252 ± 7.905
max_abs_d0 *	mm	0.254 ± 0.2454	6.448 ± 21.4	0.2073 ± 0.2492	3.318 ± 16.63
std_abs_d0 *	mm	0.08071 ± 0.09995	2.485 ± 8.287	0.06915 ± 0.1077	1.277 ± 6.684
mean_abs_dz *	mm	0.3666 ± 0.4245	3.523 ± 16.68	0.3045 ± 0.4139	1.669 ± 9.697
max_abs_dz *	mm	0.6079 ± 0.6617	9.169 ± 27.4	0.5152 ± 0.6636	4.372 ± 23.08
std_abs_dz *	mm	0.1908 ± 0.2629	3.48 ± 9.839	0.1743 ± 0.2915	1.695 ± 9.52
pt_weighted_abs_d0 *	mm	0.1572 ± 0.1504	2.225 ± 11.04	0.1308 ± 0.1672	1.193 ± 7.917
pt_weighted_abs_dz *	mm	0.3904 ± 0.4558	3.017 ± 16.44	0.332 ± 0.462	1.37 ± 8.395

* Variable utilizada como entrada del modelo.

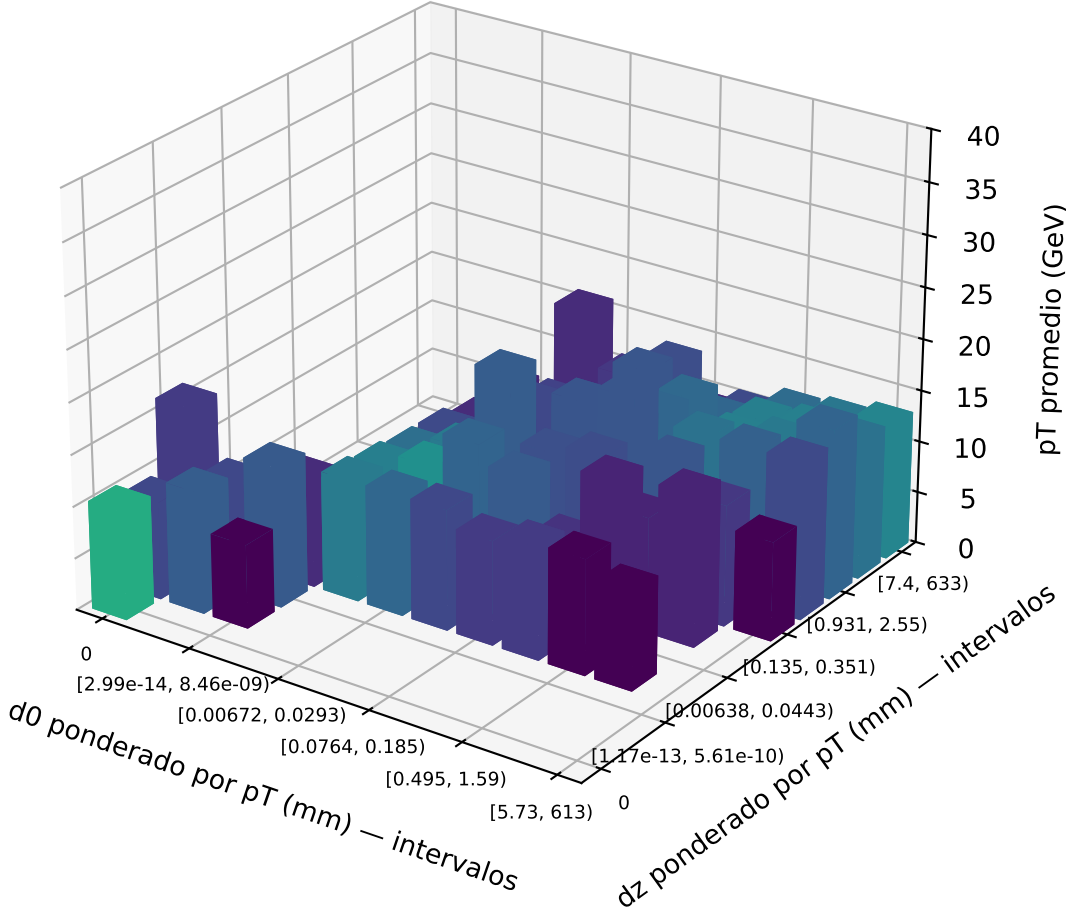
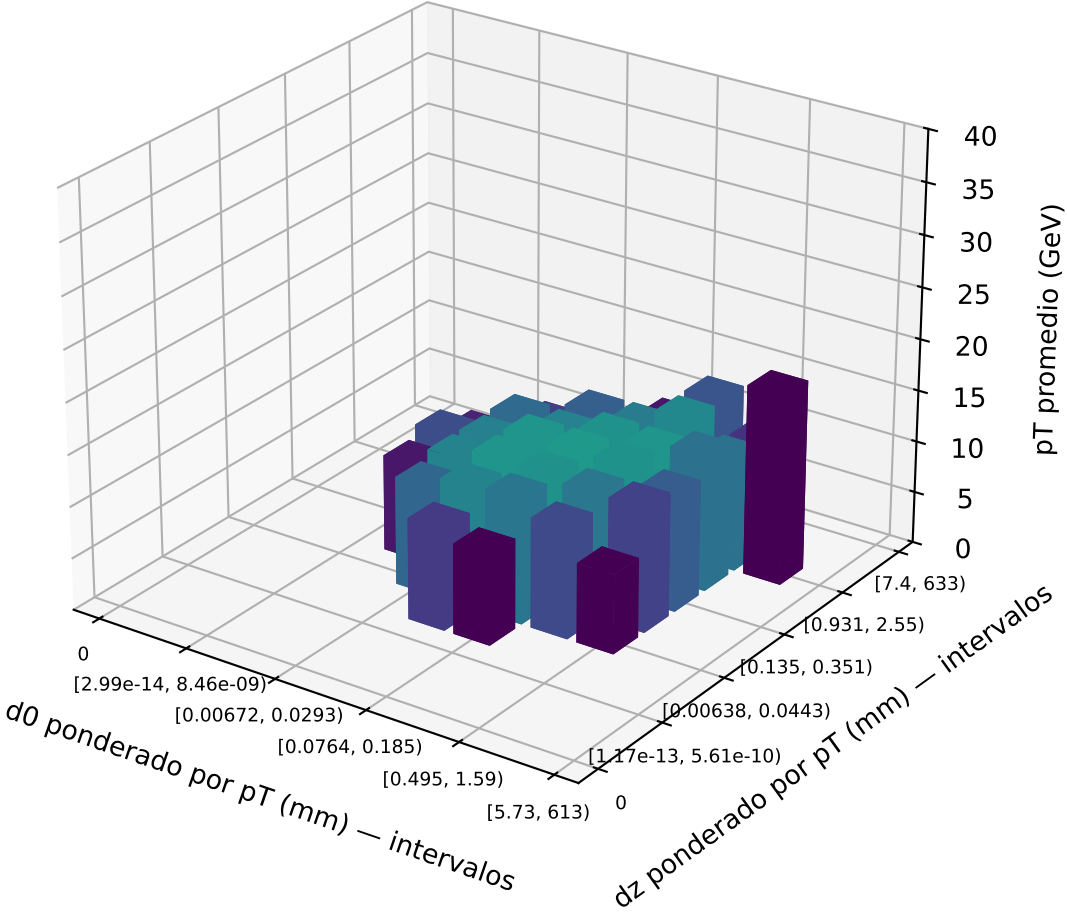
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC8 · semilla 73

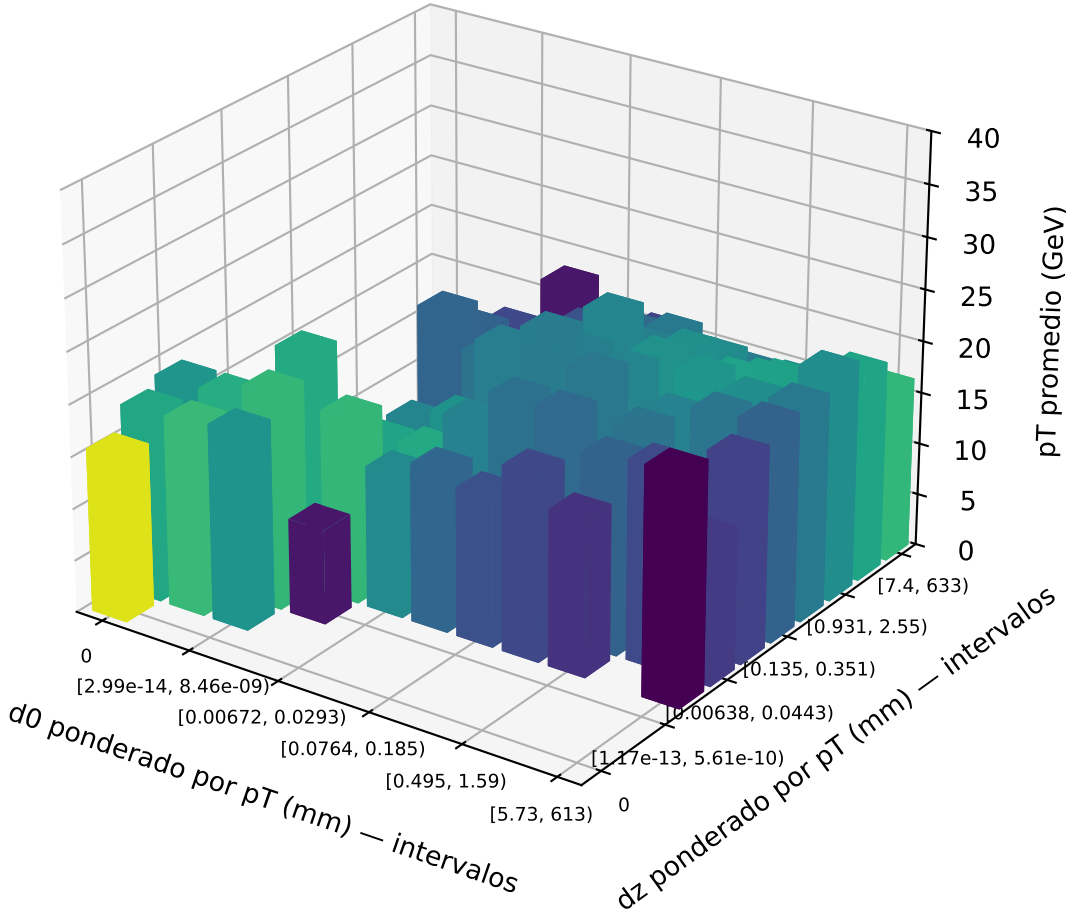
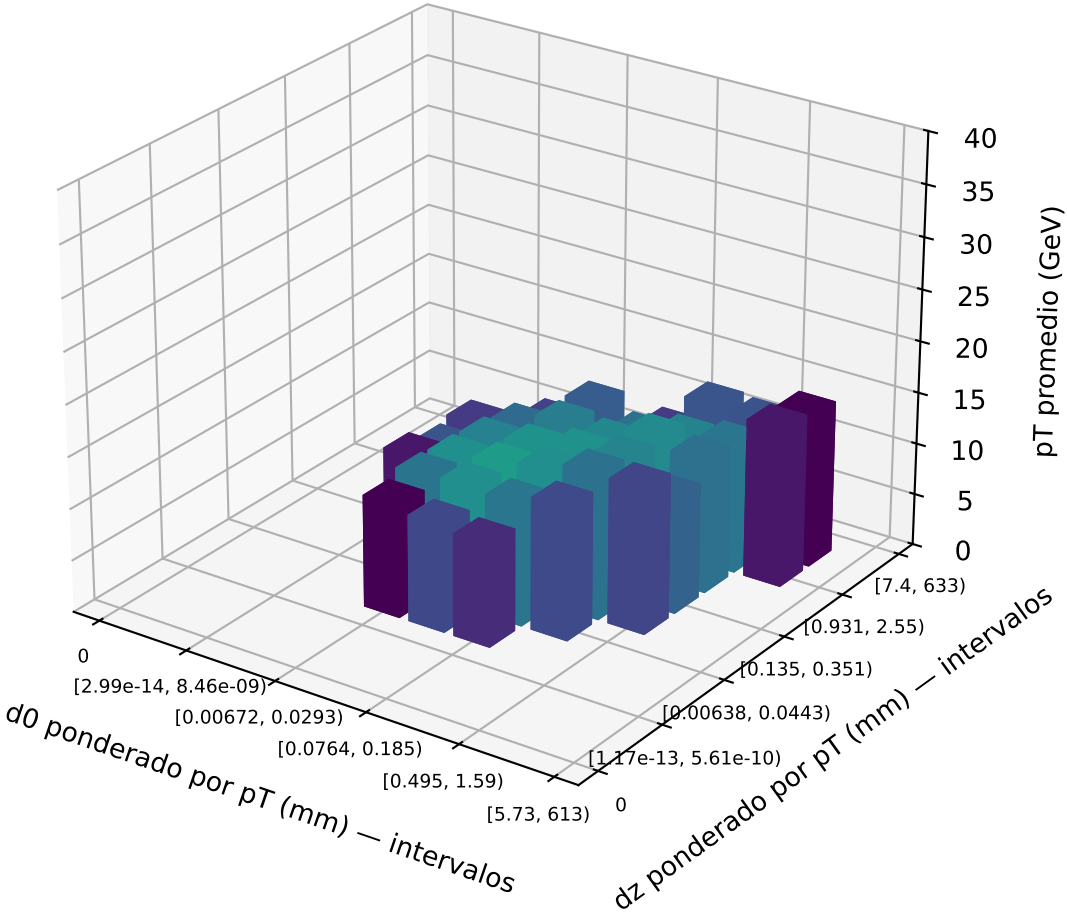
TP — N=3,562

FN — N=3,113



FP — N=3,294

TN — N=57,582



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC8 · semilla 73

Variable	Unidad	TP N=3,562	FN N=3,113	FP N=3,294	TN N=57,582
jet_pt	GeV	12.21 ± 6.285	12.32 ± 7.017	12.21 ± 5.965	16.66 ± 11.85
jet_eta	1	-0.000559 ± 1.391	0.005981 ± 1.441	-0.000785 ± 1.44	0.002892 ± 1.499
jet_abs_eta	1	1.198 ± 0.7076	1.235 ± 0.7427	1.239 ± 0.7344	1.278 ± 0.7839
jet_mass	GeV	2.512 ± 1.367	2.55 ± 1.709	2.427 ± 1.228	2.876 ± 1.894
jet_ncharged	count	2.763 ± 1.546	2.672 ± 2.015	2.468 ± 1.346	2.595 ± 1.957
jet_nneutral	count	1.847 ± 1.599	2.061 ± 1.637	1.84 ± 1.421	2.4 ± 1.8
n_tracks_cone	count	2.793 ± 1.535	2.713 ± 2.016	2.49 ± 1.332	2.624 ± 1.958
track_pt_sum	GeV	5.781 ± 4.456	5.006 ± 5.299	5.25 ± 3.999	6.84 ± 8.22
track_pt_sum_over_jet_pt	1	0.4755 ± 0.2237	0.3873 ± 0.2699	0.4298 ± 0.2257	0.3714 ± 0.2684
avg_track_pt	GeV	2.243 ± 1.4	1.602 ± 1.258	2.149 ± 1.227	2.167 ± 2.092
std_track_pt	GeV	0.7886 ± 0.8653	0.6771 ± 0.8661	0.6803 ± 0.8035	1.023 ± 1.628
n_tracks_below_avg_pt	count	1.431 ± 1.148	1.446 ± 1.375	1.206 ± 1.017	1.388 ± 1.369
n_tracks_above_avg_pt	count	1.362 ± 0.6482	1.267 ± 0.8669	1.284 ± 0.5455	1.236 ± 0.8138
max_pt_ratio	1	0.285 ± 0.1614	0.2113 ± 0.1595	0.2576 ± 0.135	0.2095 ± 0.1559
min_pt_ratio	1	0.1362 ± 0.147	0.08517 ± 0.1061	0.1337 ± 0.1151	0.08382 ± 0.0974
max_dr_ratio	1	0.1605 ± 0.1466	0.1064 ± 0.1119	0.1564 ± 0.1176	0.1025 ± 0.1068
min_dr_ratio	1	0.2469 ± 0.178	0.1764 ± 0.1624	0.2246 ± 0.1447	0.1788 ± 0.1548
dr_max_pt	1	0.1416 ± 0.08534	0.1326 ± 0.09945	0.1486 ± 0.08535	0.1259 ± 0.09924
dr_min_pt	1	0.2022 ± 0.1007	0.1904 ± 0.1195	0.2034 ± 0.09506	0.19 ± 0.1178
dr_max_dr	1	0.2402 ± 0.09672	0.2278 ± 0.1233	0.2331 ± 0.0899	0.2209 ± 0.1213
dr_min_dr	1	0.1123 ± 0.07399	0.1035 ± 0.0848	0.125 ± 0.07947	0.1031 ± 0.08829
pt_difference	GeV	1.85 ± 2.11	1.658 ± 2.269	1.576 ± 1.892	2.511 ± 4.113
r50	1	0.1477 ± 0.0826	0.1406 ± 0.09742	0.1554 ± 0.0826	0.1325 ± 0.09761
r95	1	0.2376 ± 0.09615	0.2252 ± 0.1225	0.2306 ± 0.08985	0.2146 ± 0.1199
mean_abs_d0 *	mm	0.1532 ± 0.1593	2.523 ± 11.52	0.1352 ± 0.1796	1.259 ± 7.933
max_abs_d0 *	mm	0.2672 ± 0.2749	6.877 ± 22.09	0.2561 ± 0.3228	3.338 ± 16.69
std_abs_d0 *	mm	0.08688 ± 0.1099	2.651 ± 8.555	0.09236 ± 0.1404	1.284 ± 6.707
mean_abs_dz *	mm	0.3838 ± 0.4624	3.73 ± 17.24	0.3503 ± 0.5199	1.676 ± 9.73
max_abs_dz *	mm	0.6615 ± 0.7883	9.721 ± 28.29	0.6382 ± 0.8944	4.392 ± 23.16
std_abs_dz *	mm	0.2108 ± 0.3114	3.692 ± 10.15	0.2255 ± 0.3922	1.702 ± 9.553
pt_weighted_abs_d0 *	mm	0.162 ± 0.1665	2.368 ± 11.41	0.1484 ± 0.195	1.199 ± 7.945
pt_weighted_abs_dz *	mm	0.4048 ± 0.487	3.189 ± 17.01	0.3824 ± 0.5699	1.374 ± 8.424

* Variable utilizada como entrada del modelo.

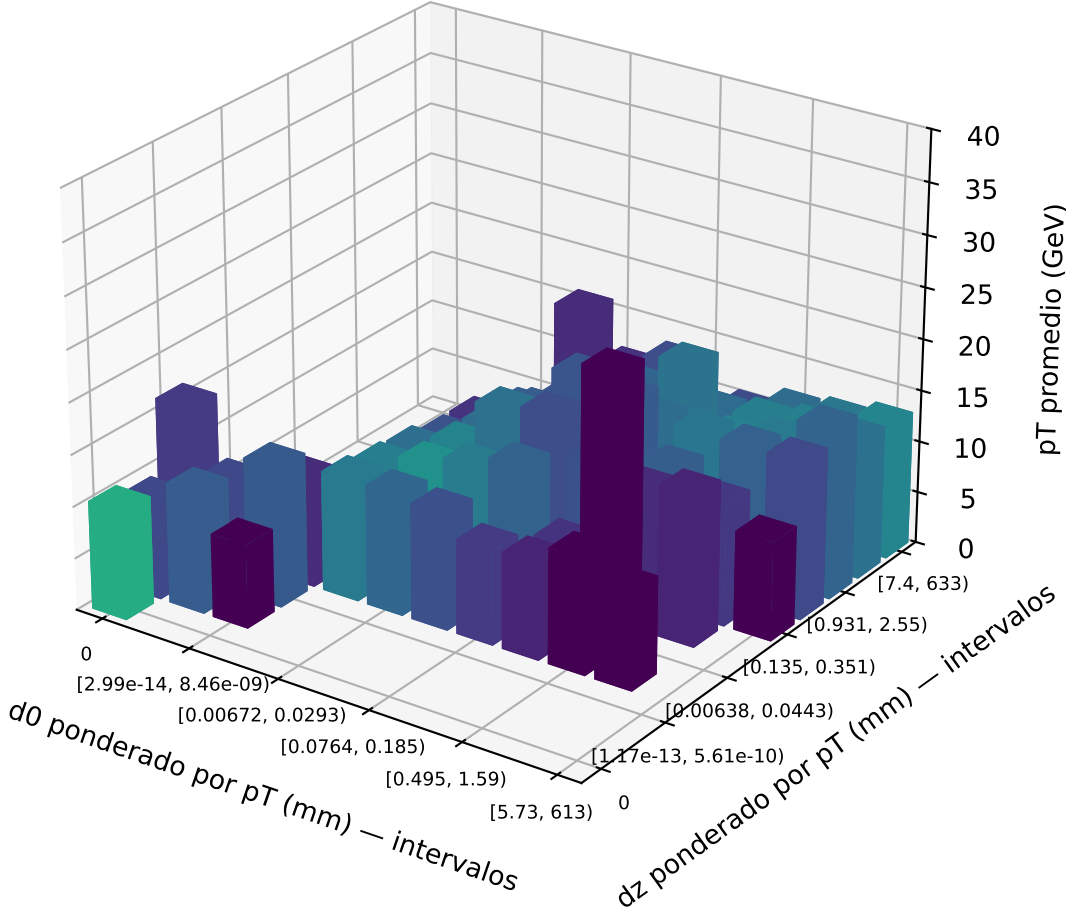
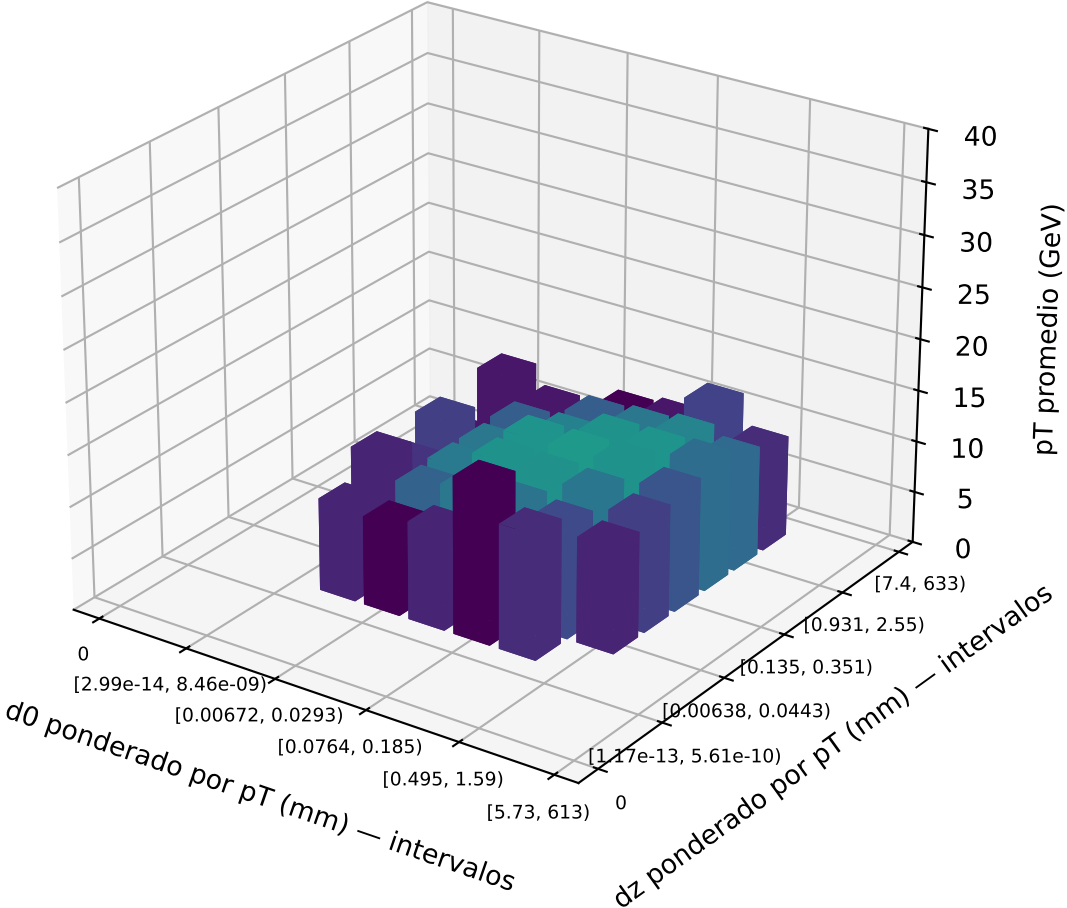
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — RF32 · referencia existente

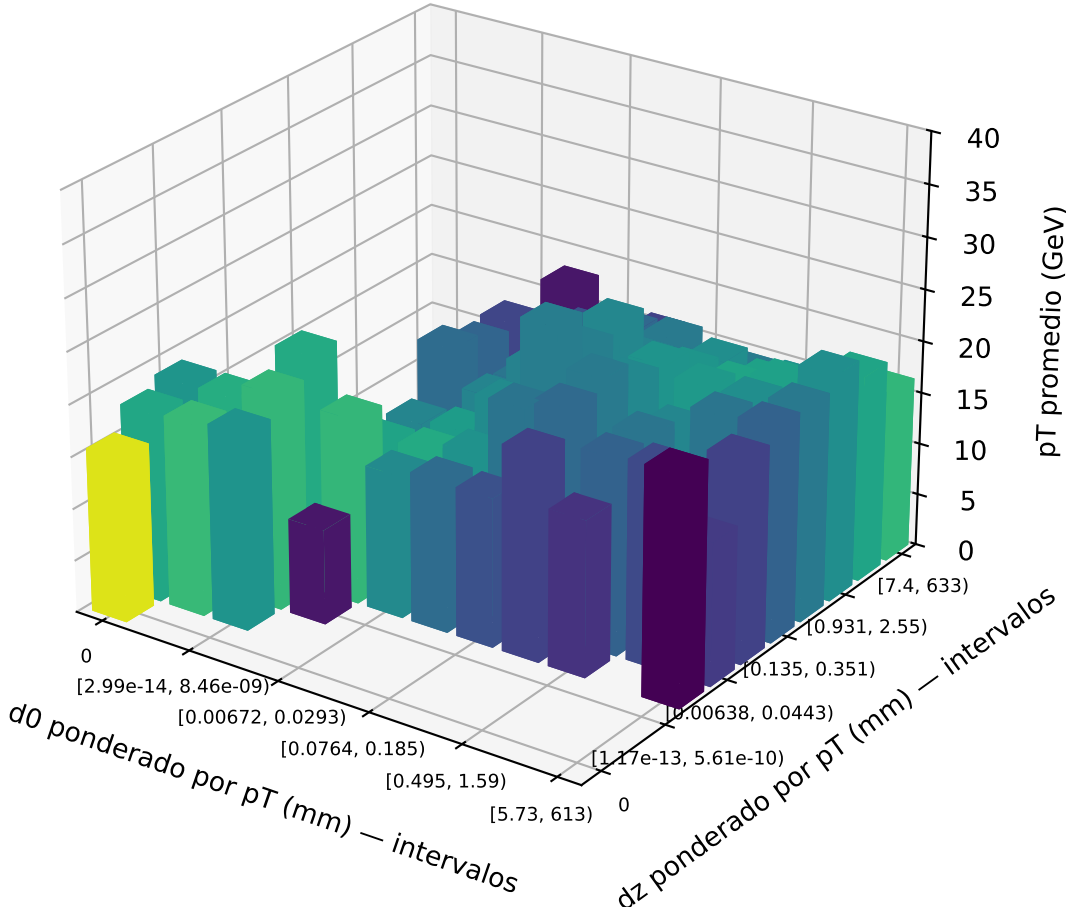
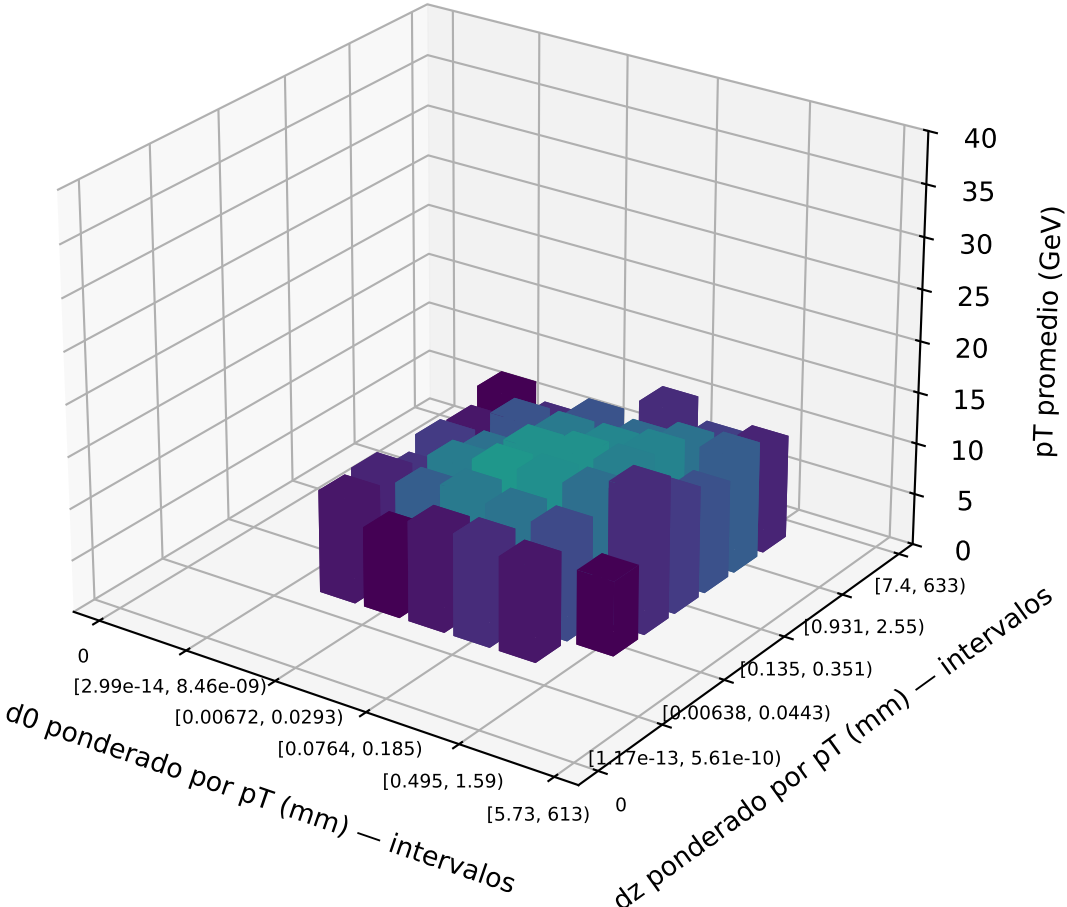
TP — N=3,204

FN — N=3,471



FP — N=2,300

TN — N=58,576



Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — RF32 · referencia existente

Variable	Unidad	TP N=3,204	FN N=3,471	FP N=2,300	TN N=58,576
jet_pt *	GeV	11.97 ± 5.062	12.53 ± 7.803	11.67 ± 4.719	16.6 ± 11.8
jet_eta *	1	0.006546 ± 1.357	-0.001252 ± 1.466	0.01184 ± 1.384	0.002334 ± 1.5
jet_abs_eta *	1	1.166 ± 0.6941	1.261 ± 0.7484	1.189 ± 0.7083	1.279 ± 0.7839
jet_mass *	GeV	2.483 ± 1.325	2.573 ± 1.707	2.359 ± 1.163	2.871 ± 1.887
jet_ncharged *	count	2.8 ± 1.55	2.647 ± 1.967	2.483 ± 1.337	2.592 ± 1.949
jet_nneutral *	count	1.763 ± 1.534	2.117 ± 1.678	1.713 ± 1.351	2.395 ± 1.796
n_tracks_cone *	count	2.825 ± 1.542	2.693 ± 1.965	2.495 ± 1.322	2.621 ± 1.95
track_pt_sum *	GeV	5.806 ± 3.56	5.063 ± 5.821	5.169 ± 3.368	6.816 ± 8.18
track_pt_sum_over_jet_pt *	1	0.4899 ± 0.2206	0.383 ± 0.2645	0.4467 ± 0.2238	0.3717 ± 0.2678
avg_track_pt *	GeV	2.305 ± 1.465	1.611 ± 1.19	2.192 ± 1.232	2.165 ± 2.081
std_track_pt *	GeV	0.7866 ± 0.8247	0.6905 ± 0.9027	0.68 ± 0.7387	1.017 ± 1.619
n_tracks_below_avg_pt *	count	1.452 ± 1.154	1.426 ± 1.348	1.223 ± 1.027	1.384 ± 1.363
n_tracks_above_avg_pt *	count	1.373 ± 0.6537	1.267 ± 0.8424	1.273 ± 0.5333	1.237 ± 0.8104
max_pt_ratio *	1	0.2952 ± 0.1681	0.2095 ± 0.1502	0.2734 ± 0.146	0.2097 ± 0.1551
min_pt_ratio *	1	0.1431 ± 0.1597	0.08404 ± 0.09115	0.1415 ± 0.1301	0.08436 ± 0.09703
max_dr_ratio *	1	0.1673 ± 0.1578	0.1058 ± 0.09933	0.1633 ± 0.131	0.1032 ± 0.1064
min_dr_ratio *	1	0.2574 ± 0.1864	0.174 ± 0.1521	0.2394 ± 0.158	0.179 ± 0.1541
dr_max_pt *	1	0.1377 ± 0.08425	0.1371 ± 0.09916	0.1459 ± 0.08183	0.1264 ± 0.09921
dr_min_pt *	1	0.2002 ± 0.1011	0.1935 ± 0.1175	0.2033 ± 0.0946	0.1902 ± 0.1175
dr_max_dr *	1	0.2383 ± 0.09862	0.2308 ± 0.1196	0.2332 ± 0.09007	0.2211 ± 0.1209
dr_min_dr *	1	0.1086 ± 0.0717	0.1078 ± 0.08578	0.122 ± 0.07548	0.1036 ± 0.08836
pt_difference *	GeV	1.837 ± 1.882	1.69 ± 2.434	1.563 ± 1.707	2.496 ± 4.091
r50 *	1	0.1442 ± 0.0817	0.1445 ± 0.09683	0.1519 ± 0.07905	0.1331 ± 0.09757
r95 *	1	0.2358 ± 0.09815	0.2281 ± 0.1188	0.2312 ± 0.09013	0.2149 ± 0.1195
mean_abs_d0 *	mm	0.1534 ± 0.1427	2.279 ± 10.94	0.1218 ± 0.126	1.241 ± 7.866
max_abs_d0 *	mm	0.263 ± 0.2354	6.199 ± 21.02	0.2171 ± 0.208	3.287 ± 16.55
std_abs_d0 *	mm	0.08284 ± 0.09344	2.39 ± 8.138	0.07275 ± 0.09013	1.265 ± 6.651
mean_abs_dz *	mm	0.3764 ± 0.4479	3.391 ± 16.36	0.3123 ± 0.4376	1.655 ± 9.649
max_abs_dz *	mm	0.6311 ± 0.6939	8.814 ± 26.92	0.5213 ± 0.6099	4.333 ± 22.97
std_abs_dz *	mm	0.1934 ± 0.2555	3.349 ± 9.668	0.1647 ± 0.235	1.68 ± 9.474
pt_weighted_abs_d0 *	mm	0.1599 ± 0.1453	2.142 ± 10.83	0.127 ± 0.1282	1.182 ± 7.878
pt_weighted_abs_dz *	mm	0.3928 ± 0.461	2.913 ± 16.13	0.3244 ± 0.4428	1.359 ± 8.354

* Variable utilizada como entrada del modelo.

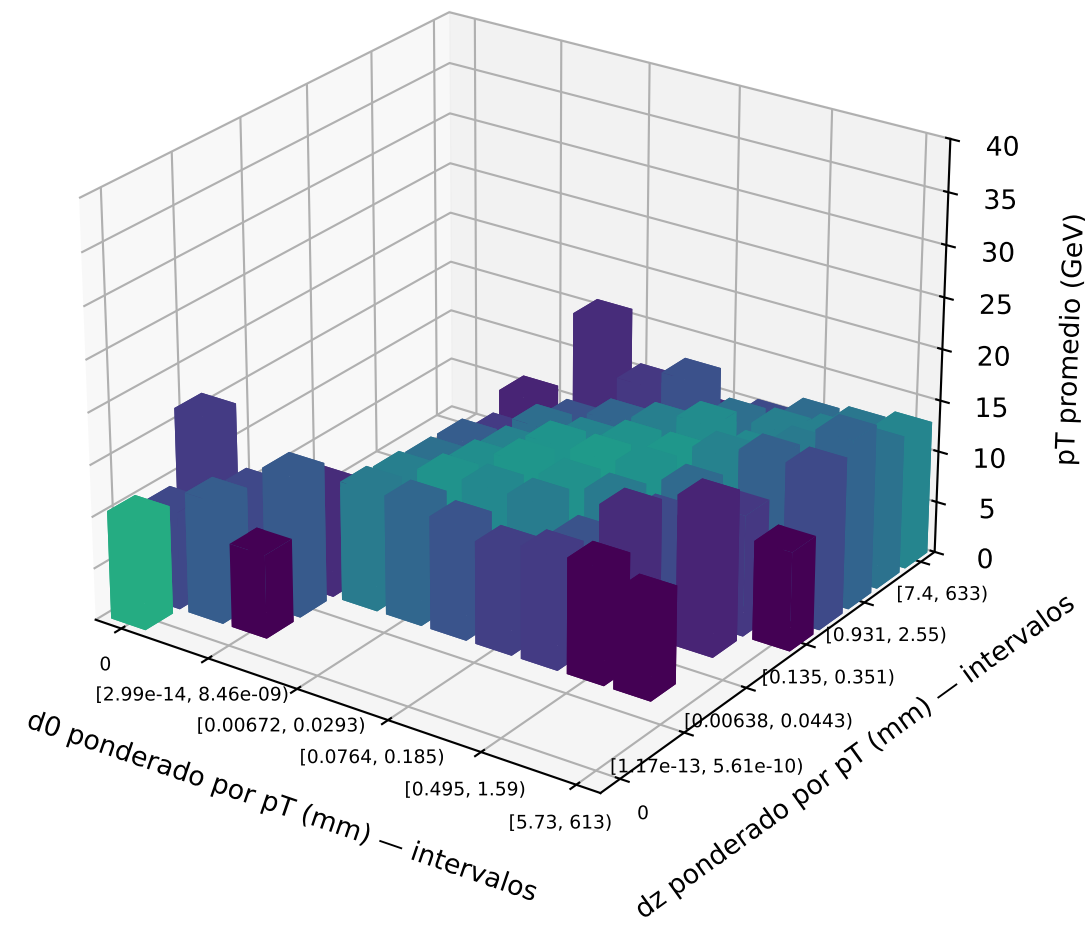
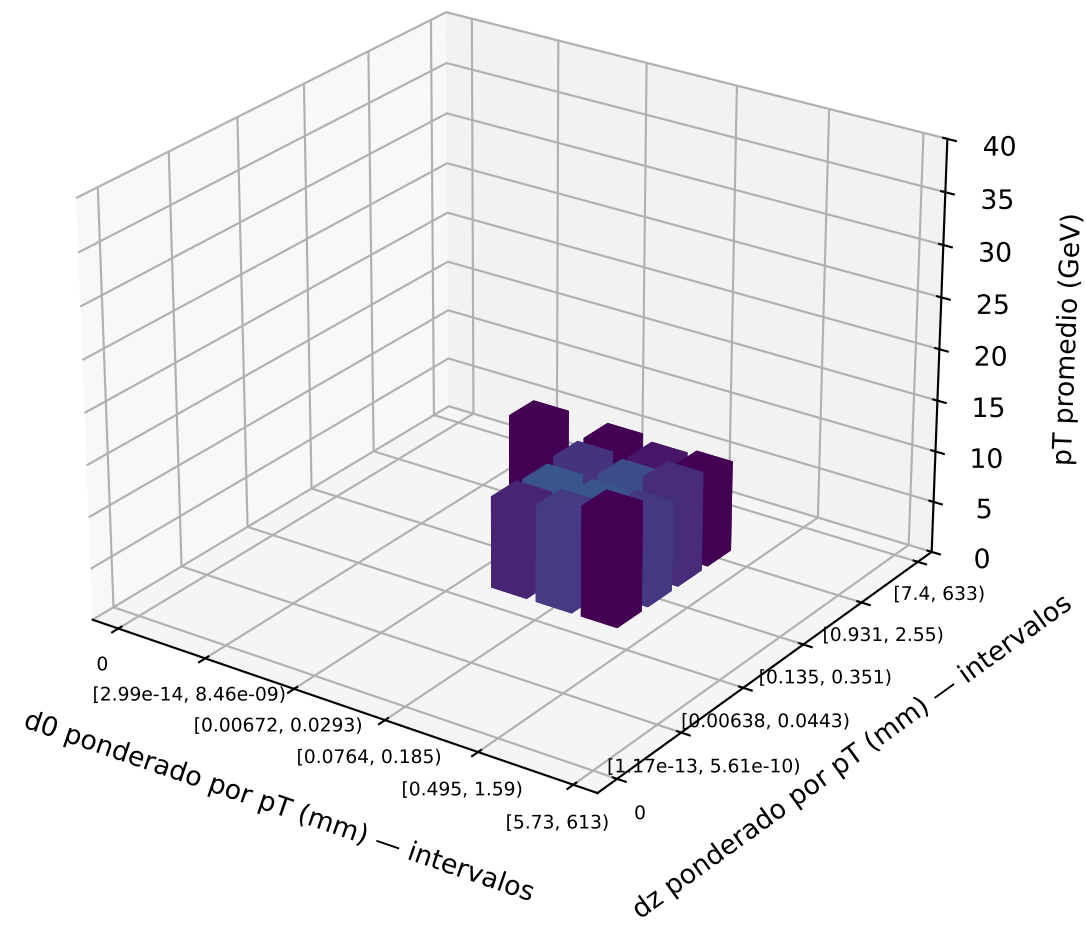
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC32 · ensemble

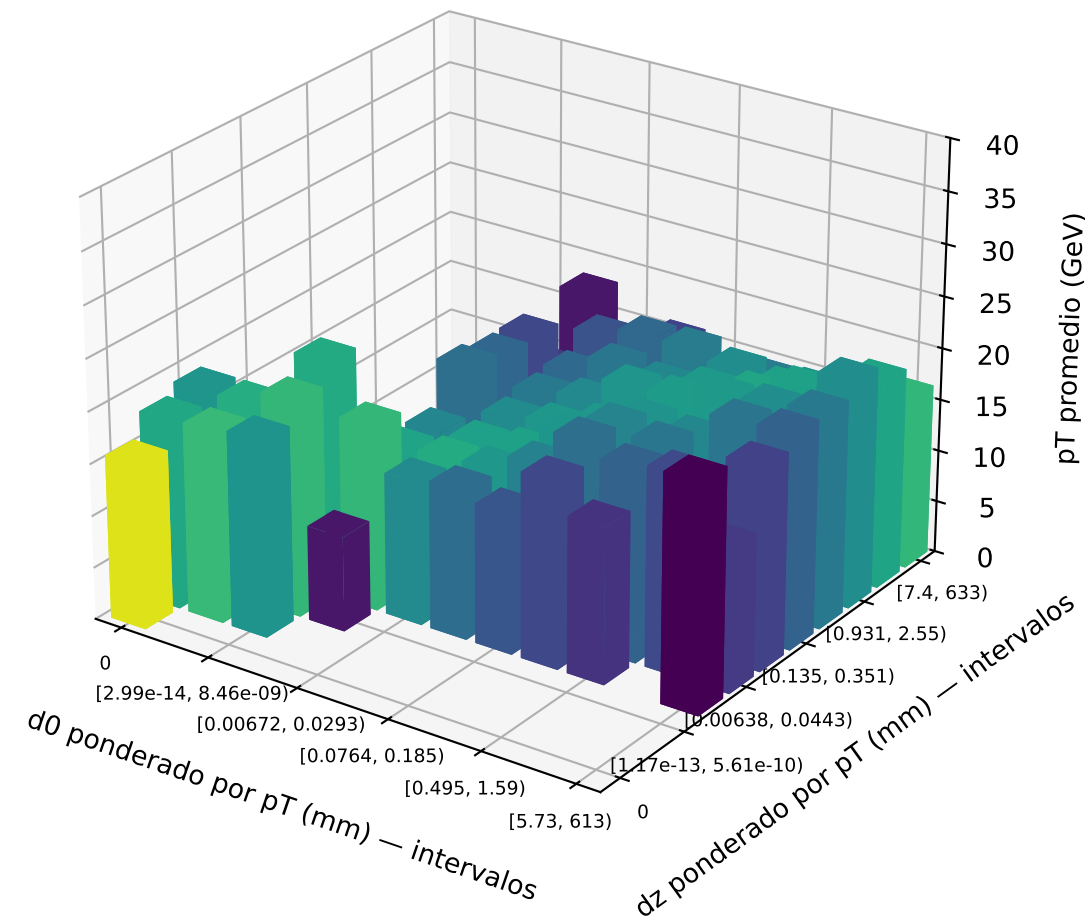
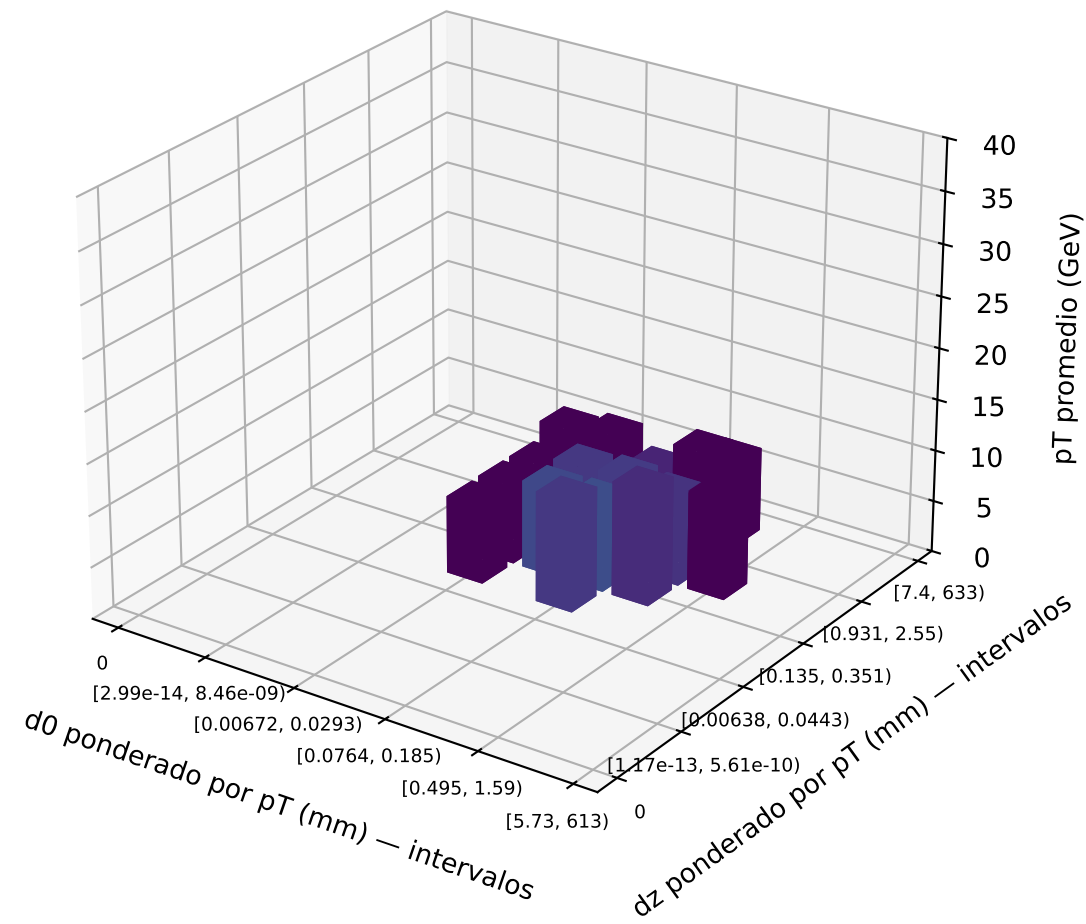
TP — N=80

FN — N=6,595



FP — N=63

TN — N=60,813



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC32 · ensemble

Variable	Unidad	TP N=80	FN N=6,595	FP N=63	TN N=60,813
jet_pt *	GeV	8.9 ± 1.541	12.3 ± 6.664	9.016 ± 1.886	16.42 ± 11.66
jet_eta *	1	0.02875 ± 1.119	0.002173 ± 1.418	-0.1885 ± 1.431	0.002891 ± 1.496
jet_abs_eta *	1	0.9278 ± 0.6267	1.219 ± 0.7248	1.303 ± 0.6194	1.275 ± 0.7815
jet_mass *	GeV	1.851 ± 0.6772	2.538 ± 1.542	2.149 ± 0.7905	2.852 ± 1.868
jet_ncharged *	count	1.962 ± 0.8866	2.73 ± 1.787	2 ± 0.9759	2.588 ± 1.93
jet_nneutral *	count	1.562 ± 0.7219	1.952 ± 1.627	1.524 ± 0.7095	2.37 ± 1.787
n_tracks_cone *	count	1.95 ± 0.8646	2.766 ± 1.782	1.984 ± 0.9511	2.617 ± 1.93
track_pt_sum *	GeV	3.761 ± 1.304	5.44 ± 4.907	3.482 ± 1.281	6.757 ± 8.06
track_pt_sum_over_jet_pt *	1	0.4227 ± 0.1334	0.4345 ± 0.2513	0.3891 ± 0.1395	0.3745 ± 0.2667
avg_track_pt *	GeV	2.147 ± 0.9364	1.942 ± 1.378	1.908 ± 0.7414	2.166 ± 2.056
std_track_pt *	GeV	0.2693 ± 0.2781	0.7423 ± 0.8706	0.2955 ± 0.3562	1.005 ± 1.597
n_tracks_below_avg_pt *	count	0.875 ± 0.7479	1.445 ± 1.262	0.873 ± 0.7662	1.379 ± 1.353
n_tracks_above_avg_pt *	count	1.075 ± 0.2634	1.321 ± 0.7631	1.111 ± 0.3143	1.238 ± 0.802
max_pt_ratio *	1	0.283 ± 0.1116	0.2502 ± 0.1652	0.2515 ± 0.09436	0.2121 ± 0.1552
min_pt_ratio *	1	0.2192 ± 0.1389	0.1111 ± 0.1314	0.1817 ± 0.1073	0.08642 ± 0.09903
max_dr_ratio *	1	0.2341 ± 0.1337	0.1341 ± 0.1338	0.1985 ± 0.1028	0.1053 ± 0.108
min_dr_ratio *	1	0.269 ± 0.12	0.2133 ± 0.1749	0.2341 ± 0.1049	0.1812 ± 0.1547
dr_max_pt *	1	0.1345 ± 0.06329	0.1374 ± 0.0926	0.1506 ± 0.06423	0.1271 ± 0.0987
dr_min_pt *	1	0.1745 ± 0.06859	0.197 ± 0.1104	0.1952 ± 0.07683	0.1907 ± 0.1168
dr_max_dr *	1	0.1878 ± 0.07379	0.235 ± 0.1104	0.2138 ± 0.07061	0.2216 ± 0.1199
dr_min_dr *	1	0.1285 ± 0.06107	0.1079 ± 0.0795	0.136 ± 0.06543	0.1043 ± 0.08799
pt_difference *	GeV	0.5919 ± 0.6306	1.775 ± 2.196	0.6399 ± 0.7496	2.462 ± 4.032
r50 *	1	0.1429 ± 0.06298	0.1444 ± 0.09016	0.1575 ± 0.06083	0.1337 ± 0.09703
r95 *	1	0.1878 ± 0.07379	0.2323 ± 0.1097	0.2138 ± 0.07061	0.2155 ± 0.1186
mean_abs_d0 *	mm	0.1191 ± 0.07604	1.272 ± 8.004	0.1255 ± 0.09311	1.2 ± 7.723
max_abs_d0 *	mm	0.1797 ± 0.133	3.388 ± 15.54	0.2361 ± 0.282	3.174 ± 16.25
std_abs_d0 *	mm	0.04999 ± 0.05975	1.298 ± 6.016	0.08339 ± 0.1412	1.221 ± 6.532
mean_abs_dz *	mm	0.3412 ± 0.2811	1.964 ± 11.97	0.4868 ± 0.5319	1.606 ± 9.474
max_abs_dz *	mm	0.551 ± 0.6172	4.939 ± 19.96	0.8873 ± 1.21	4.193 ± 22.55
std_abs_dz *	mm	0.1777 ± 0.2886	1.855 ± 7.191	0.3151 ± 0.604	1.624 ± 9.302
pt_weighted_abs_d0 *	mm	0.1208 ± 0.07972	1.204 ± 7.92	0.1211 ± 0.08851	1.143 ± 7.734
pt_weighted_abs_dz *	mm	0.3393 ± 0.2836	1.72 ± 11.77	0.4557 ± 0.4588	1.321 ± 8.202

* Variable utilizada como entrada del modelo.

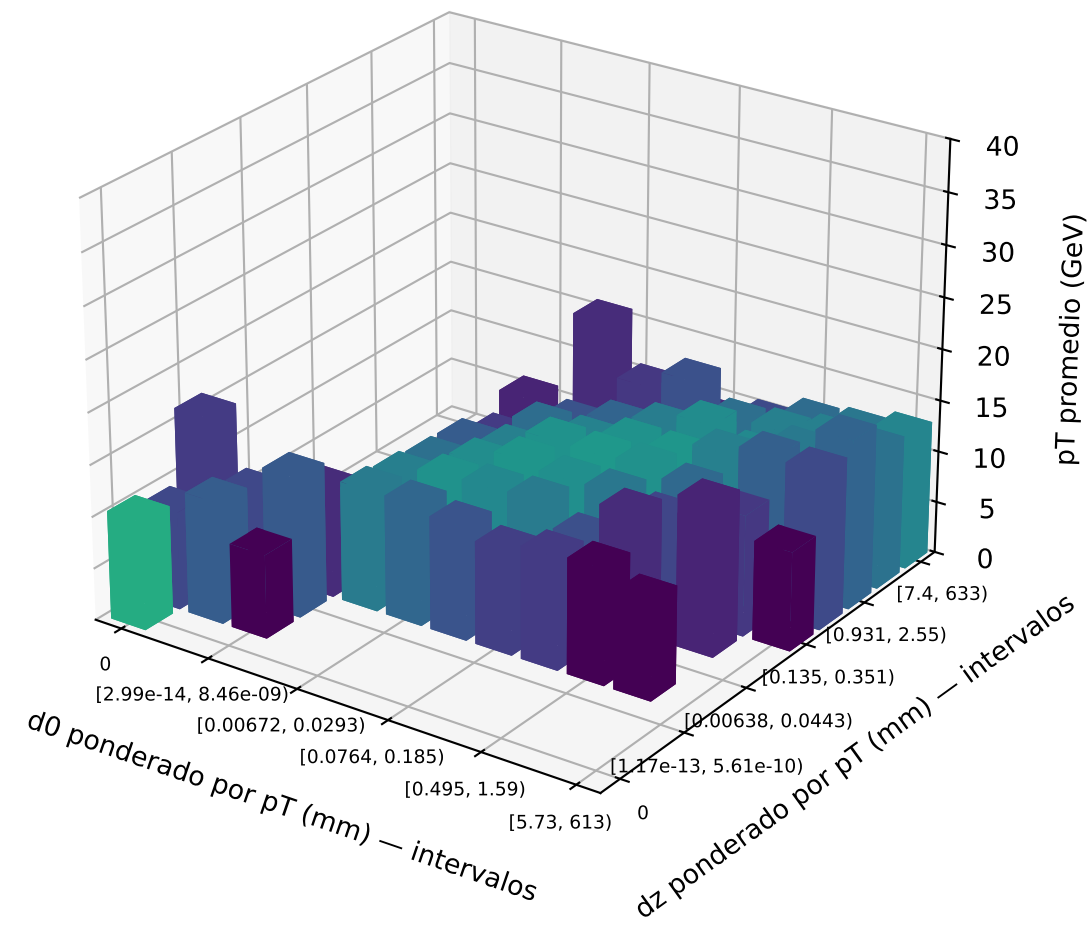
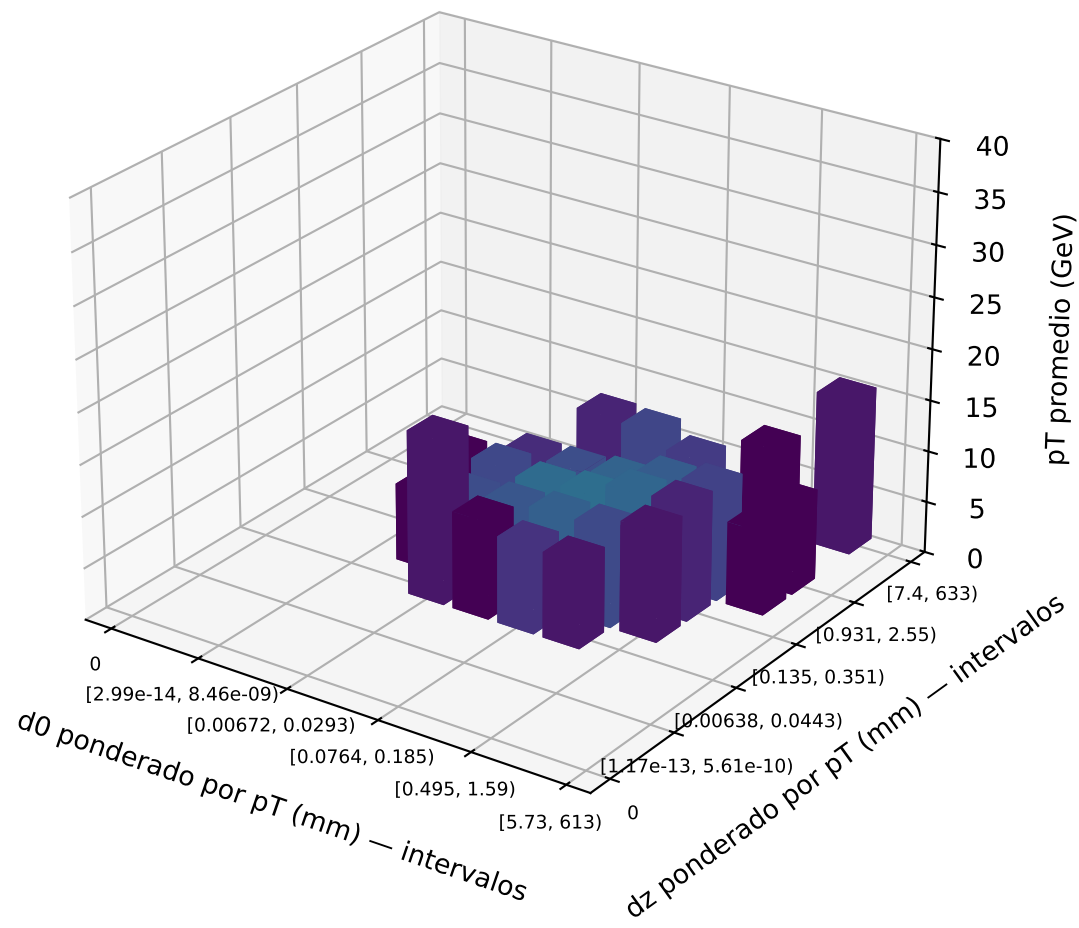
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC32 · semilla 101

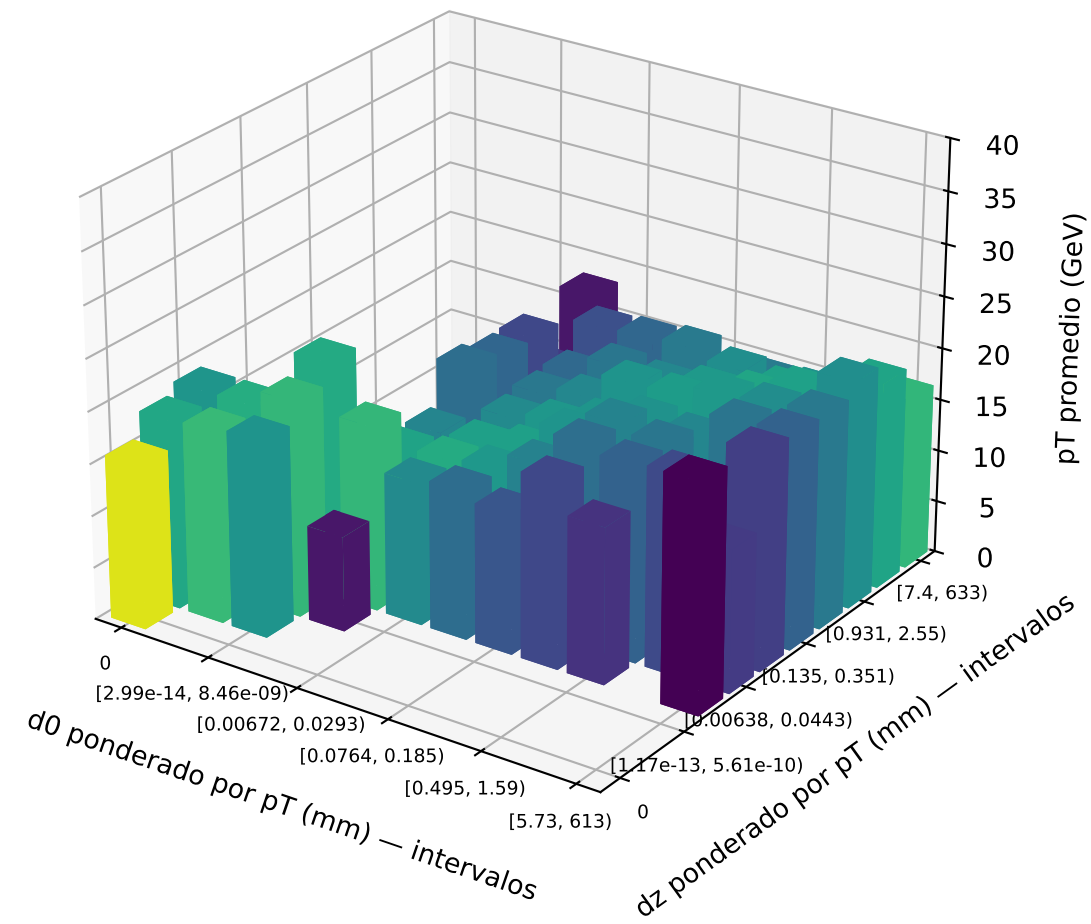
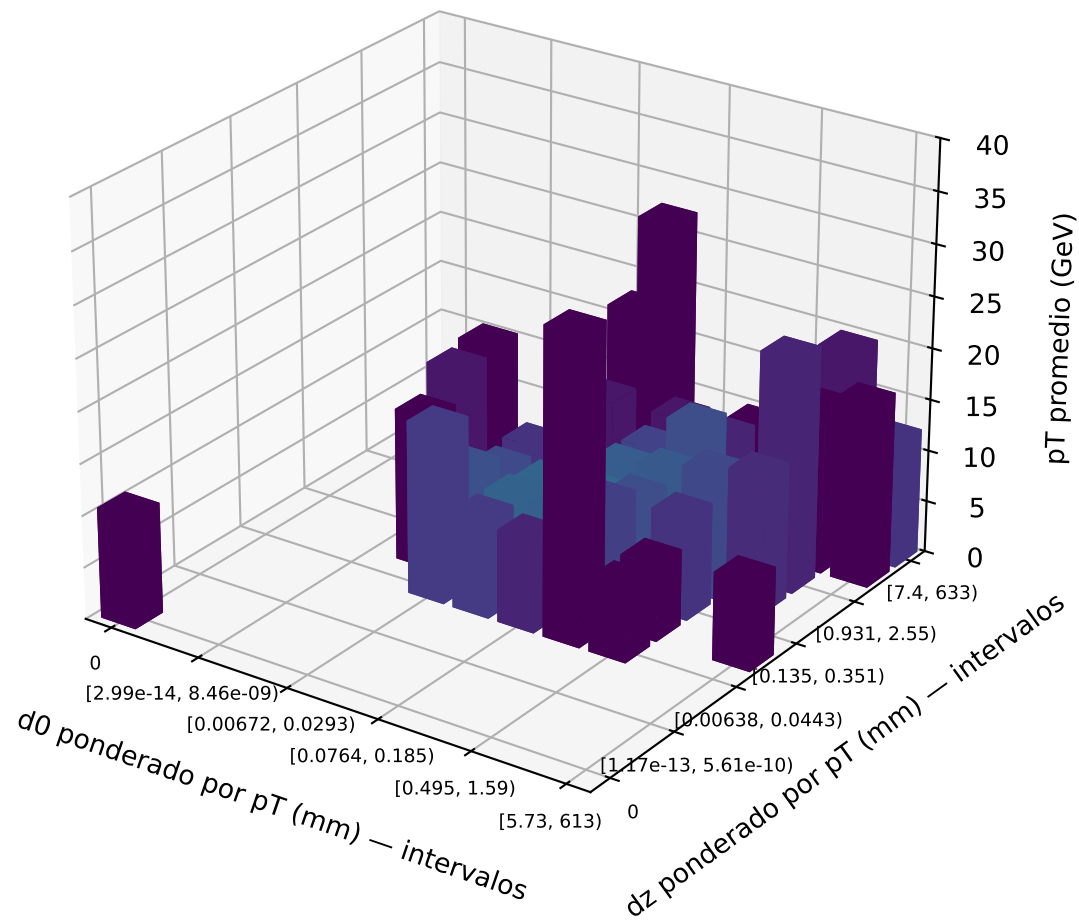
TP — N=350

FN — N=6,325



FP — N=356

TN — N=60,520



Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC32 · semilla 101

Variable	Unidad	TP N=350	FN N=6,325	FP N=356	TN N=60,520
jet_pt *	GeV	9.643 ± 2.776	12.41 ± 6.757	10.71 ± 4.418	16.45 ± 11.67
jet_eta *	1	0.1605 ± 1.301	-0.00625 ± 1.42	0.09337 ± 1.414	0.00216 ± 1.496
jet_abs_eta *	1	1.112 ± 0.6949	1.221 ± 0.7256	1.217 ± 0.727	1.276 ± 0.7816
jet_mass *	GeV	1.913 ± 1.048	2.564 ± 1.552	2.27 ± 1.229	2.855 ± 1.87
jet_ncharged *	count	2.143 ± 1.165	2.753 ± 1.804	2.374 ± 1.363	2.589 ± 1.932
jet_nneutral *	count	1.414 ± 1.153	1.976 ± 1.637	1.545 ± 1.099	2.374 ± 1.788
n_tracks_cone *	count	2.109 ± 1.144	2.792 ± 1.798	2.376 ± 1.353	2.618 ± 1.932
track_pt_sum *	GeV	4.536 ± 2.211	5.469 ± 4.984	4.927 ± 3.193	6.765 ± 8.075
track_pt_sum_over_jet_pt *	1	0.4737 ± 0.189	0.4322 ± 0.253	0.4594 ± 0.1957	0.374 ± 0.2669
avg_track_pt *	GeV	2.534 ± 1.483	1.911 ± 1.36	2.293 ± 1.198	2.165 ± 2.059
std_track_pt *	GeV	0.4045 ± 0.4765	0.755 ± 0.8804	0.496 ± 0.5555	1.007 ± 1.6
n_tracks_below_avg_pt *	count	0.96 ± 0.9493	1.465 ± 1.268	1.163 ± 1.074	1.379 ± 1.354
n_tracks_above_avg_pt *	count	1.149 ± 0.379	1.327 ± 0.7741	1.213 ± 0.4851	1.238 ± 0.8032
max_pt_ratio *	1	0.3279 ± 0.1663	0.2463 ± 0.1636	0.291 ± 0.1397	0.2117 ± 0.1552
min_pt_ratio *	1	0.2374 ± 0.201	0.1055 ± 0.1235	0.1879 ± 0.1564	0.08593 ± 0.09834
max_dr_ratio *	1	0.2516 ± 0.195	0.1288 ± 0.127	0.2055 ± 0.1498	0.1048 ± 0.1075
min_dr_ratio *	1	0.3086 ± 0.179	0.2088 ± 0.1727	0.2736 ± 0.1525	0.1807 ± 0.1545
dr_max_pt *	1	0.1217 ± 0.07971	0.1383 ± 0.09287	0.1306 ± 0.07427	0.1271 ± 0.0988
dr_min_pt *	1	0.1685 ± 0.09677	0.1983 ± 0.1105	0.1858 ± 0.09144	0.1908 ± 0.1169
dr_max_dr *	1	0.1869 ± 0.1002	0.237 ± 0.11	0.2128 ± 0.09444	0.2216 ± 0.12
dr_min_dr *	1	0.1097 ± 0.07498	0.1081 ± 0.07957	0.1139 ± 0.06867	0.1043 ± 0.08807
pt_difference *	GeV	0.9203 ± 1.111	1.807 ± 2.223	1.146 ± 1.328	2.468 ± 4.04
r50 *	1	0.1298 ± 0.07738	0.1452 ± 0.09046	0.1396 ± 0.07342	0.1337 ± 0.09712
r95 *	1	0.1869 ± 0.1002	0.2343 ± 0.1094	0.2119 ± 0.09422	0.2155 ± 0.1187
mean_abs_d0 *	mm	0.2017 ± 0.4021	1.317 ± 8.17	0.986 ± 6.227	1.2 ± 7.727
max_abs_d0 *	mm	0.3772 ± 1.124	3.514 ± 15.85	2.013 ± 10.91	3.178 ± 16.27
std_abs_d0 *	mm	0.1262 ± 0.4579	1.347 ± 6.137	0.8631 ± 4.734	1.222 ± 6.538
mean_abs_dz *	mm	0.4162 ± 1.118	2.029 ± 12.21	1.383 ± 5.559	1.606 ± 9.487
max_abs_dz *	mm	0.795 ± 3.272	5.113 ± 20.35	3.14 ± 12.13	4.195 ± 22.59
std_abs_dz *	mm	0.2736 ± 1.426	1.921 ± 7.328	1.328 ± 5.574	1.624 ± 9.315
pt_weighted_abs_d0 *	mm	0.1903 ± 0.3415	1.246 ± 8.084	0.9838 ± 6.466	1.143 ± 7.737
pt_weighted_abs_dz *	mm	0.4642 ± 1.818	1.772 ± 12.01	1.256 ± 5.211	1.321 ± 8.212

* Variable utilizada como entrada del modelo.

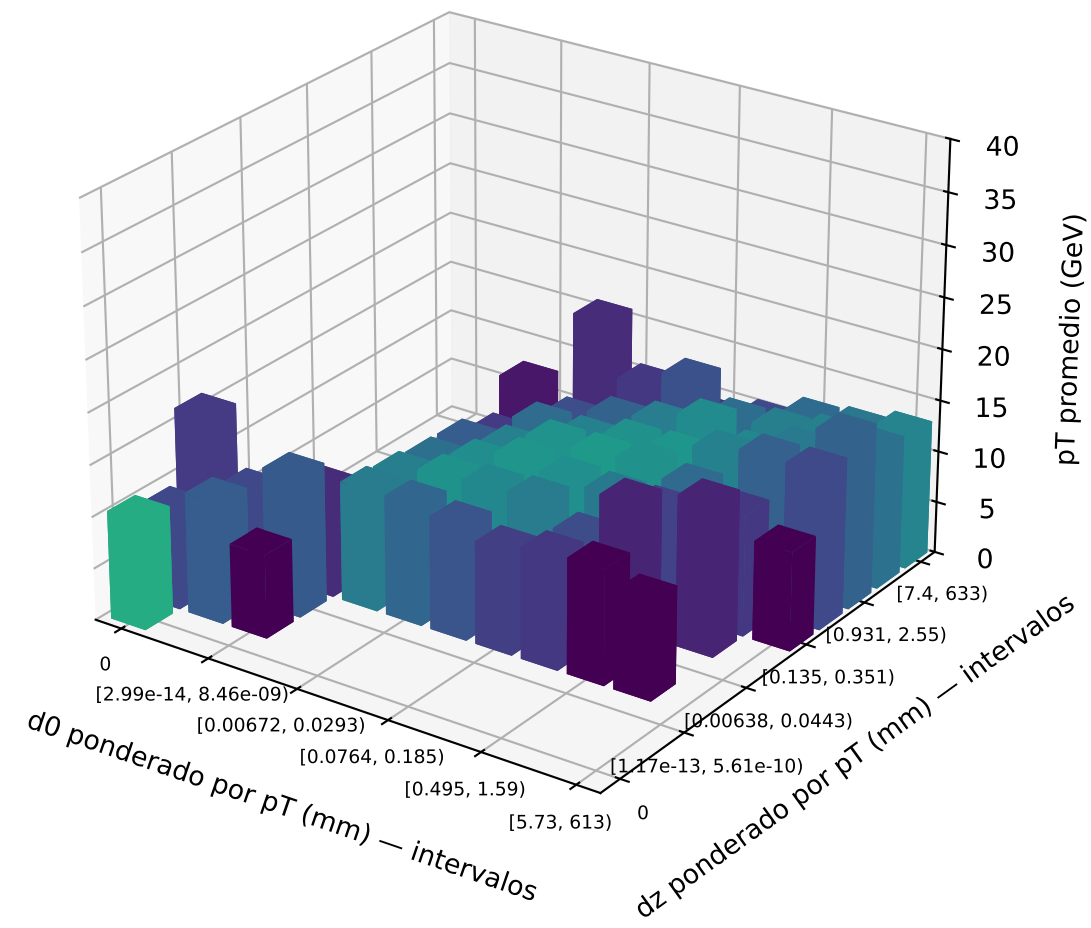
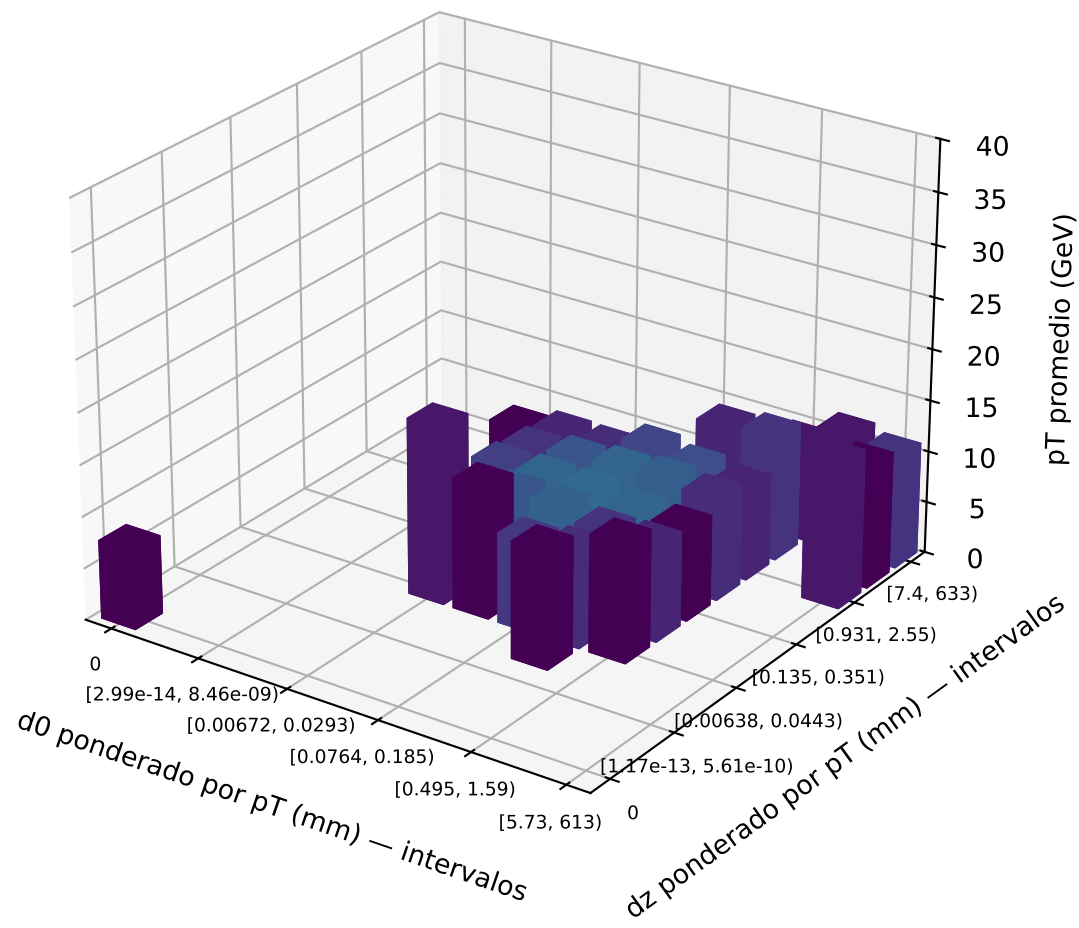
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC32 · semilla 42

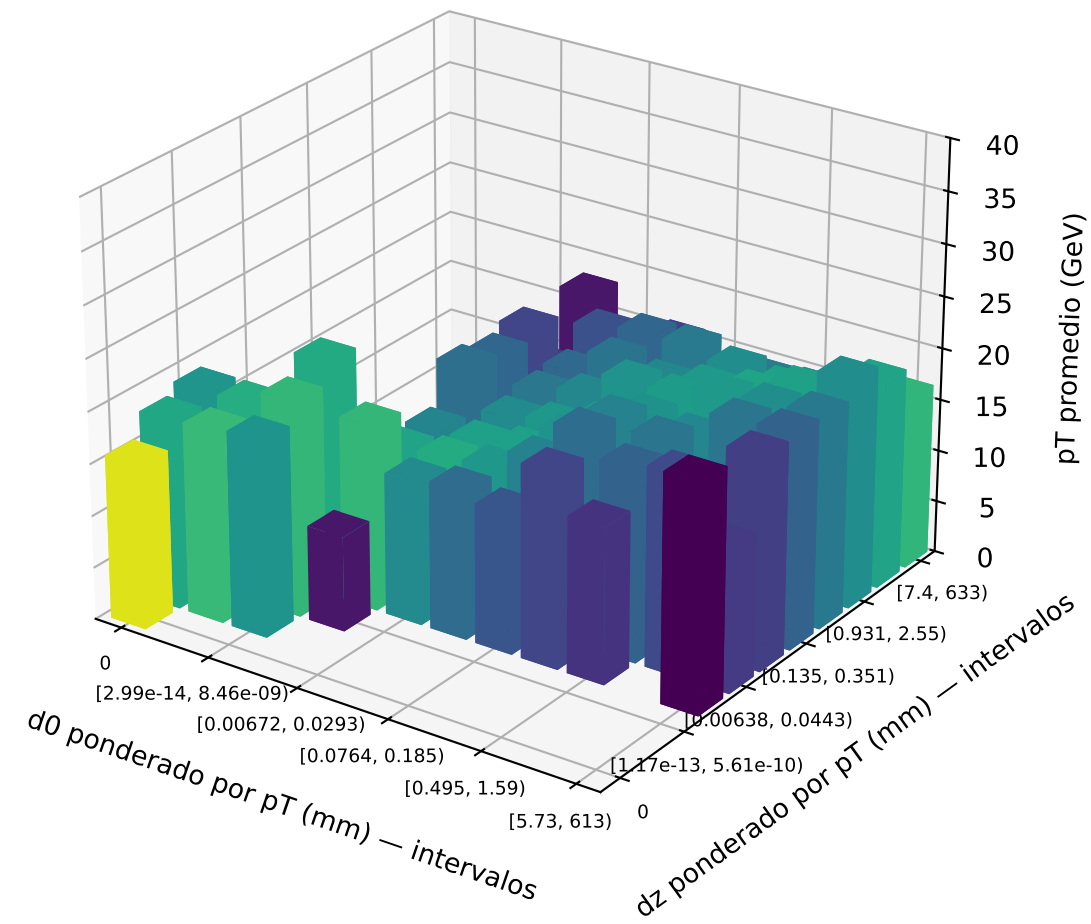
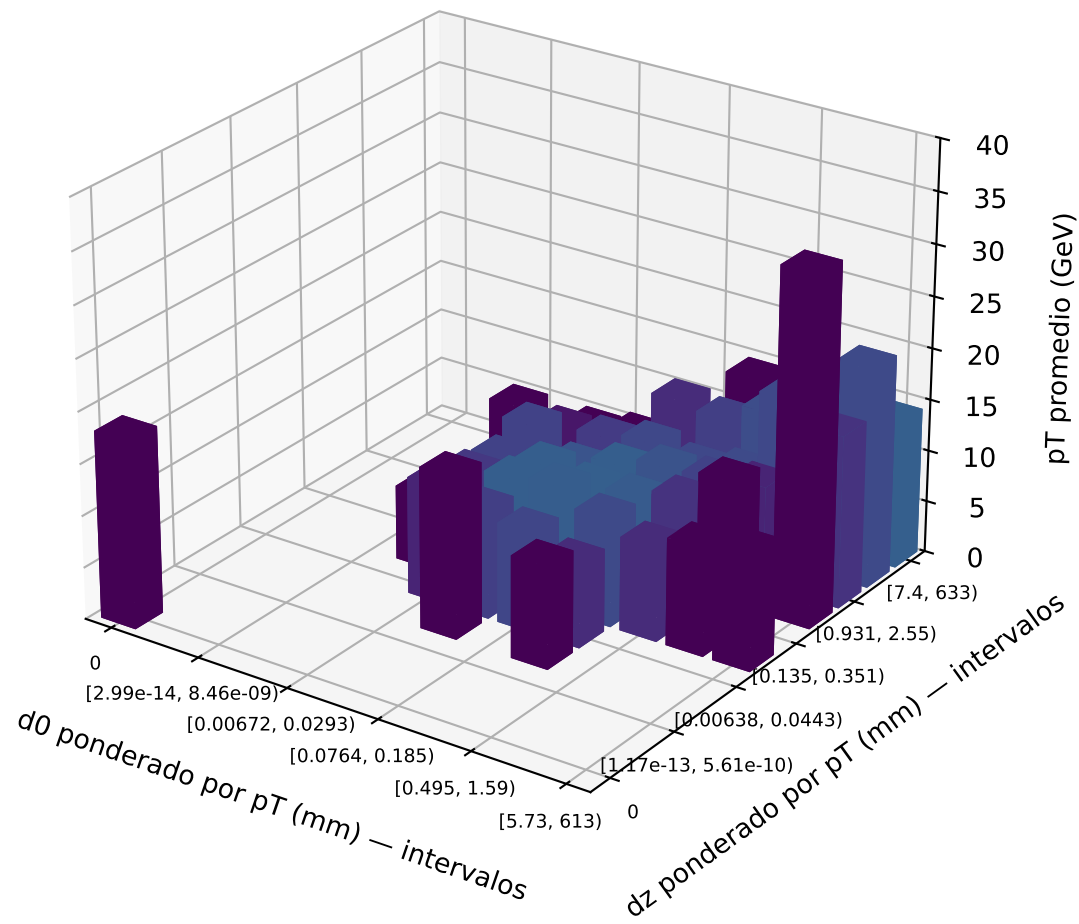
TP — N=349

FN — N=6,326



FP — N=435

TN — N=60,441



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC32 · semilla 42

Variable	Unidad	TP N=349	FN N=6,326	FP N=435	TN N=60,441
jet_pt *	GeV	9.913 ± 2.296	12.39 ± 6.772	11.75 ± 4.869	16.45 ± 11.68
jet_eta *	1	-0.07897 ± 1.391	0.006985 ± 1.416	0.01054 ± 1.453	0.002637 ± 1.496
jet_abs_eta *	1	1.19 ± 0.7251	1.217 ± 0.7243	1.254 ± 0.733	1.276 ± 0.7817
jet_mass *	GeV	2.013 ± 1.041	2.559 ± 1.554	2.637 ± 1.254	2.853 ± 1.871
jet_ncharged *	count	2.464 ± 1.342	2.735 ± 1.801	3.021 ± 1.595	2.585 ± 1.931
jet_nneutral *	count	1.573 ± 1.119	1.968 ± 1.641	1.736 ± 1.232	2.374 ± 1.789
n_tracks_cone *	count	2.481 ± 1.362	2.771 ± 1.795	3.057 ± 1.626	2.613 ± 1.931
track_pt_sum *	GeV	4.756 ± 2.061	5.456 ± 4.99	5.671 ± 3.721	6.762 ± 8.079
track_pt_sum_over_jet_pt *	1	0.4853 ± 0.1941	0.4315 ± 0.2526	0.4799 ± 0.2047	0.3738 ± 0.2669
avg_track_pt *	GeV	2.341 ± 1.54	1.922 ± 1.361	2.037 ± 1.115	2.167 ± 2.06
std_track_pt *	GeV	0.5339 ± 0.5834	0.7478 ± 0.8791	0.7149 ± 0.7426	1.006 ± 1.601
n_tracks_below_avg_pt *	count	1.192 ± 1.024	1.452 ± 1.269	1.589 ± 1.178	1.377 ± 1.354
n_tracks_above_avg_pt *	count	1.289 ± 0.5347	1.319 ± 0.77	1.469 ± 0.6615	1.237 ± 0.8024
max_pt_ratio *	1	0.3167 ± 0.1814	0.247 ± 0.163	0.2684 ± 0.1348	0.2117 ± 0.1553
min_pt_ratio *	1	0.196 ± 0.207	0.1078 ± 0.125	0.1283 ± 0.138	0.08622 ± 0.09869
max_dr_ratio *	1	0.2162 ± 0.2027	0.1308 ± 0.128	0.151 ± 0.1391	0.1051 ± 0.1077
min_dr_ratio *	1	0.2898 ± 0.1959	0.2098 ± 0.1723	0.2252 ± 0.1465	0.1809 ± 0.1547
dr_max_pt *	1	0.1221 ± 0.07832	0.1382 ± 0.09294	0.1379 ± 0.07668	0.127 ± 0.09881
dr_min_pt *	1	0.1826 ± 0.09934	0.1975 ± 0.1105	0.2148 ± 0.1005	0.1906 ± 0.1168
dr_max_dr *	1	0.2088 ± 0.104	0.2358 ± 0.1103	0.2492 ± 0.09326	0.2214 ± 0.12
dr_min_dr *	1	0.1041 ± 0.06952	0.1084 ± 0.07983	0.1098 ± 0.0665	0.1043 ± 0.08811
pt_difference *	GeV	1.221 ± 1.278	1.79 ± 2.223	1.742 ± 1.916	2.466 ± 4.041
r50 *	1	0.1321 ± 0.07869	0.1451 ± 0.09041	0.1486 ± 0.0722	0.1337 ± 0.09715
r95 *	1	0.2084 ± 0.1035	0.2331 ± 0.1096	0.2463 ± 0.09207	0.2153 ± 0.1187
mean_abs_d0 *	mm	0.9043 ± 5.837	1.278 ± 8.058	3.366 ± 12.78	1.183 ± 7.669
max_abs_d0 *	mm	1.984 ± 11.78	3.425 ± 15.62	10.37 ± 33.85	3.119 ± 16.04
std_abs_d0 *	mm	0.8516 ± 5.339	1.306 ± 6.014	4.255 ± 14.01	1.198 ± 6.438
mean_abs_dz *	mm	1.054 ± 3.569	1.993 ± 12.19	3.64 ± 9.674	1.59 ± 9.466
max_abs_dz *	mm	2.767 ± 10.52	5.003 ± 20.23	11.77 ± 26.86	4.135 ± 22.5
std_abs_dz *	mm	1.111 ± 4.344	1.874 ± 7.271	4.937 ± 11.58	1.599 ± 9.275
pt_weighted_abs_d0 *	mm	0.9683 ± 6.482	1.203 ± 7.943	3.336 ± 12.56	1.127 ± 7.682
pt_weighted_abs_dz *	mm	1.017 ± 3.751	1.741 ± 11.99	2.942 ± 7.865	1.309 ± 8.199

* Variable utilizada como entrada del modelo.

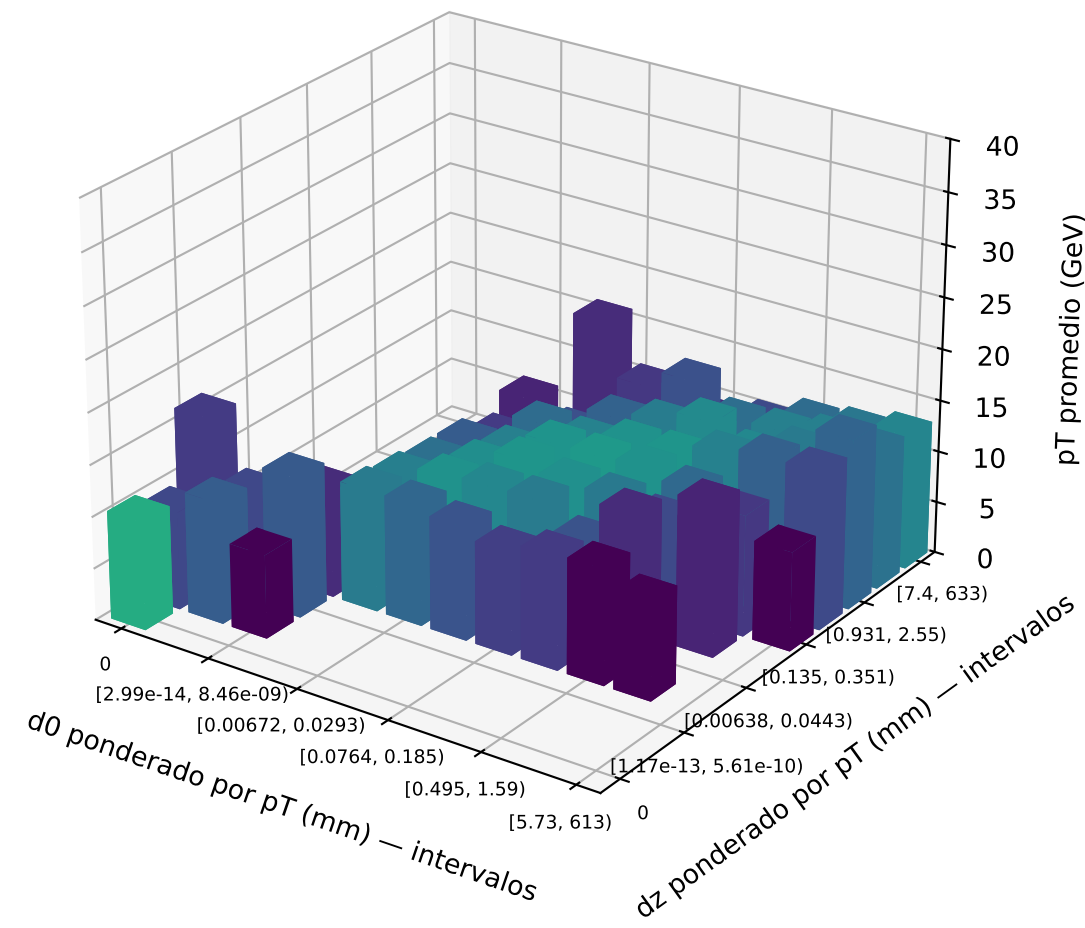
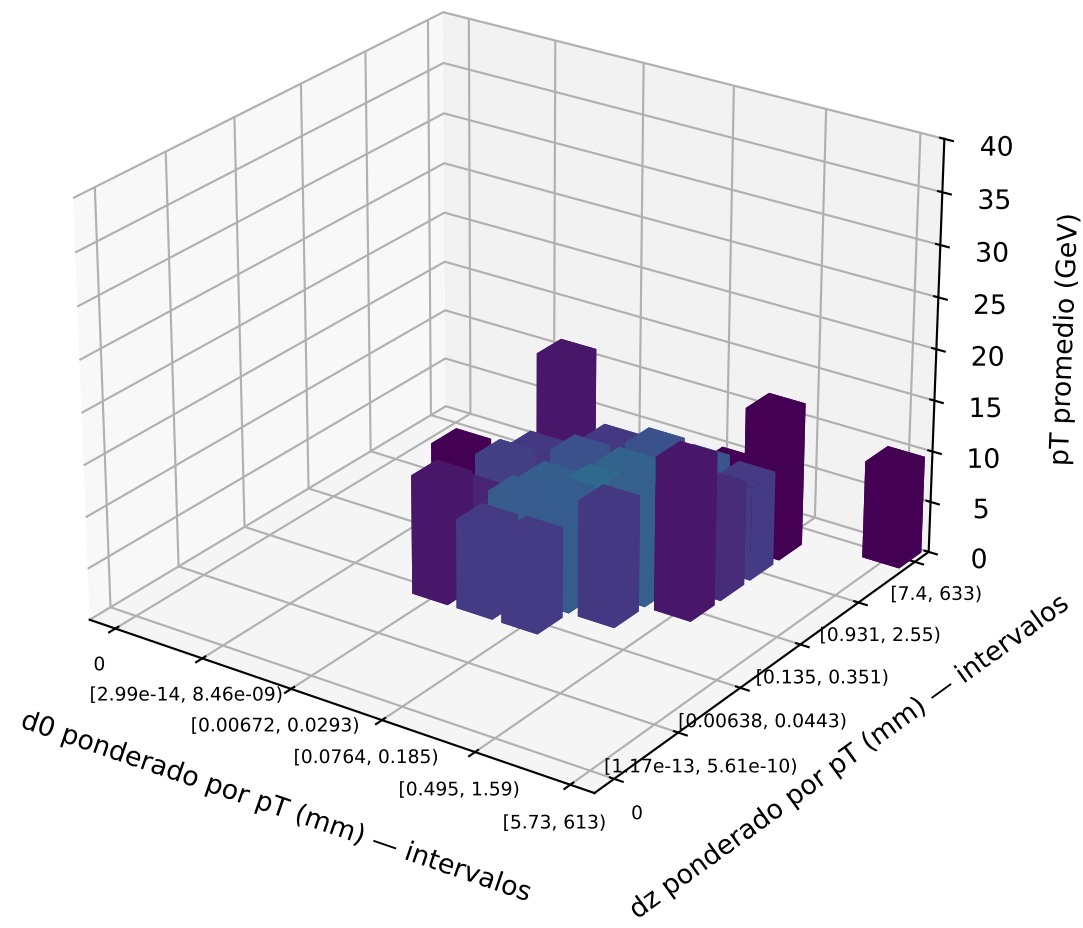
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC32 · semilla 73

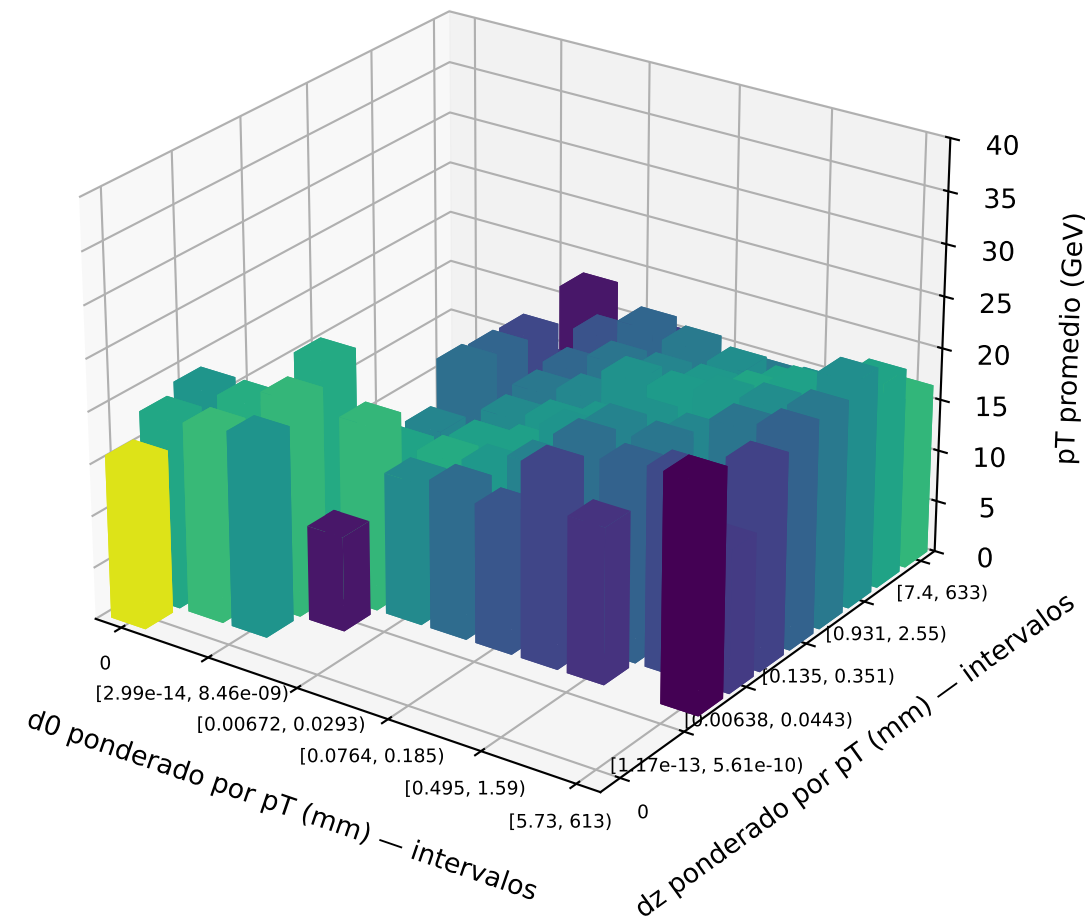
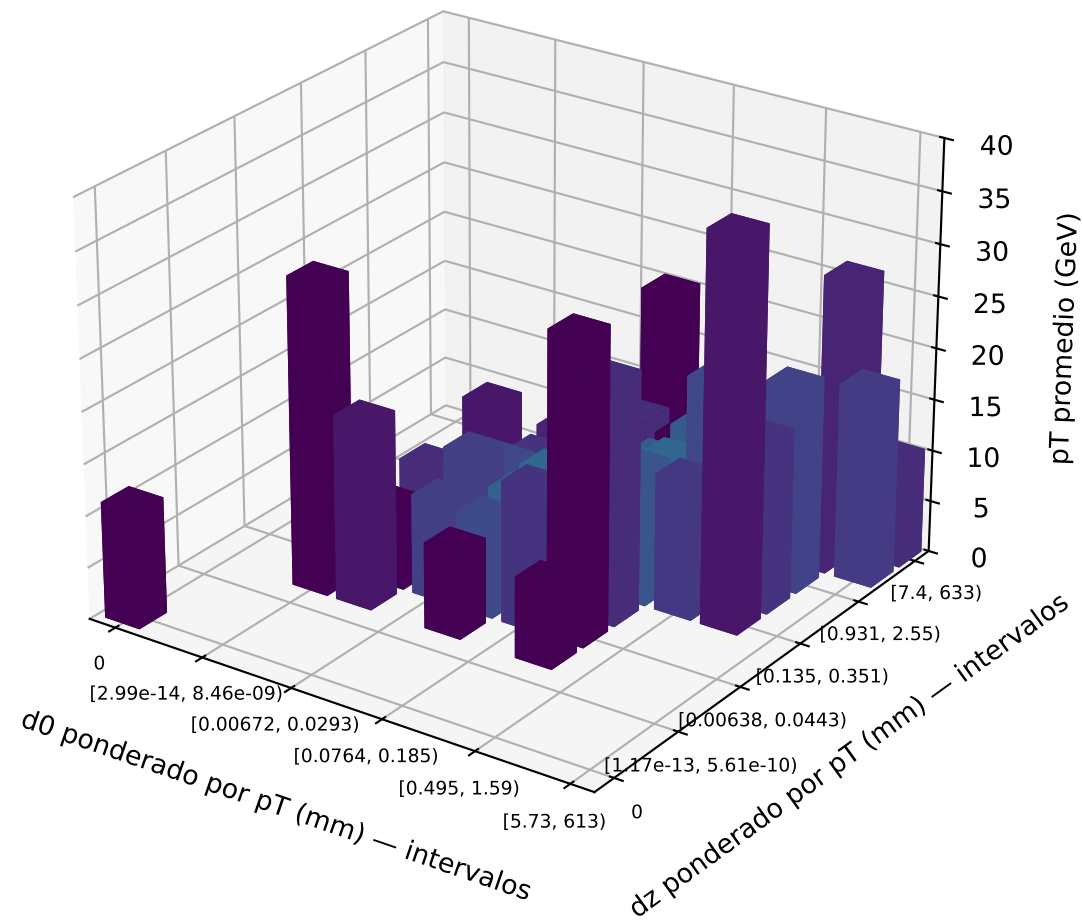
TP — N=324

FN — N=6,351



FP — N=425

TN — N=60,451



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC32 · semilla 73

Variable	Unidad	TP N=324	FN N=6,351	FP N=425	TN N=60,451
jet_pt *	GeV	10.03 ± 2.808	12.38 ± 6.754	11.99 ± 7.591	16.45 ± 11.67
jet_eta *	1	-0.04922 ± 1.401	0.005129 ± 1.415	-0.1387 ± 1.516	0.003688 ± 1.496
jet_abs_eta *	1	1.198 ± 0.7279	1.216 ± 0.7242	1.323 ± 0.7541	1.275 ± 0.7815
jet_mass *	GeV	2.183 ± 0.8589	2.548 ± 1.561	2.591 ± 1.565	2.854 ± 1.869
jet_ncharged *	count	2.438 ± 1.237	2.735 ± 1.803	2.739 ± 1.642	2.587 ± 1.931
jet_nneutral *	count	1.565 ± 1.191	1.966 ± 1.637	1.807 ± 1.447	2.373 ± 1.788
n_tracks_cone *	count	2.448 ± 1.26	2.772 ± 1.797	2.746 ± 1.638	2.615 ± 1.932
track_pt_sum *	GeV	4.662 ± 2.691	5.458 ± 4.966	5.563 ± 4.77	6.762 ± 8.074
track_pt_sum_over_jet_pt *	1	0.4598 ± 0.2076	0.433 ± 0.2521	0.4538 ± 0.2028	0.374 ± 0.2669
avg_track_pt *	GeV	2.008 ± 0.9372	1.941 ± 1.392	2.04 ± 0.9765	2.167 ± 2.06
std_track_pt *	GeV	0.6472 ± 0.6449	0.7412 ± 0.877	0.6756 ± 0.832	1.006 ± 1.6
n_tracks_below_avg_pt *	count	1.222 ± 0.9876	1.449 ± 1.27	1.409 ± 1.22	1.378 ± 1.353
n_tracks_above_avg_pt *	count	1.225 ± 0.4732	1.322 ± 0.771	1.336 ± 0.6076	1.238 ± 0.8029
max_pt_ratio *	1	0.2841 ± 0.1313	0.2489 ± 0.166	0.2623 ± 0.1156	0.2118 ± 0.1554
min_pt_ratio *	1	0.1415 ± 0.09935	0.1109 ± 0.1333	0.1372 ± 0.1044	0.08617 ± 0.09896
max_dr_ratio *	1	0.1551 ± 0.09677	0.1343 ± 0.1358	0.1551 ± 0.1033	0.1051 ± 0.108
min_dr_ratio *	1	0.2636 ± 0.1417	0.2115 ± 0.1756	0.2298 ± 0.1285	0.1809 ± 0.1548
dr_max_pt *	1	0.1419 ± 0.07037	0.1372 ± 0.09328	0.1503 ± 0.06857	0.127 ± 0.09884
dr_min_pt *	1	0.223 ± 0.07804	0.1954 ± 0.1112	0.21 ± 0.08244	0.1906 ± 0.1169
dr_max_dr *	1	0.2433 ± 0.07494	0.2339 ± 0.1116	0.2414 ± 0.077	0.2215 ± 0.1201
dr_min_dr *	1	0.1249 ± 0.06576	0.1073 ± 0.07987	0.1257 ± 0.06438	0.1042 ± 0.0881
pt_difference *	GeV	1.451 ± 1.444	1.776 ± 2.218	1.626 ± 2.164	2.466 ± 4.04
r50 *	1	0.1469 ± 0.06617	0.1443 ± 0.09093	0.1569 ± 0.06549	0.1336 ± 0.09716
r95 *	1	0.2422 ± 0.07436	0.2313 ± 0.1109	0.2402 ± 0.07667	0.2153 ± 0.1188
mean_abs_d0 *	mm	0.263 ± 1.632	1.309 ± 8.146	0.8379 ± 3.557	1.201 ± 7.741
max_abs_d0 *	mm	0.7265 ± 6.546	3.484 ± 15.75	2.998 ± 13.54	3.172 ± 16.26
std_abs_d0 *	mm	0.2841 ± 2.831	1.333 ± 6.094	1.232 ± 5.697	1.22 ± 6.534
mean_abs_dz *	mm	0.6436 ± 2.291	2.011 ± 12.18	1.146 ± 3.001	1.608 ± 9.499
max_abs_dz *	mm	1.484 ± 9.038	5.06 ± 20.23	3.513 ± 10.84	4.194 ± 22.6
std_abs_dz *	mm	0.545 ± 3.908	1.9 ± 7.27	1.382 ± 4.457	1.624 ± 9.323
pt_weighted_abs_d0 *	mm	0.2077 ± 0.703	1.241 ± 8.067	0.9241 ± 4.734	1.144 ± 7.747
pt_weighted_abs_dz *	mm	0.5364 ± 1.059	1.763 ± 11.99	1.119 ± 3.513	1.322 ± 8.221

* Variable utilizada como entrada del modelo.

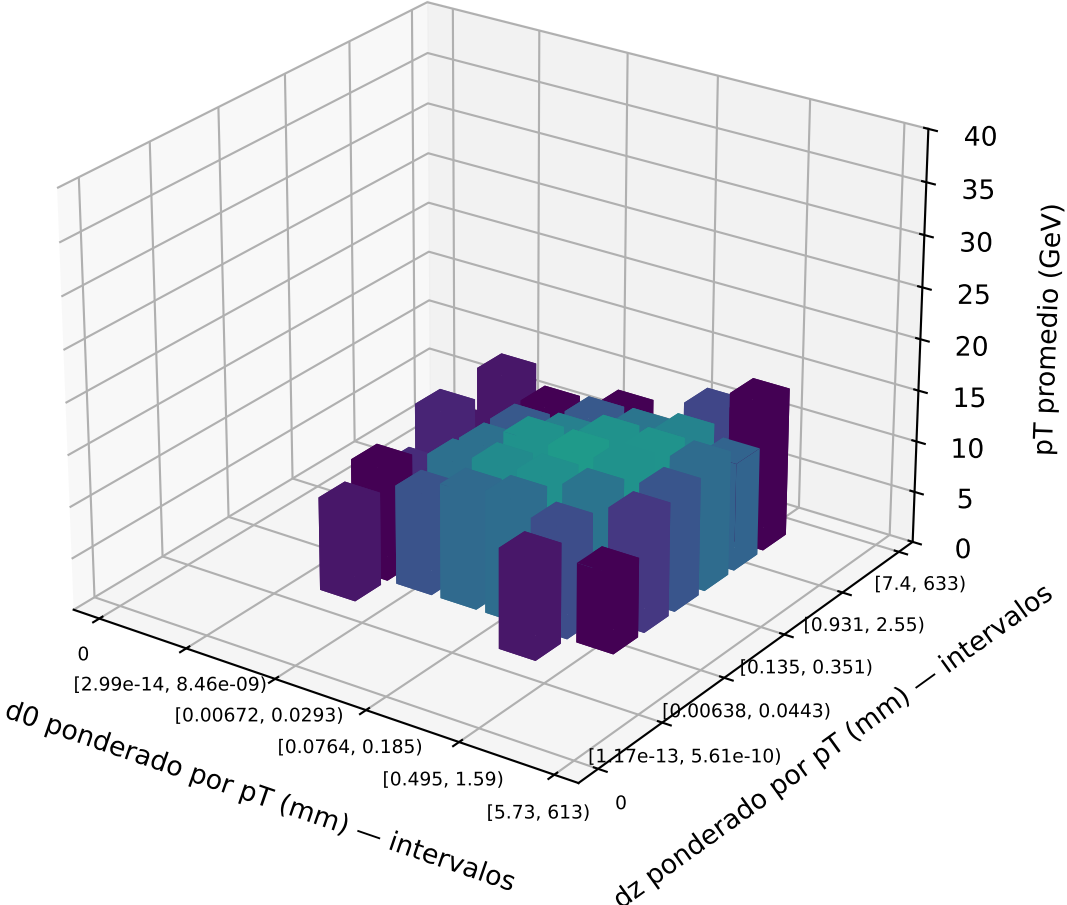
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

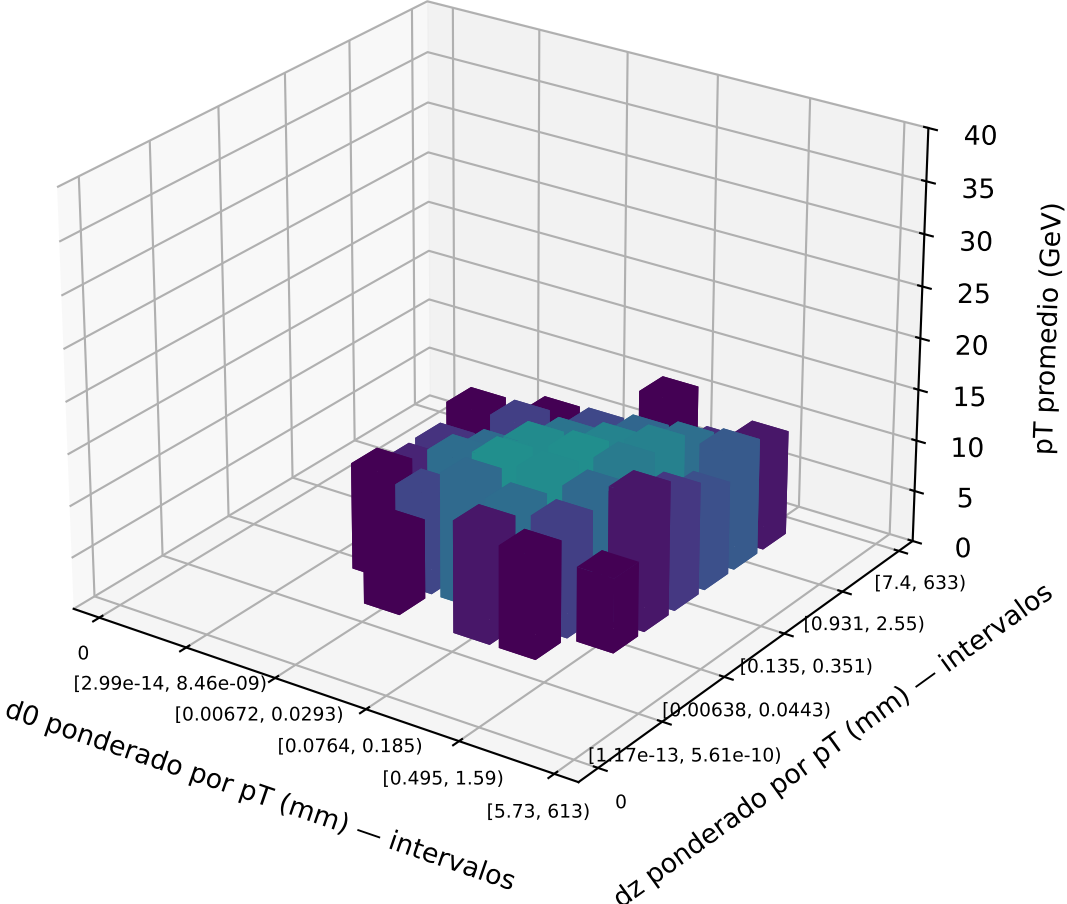
Perfil de pT medio en intervalos de d0/dz — RF16

TP — N=2,819

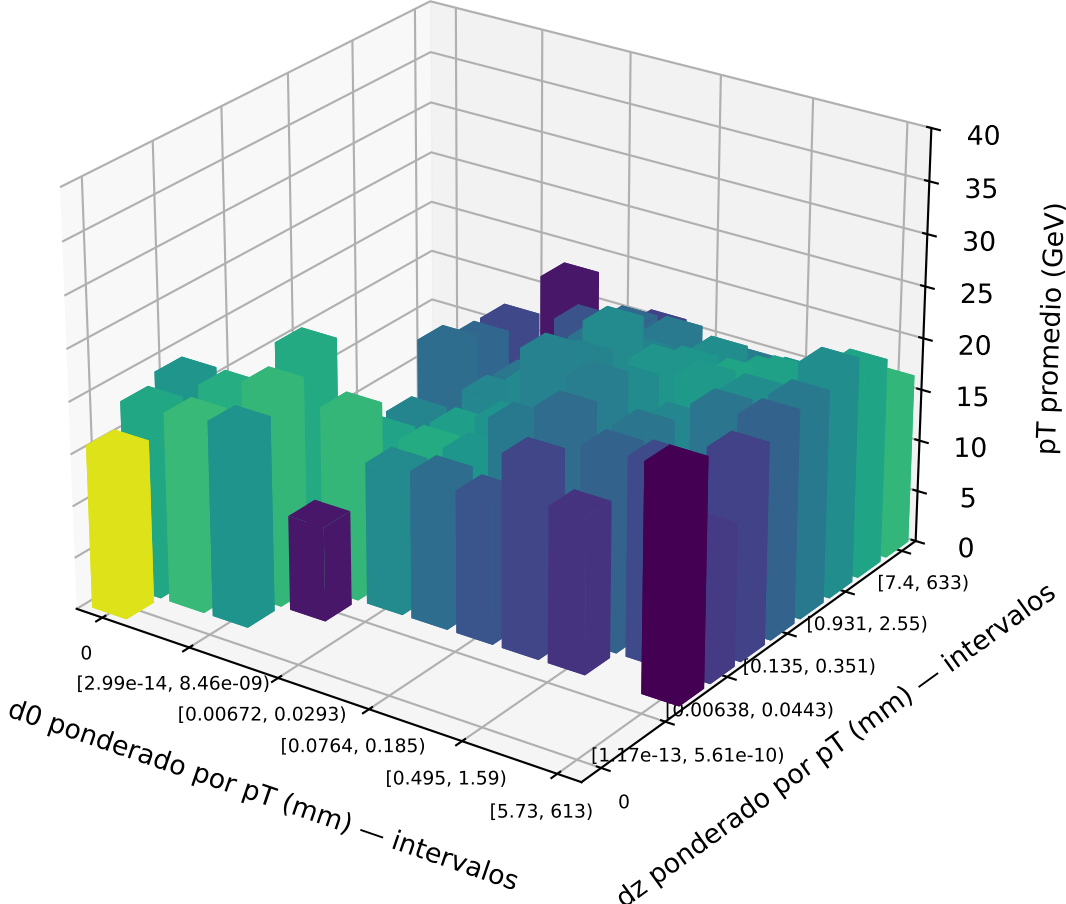
FN — N=3,856



FP — N=1,769



TN — N=59,107



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — RF16

Variable	Unidad	TP N=2,819	FN N=3,856	FP N=1,769	TN N=59,107
jet_pt *	GeV	11.92 ± 5.068	12.51 ± 7.571	11.7 ± 4.989	16.56 ± 11.76
jet_eta *	1	0.003177 ± 1.349	0.00199 ± 1.461	0.02252 ± 1.355	0.0021 ± 1.5
jet_abs_eta *	1	1.161 ± 0.6869	1.255 ± 0.7481	1.168 ± 0.6876	1.279 ± 0.7837
jet_mass *	GeV	2.46 ± 1.314	2.581 ± 1.678	2.296 ± 1.12	2.868 ± 1.882
jet_ncharged	count	2.783 ± 1.528	2.675 ± 1.944	2.4 ± 1.284	2.593 ± 1.945
jet_nneutral	count	1.737 ± 1.524	2.1 ± 1.671	1.679 ± 1.313	2.39 ± 1.794
n_tracks_cone	count	2.807 ± 1.516	2.719 ± 1.943	2.417 ± 1.268	2.622 ± 1.946
track_pt_sum *	GeV	5.847 ± 3.52	5.107 ± 5.655	5.196 ± 3.524	6.801 ± 8.149
track_pt_sum_over_jet_pt *	1	0.4953 ± 0.2184	0.3898 ± 0.2623	0.4464 ± 0.2242	0.3724 ± 0.2675
avg_track_pt *	GeV	2.339 ± 1.472	1.656 ± 1.219	2.238 ± 1.236	2.164 ± 2.074
std_track_pt	GeV	0.7818 ± 0.809	0.7036 ± 0.9064	0.6719 ± 0.7267	1.014 ± 1.614
n_tracks_below_avg_pt	count	1.437 ± 1.14	1.439 ± 1.339	1.162 ± 0.9911	1.385 ± 1.361
n_tracks_above_avg_pt	count	1.37 ± 0.6445	1.28 ± 0.8317	1.256 ± 0.5124	1.238 ± 0.8088
max_pt_ratio	1	0.2982 ± 0.1662	0.2158 ± 0.1546	0.2741 ± 0.1385	0.2103 ± 0.1553
min_pt_ratio	1	0.1473 ± 0.1624	0.08692 ± 0.09672	0.1454 ± 0.1256	0.08476 ± 0.09764
max_dr_ratio	1	0.1716 ± 0.1605	0.1087 ± 0.1035	0.1676 ± 0.1267	0.1036 ± 0.1069
min_dr_ratio	1	0.2599 ± 0.1847	0.1805 ± 0.1585	0.2415 ± 0.1506	0.1794 ± 0.1544
dr_max_pt	1	0.1362 ± 0.0837	0.1383 ± 0.0981	0.1454 ± 0.08257	0.1266 ± 0.09906
dr_min_pt	1	0.1988 ± 0.1016	0.1952 ± 0.1157	0.1998 ± 0.09415	0.1905 ± 0.1173
dr_max_dr	1	0.2361 ± 0.09922	0.2332 ± 0.1174	0.2276 ± 0.08975	0.2214 ± 0.1207
dr_min_dr	1	0.1078 ± 0.07205	0.1084 ± 0.08426	0.1231 ± 0.07788	0.1038 ± 0.0882
pt_difference *	GeV	1.828 ± 1.862	1.711 ± 2.397	1.541 ± 1.691	2.488 ± 4.076
r50	1	0.1429 ± 0.08151	0.1455 ± 0.09553	0.151 ± 0.07925	0.1332 ± 0.09743
r95	1	0.2335 ± 0.09876	0.2306 ± 0.1166	0.2255 ± 0.08984	0.2152 ± 0.1193
mean_abs_d0 *	mm	0.1608 ± 0.1406	2.061 ± 10.4	0.1292 ± 0.1278	1.231 ± 7.832
max_abs_d0 *	mm	0.2705 ± 0.2311	5.601 ± 20.02	0.212 ± 0.1937	3.259 ± 16.48
std_abs_d0 *	mm	0.08402 ± 0.09225	2.159 ± 7.753	0.06593 ± 0.08172	1.254 ± 6.622
mean_abs_dz *	mm	0.3863 ± 0.4291	3.083 ± 15.55	0.3146 ± 0.3957	1.643 ± 9.607
max_abs_dz *	mm	0.6368 ± 0.6515	7.993 ± 25.66	0.4992 ± 0.5478	4.3 ± 22.87
std_abs_dz *	mm	0.1931 ± 0.2434	3.034 ± 9.222	0.1505 ± 0.2129	1.666 ± 9.432
pt_weighted_abs_d0 *	mm	0.1673 ± 0.1421	1.939 ± 10.29	0.135 ± 0.1278	1.172 ± 7.843
pt_weighted_abs_dz *	mm	0.4056 ± 0.4478	2.652 ± 15.32	0.3295 ± 0.4002	1.35 ± 8.317

* Variable utilizada como entrada del modelo.

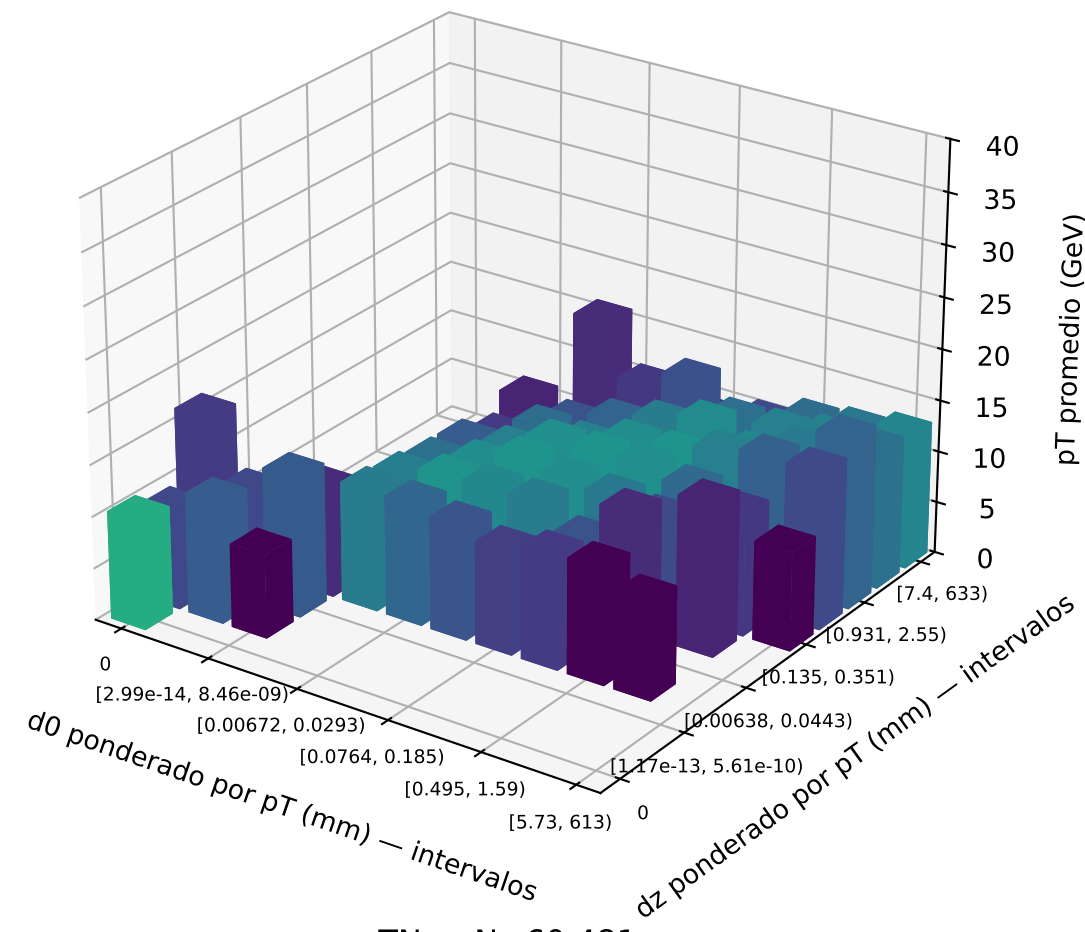
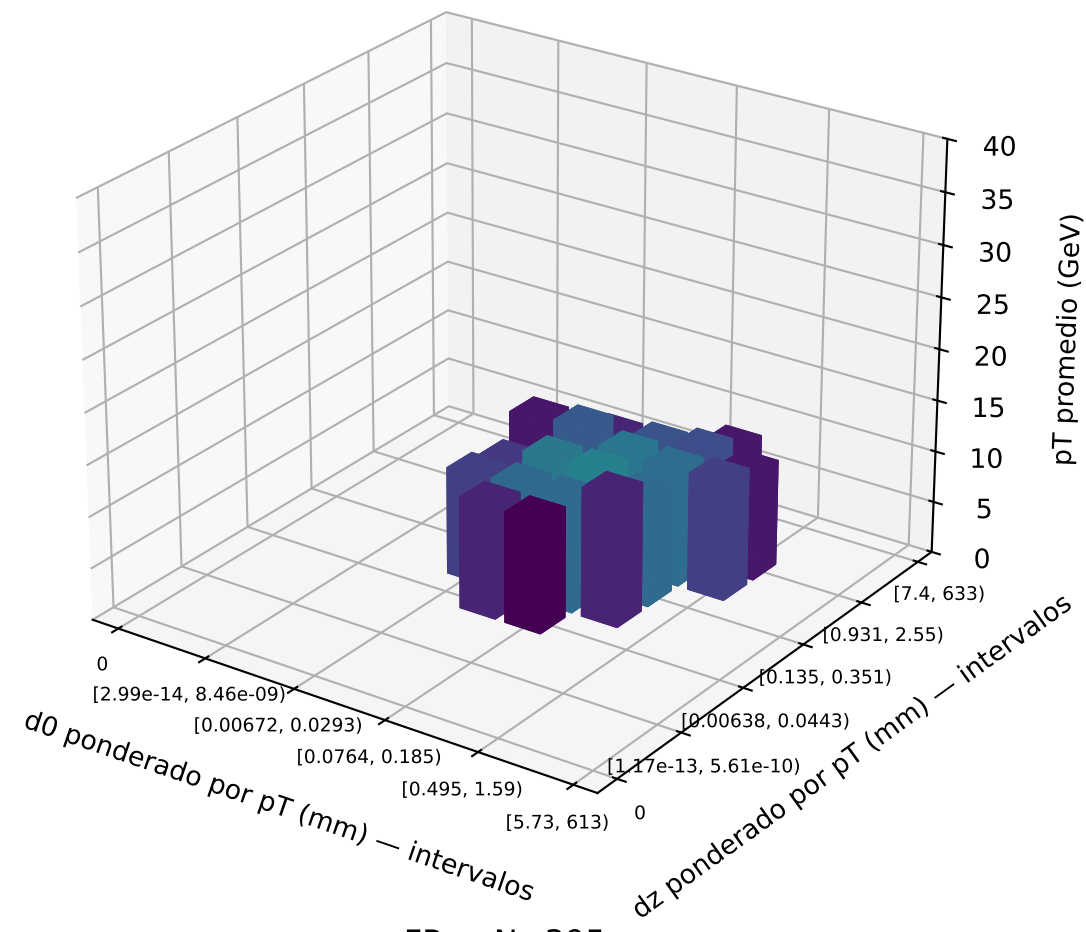
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC16 · ensemble

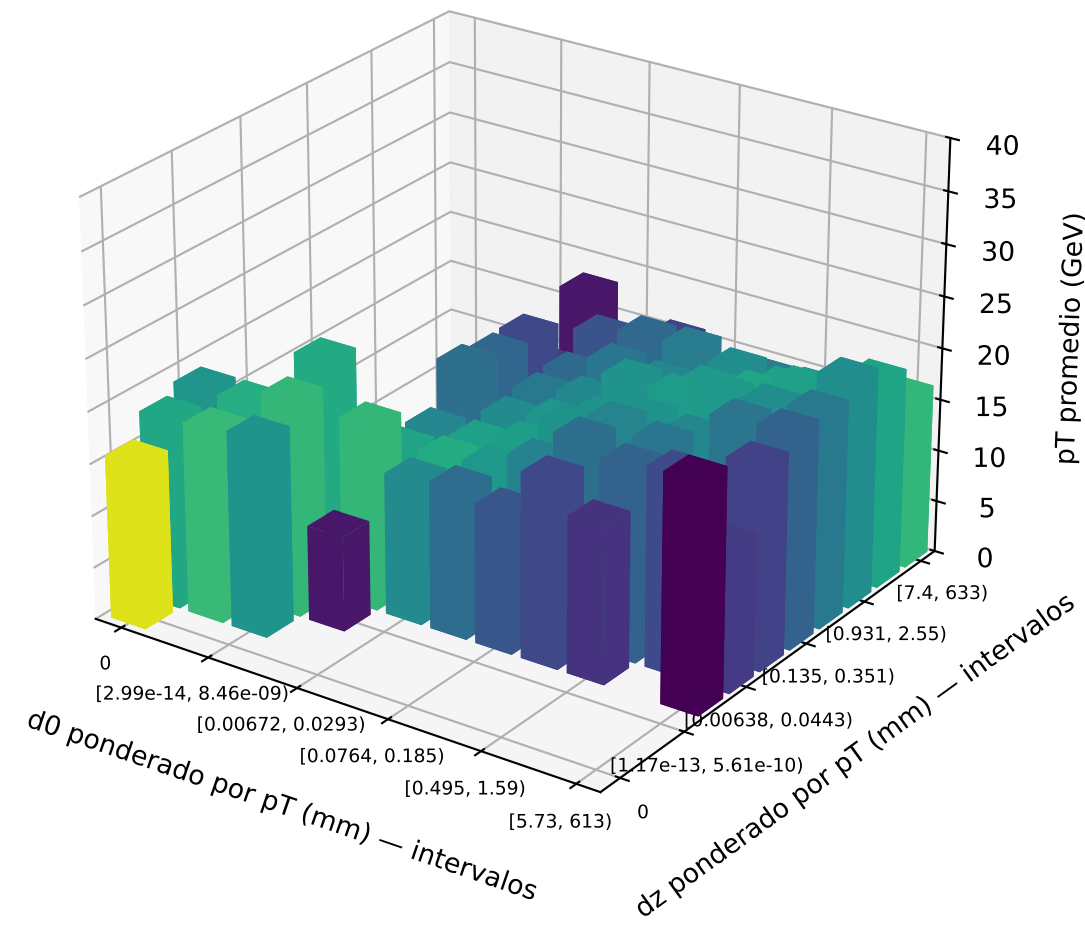
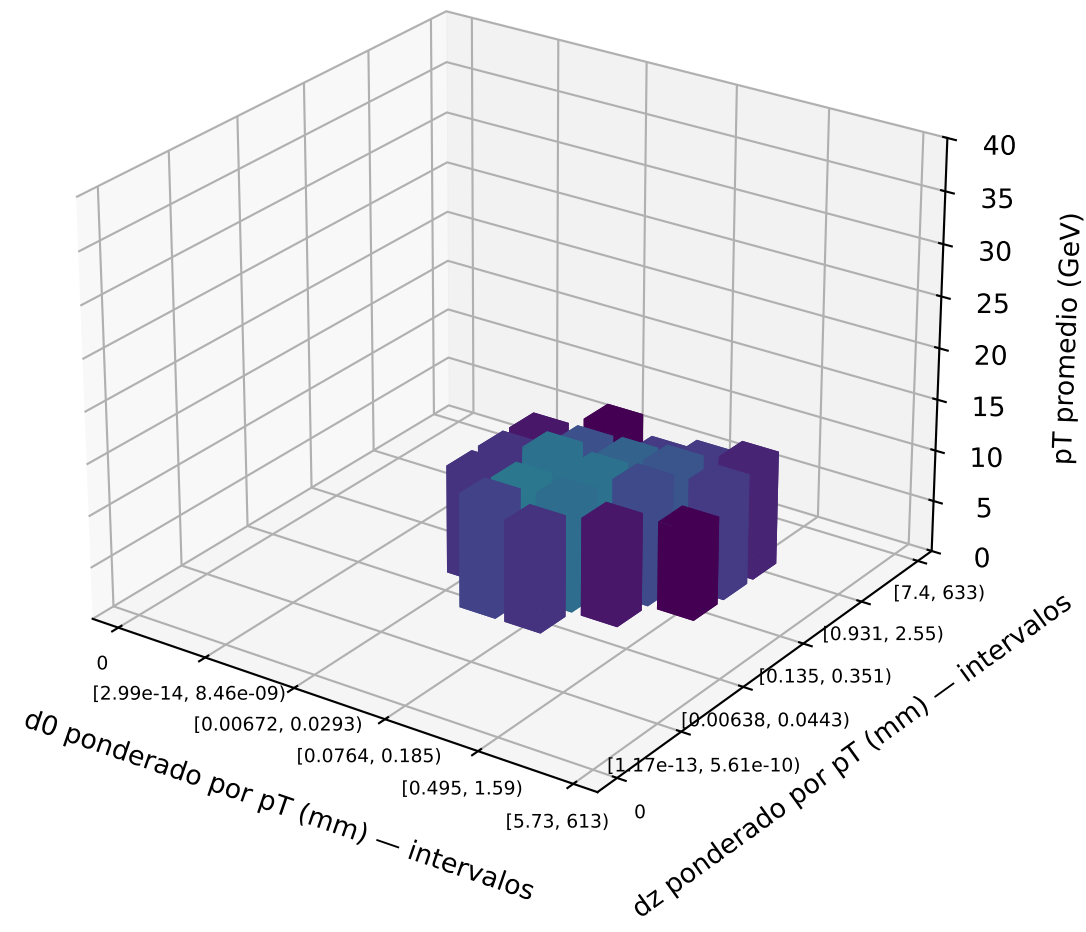
TP — N=562

FN — N=6,113



FP — N=395

TN — N=60,481



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC16 · ensemble

Variable	Unidad	TP N=562	FN N=6,113	FP N=395	TN N=60,481
jet_pt *	GeV	11.29 ± 3.018	12.35 ± 6.867	10.95 ± 2.716	16.45 ± 11.68
jet_eta *	1	0.1264 ± 1.395	-0.008904 ± 1.416	0.2734 ± 1.347	0.000925 ± 1.497
jet_abs_eta *	1	1.266 ± 0.6003	1.211 ± 0.7346	1.237 ± 0.601	1.276 ± 0.7824
jet_mass *	GeV	2.483 ± 0.8399	2.534 ± 1.585	2.311 ± 0.7828	2.855 ± 1.871
jet_ncharged	count	3.319 ± 1.147	2.666 ± 1.818	2.886 ± 1.05	2.586 ± 1.934
jet_nneutral	count	1.6 ± 1.154	1.979 ± 1.653	1.453 ± 1.009	2.375 ± 1.788
n_tracks_cone	count	3.368 ± 1.131	2.7 ± 1.813	2.942 ± 1.023	2.614 ± 1.934
track_pt_sum *	GeV	6.095 ± 2.286	5.358 ± 5.051	5.692 ± 1.961	6.761 ± 8.081
track_pt_sum_over_jet_pt *	1	0.5417 ± 0.1515	0.4245 ± 0.2551	0.5243 ± 0.1447	0.3735 ± 0.267
avg_track_pt *	GeV	1.876 ± 0.6023	1.95 ± 1.424	2.046 ± 0.6941	2.167 ± 2.061
std_track_pt	GeV	0.8775 ± 0.5582	0.7237 ± 0.8894	0.7834 ± 0.5459	1.006 ± 1.601
n_tracks_below_avg_pt	count	1.865 ± 0.9449	1.399 ± 1.277	1.587 ± 0.8445	1.377 ± 1.355
n_tracks_above_avg_pt	count	1.504 ± 0.6032	1.301 ± 0.7701	1.354 ± 0.5656	1.238 ± 0.803
max_pt_ratio	1	0.2761 ± 0.0991	0.2483 ± 0.1693	0.2829 ± 0.09322	0.2117 ± 0.1554
min_pt_ratio	1	0.08893 ± 0.05729	0.1146 ± 0.1367	0.1157 ± 0.07421	0.08633 ± 0.0992
max_dr_ratio	1	0.1232 ± 0.0749	0.1364 ± 0.1384	0.1529 ± 0.08922	0.1051 ± 0.1081
min_dr_ratio	1	0.2136 ± 0.1253	0.214 ± 0.1783	0.224 ± 0.1191	0.181 ± 0.1548
dr_max_pt	1	0.1396 ± 0.06974	0.1372 ± 0.0941	0.1352 ± 0.07285	0.1271 ± 0.09882
dr_min_pt	1	0.2182 ± 0.09312	0.1947 ± 0.1112	0.2034 ± 0.09768	0.1907 ± 0.1168
dr_max_dr	1	0.2681 ± 0.06904	0.2313 ± 0.1126	0.2454 ± 0.07984	0.2214 ± 0.1201
dr_min_dr	1	0.1011 ± 0.05637	0.1088 ± 0.08109	0.09874 ± 0.05622	0.1044 ± 0.08814
pt_difference *	GeV	2.104 ± 1.32	1.729 ± 2.248	1.829 ± 1.296	2.465 ± 4.042
r50	1	0.1481 ± 0.06437	0.1441 ± 0.09187	0.1472 ± 0.06743	0.1337 ± 0.09716
r95	1	0.2649 ± 0.06846	0.2288 ± 0.112	0.2421 ± 0.07985	0.2153 ± 0.1187
mean_abs_d0 *	mm	0.1354 ± 0.1104	1.362 ± 8.307	0.1138 ± 0.1438	1.206 ± 7.744
max_abs_d0 *	mm	0.2689 ± 0.2289	3.633 ± 16.11	0.2638 ± 0.4147	3.19 ± 16.3
std_abs_d0 *	mm	0.09849 ± 0.09135	1.391 ± 6.239	0.1082 ± 0.1799	1.227 ± 6.549
mean_abs_dz *	mm	0.3598 ± 0.3558	2.09 ± 12.42	0.2614 ± 0.2726	1.613 ± 9.499
max_abs_dz *	mm	0.706 ± 0.6515	5.271 ± 20.7	0.6031 ± 0.726	4.213 ± 22.61
std_abs_dz *	mm	0.2579 ± 0.252	1.979 ± 7.454	0.2507 ± 0.3238	1.631 ± 9.327
pt_weighted_abs_d0 *	mm	0.1419 ± 0.1174	1.287 ± 8.22	0.1189 ± 0.1347	1.149 ± 7.755
pt_weighted_abs_dz *	mm	0.3794 ± 0.3801	1.825 ± 12.22	0.2787 ± 0.3025	1.327 ± 8.224

* Variable utilizada como entrada del modelo.

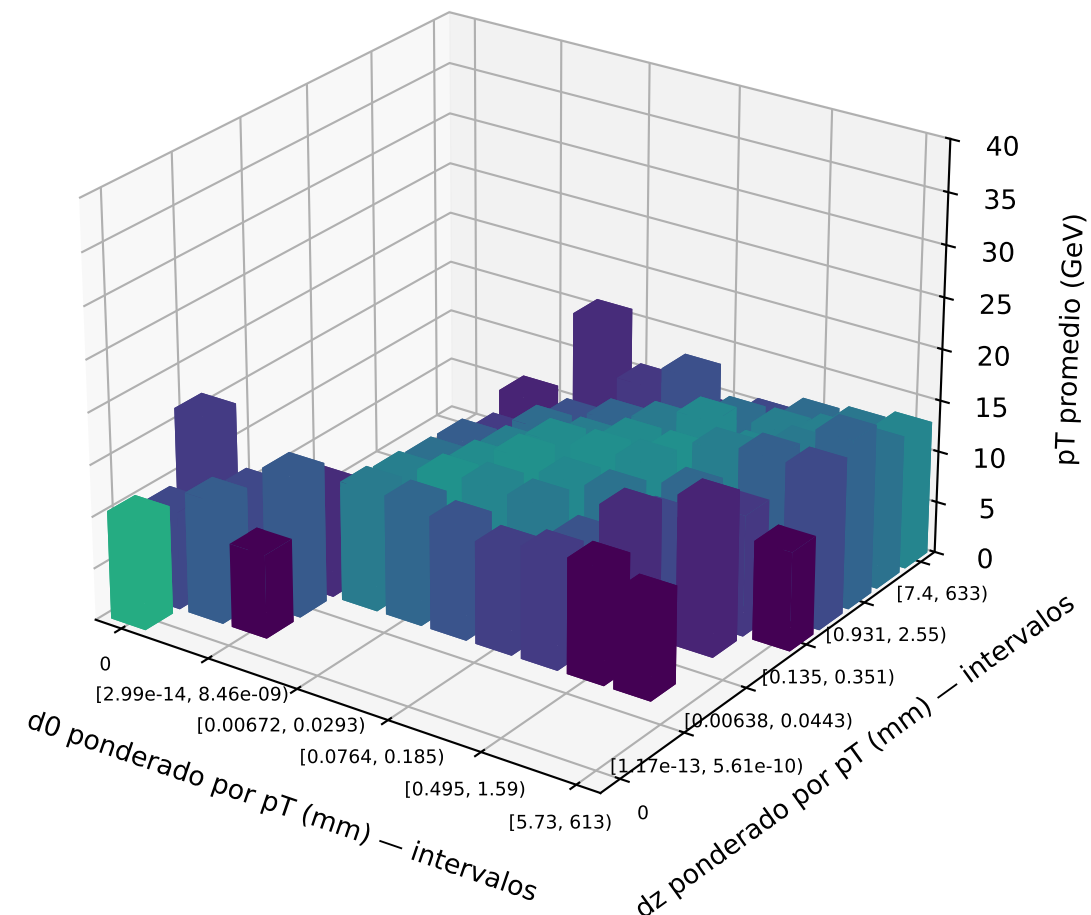
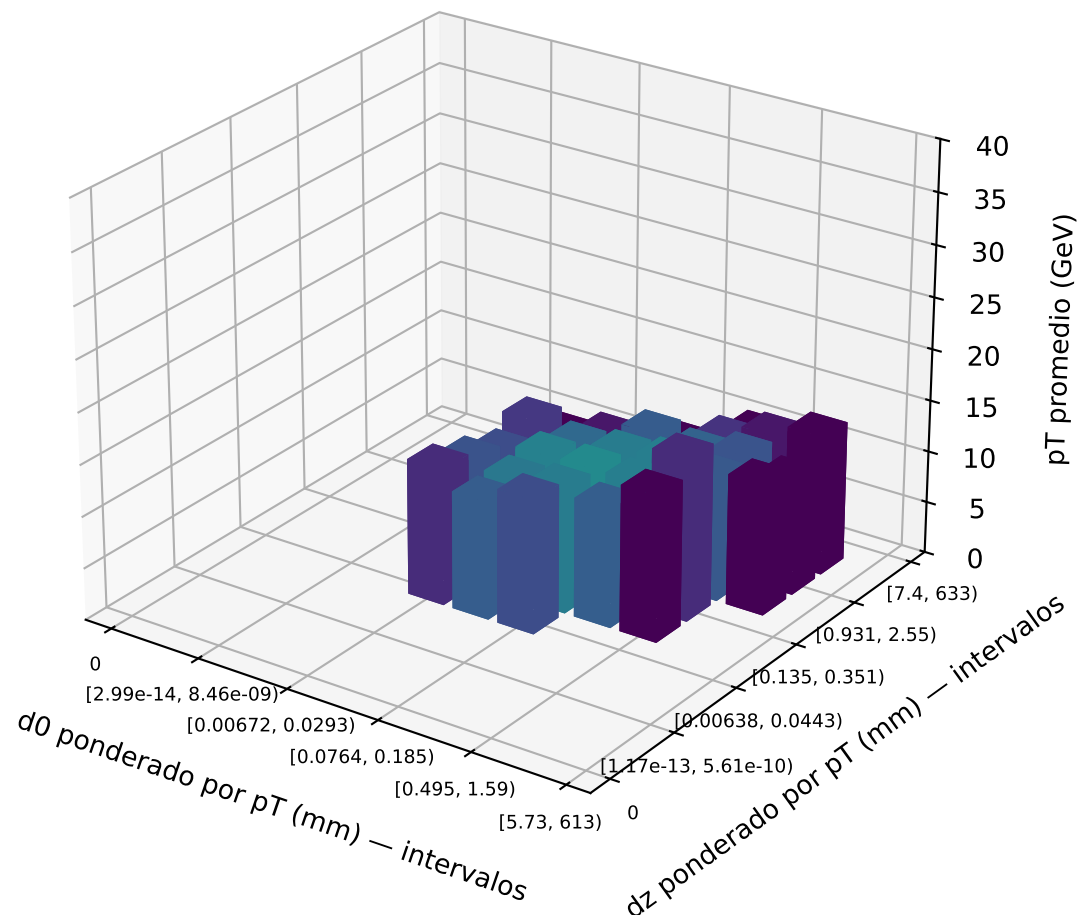
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC16 · semilla 101

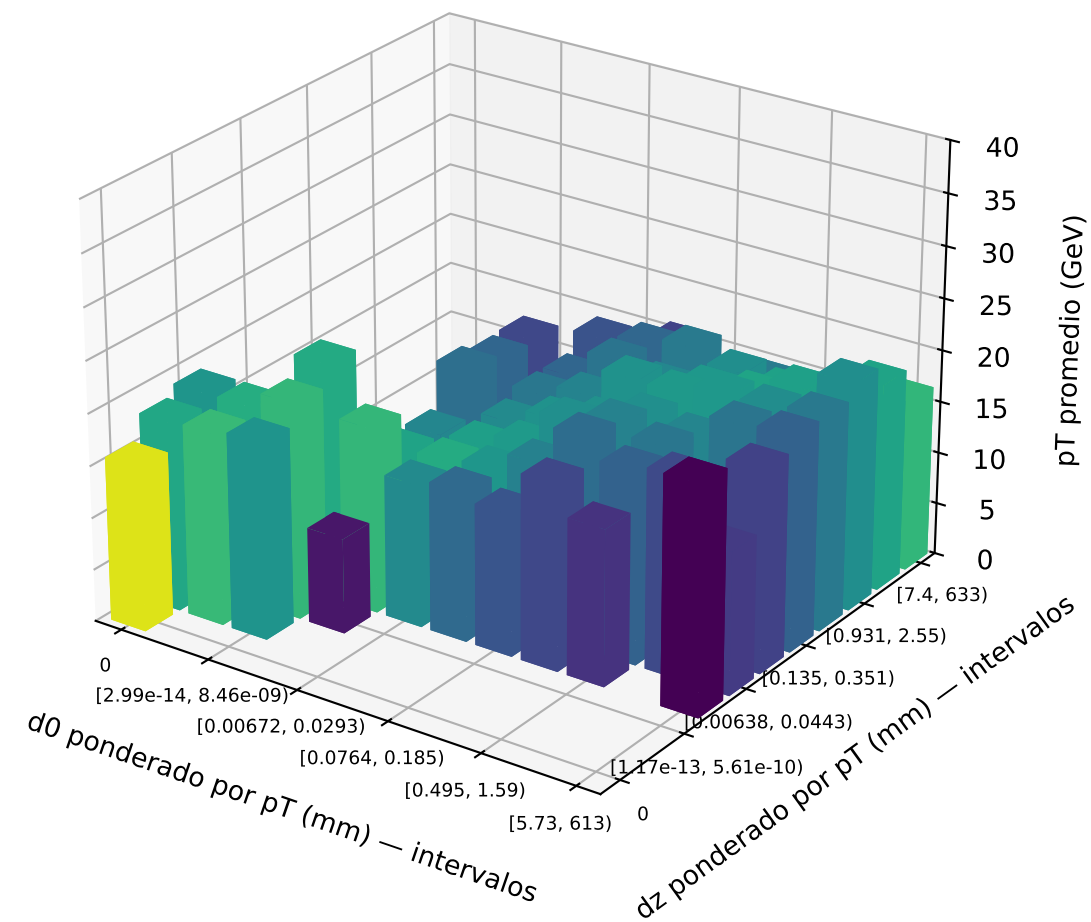
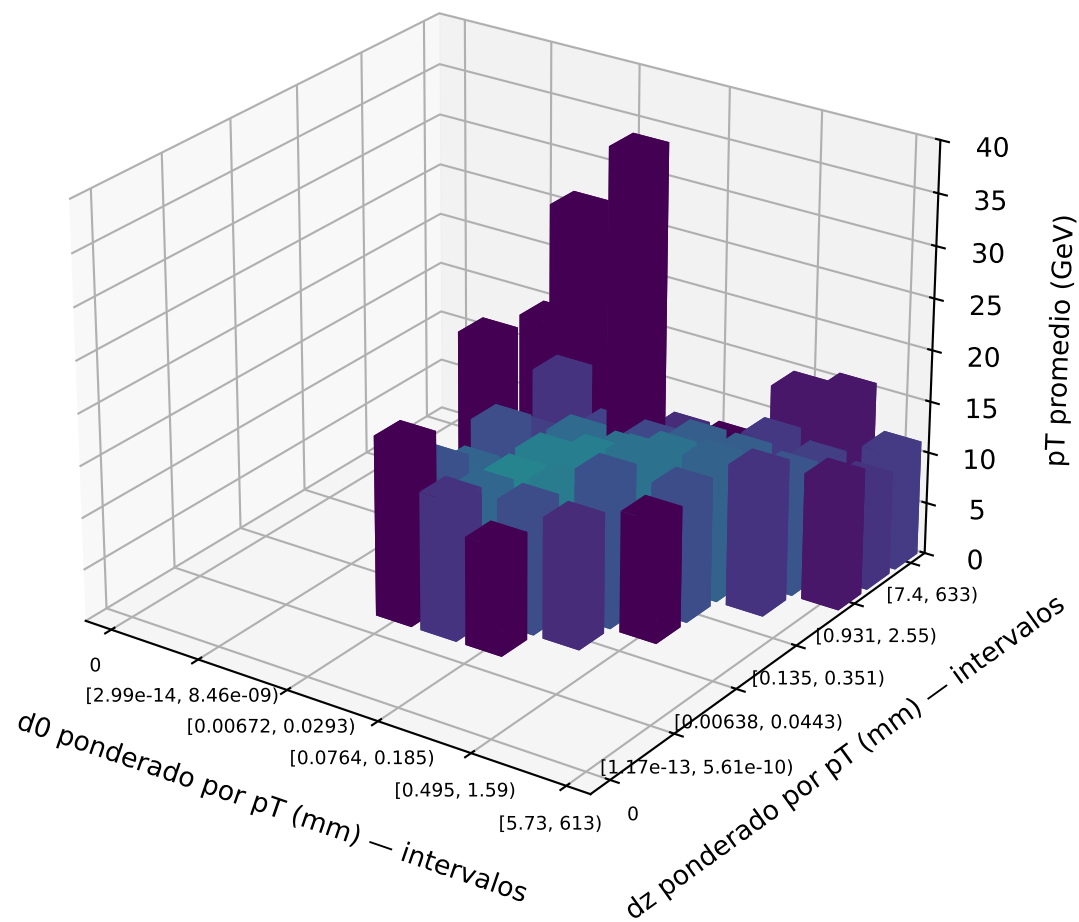
TP — N=1,096

FN — N=5,579



FP — N=1,039

TN — N=59,837



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC16 · semilla 101

Variable	Unidad	TP N=1,096	FN N=5,579	FP N=1,039	TN N=59,837
jet_pt *	GeV	11.88 ± 3.863	12.34 ± 7.052	12.38 ± 4.394	16.48 ± 11.73
jet_eta *	1	0.01727 ± 1.263	-0.000412 ± 1.443	0.05285 ± 1.288	0.001822 ± 1.499
jet_abs_eta *	1	1.096 ± 0.6274	1.239 ± 0.7397	1.118 ± 0.6427	1.278 ± 0.7832
jet_mass *	GeV	2.617 ± 1.053	2.513 ± 1.614	2.689 ± 1.042	2.855 ± 1.878
jet_ncharged	count	3.222 ± 1.422	2.622 ± 1.827	3.163 ± 1.443	2.578 ± 1.935
jet_nneutral	count	1.625 ± 1.252	2.01 ± 1.676	1.659 ± 1.185	2.382 ± 1.792
n_tracks_cone	count	3.255 ± 1.408	2.658 ± 1.824	3.202 ± 1.427	2.606 ± 1.936
track_pt_sum *	GeV	6.216 ± 2.742	5.263 ± 5.186	6.212 ± 3.007	6.763 ± 8.116
track_pt_sum_over_jet_pt *	1	0.5263 ± 0.1728	0.4163 ± 0.2589	0.5051 ± 0.1723	0.3722 ± 0.2674
avg_track_pt *	GeV	2.049 ± 0.8451	1.923 ± 1.454	2.046 ± 0.74	2.168 ± 2.07
std_track_pt	GeV	0.8298 ± 0.6397	0.7183 ± 0.9044	0.8065 ± 0.6272	1.008 ± 1.608
n_tracks_below_avg_pt	count	1.779 ± 1.081	1.371 ± 1.28	1.746 ± 1.109	1.372 ± 1.356
n_tracks_above_avg_pt	count	1.476 ± 0.6502	1.287 ± 0.7754	1.456 ± 0.6311	1.234 ± 0.8038
max_pt_ratio	1	0.2782 ± 0.1156	0.2452 ± 0.1722	0.2643 ± 0.09797	0.2112 ± 0.1558
min_pt_ratio	1	0.1091 ± 0.09656	0.1131 ± 0.1379	0.1074 ± 0.08065	0.08616 ± 0.09934
max_dr_ratio	1	0.1392 ± 0.1016	0.1345 ± 0.1398	0.136 ± 0.08915	0.1049 ± 0.1083
min_dr_ratio	1	0.2272 ± 0.1395	0.2114 ± 0.1805	0.2147 ± 0.1175	0.1807 ± 0.1552
dr_max_pt	1	0.1405 ± 0.07393	0.1368 ± 0.09548	0.1411 ± 0.07776	0.1269 ± 0.09898
dr_min_pt	1	0.2118 ± 0.09718	0.1938 ± 0.1121	0.216 ± 0.09535	0.1903 ± 0.117
dr_max_dr	1	0.2602 ± 0.08239	0.2293 ± 0.1141	0.2592 ± 0.08224	0.2209 ± 0.1203
dr_min_dr	1	0.1048 ± 0.0599	0.1089 ± 0.0826	0.1065 ± 0.06392	0.1043 ± 0.08833
pt_difference *	GeV	1.992 ± 1.508	1.715 ± 2.295	1.951 ± 1.552	2.469 ± 4.059
r50	1	0.1492 ± 0.06992	0.1435 ± 0.09328	0.1524 ± 0.07291	0.1334 ± 0.09733
r95	1	0.2571 ± 0.08189	0.2268 ± 0.1134	0.2558 ± 0.08283	0.2148 ± 0.1189
mean_abs_d0 *	mm	0.1728 ± 0.2067	1.472 ± 8.687	0.4898 ± 3.128	1.211 ± 7.775
max_abs_d0 *	mm	0.3893 ± 0.8267	3.931 ± 16.83	1.413 ± 9.111	3.201 ± 16.34
std_abs_d0 *	mm	0.1408 ± 0.3213	1.507 ± 6.517	0.6112 ± 4.262	1.23 ± 6.56
mean_abs_dz *	mm	0.3905 ± 0.5804	2.249 ± 12.99	0.623 ± 1.65	1.621 ± 9.547
max_abs_dz *	mm	0.7954 ± 1.42	5.69 ± 21.61	1.654 ± 4.225	4.233 ± 22.73
std_abs_dz *	mm	0.2858 ± 0.6152	2.139 ± 7.78	0.6958 ± 1.912	1.639 ± 9.374
pt_weighted_abs_d0 *	mm	0.1765 ± 0.1999	1.39 ± 8.597	0.4691 ± 2.532	1.154 ± 7.79
pt_weighted_abs_dz *	mm	0.3928 ± 0.5542	1.96 ± 12.78	0.6024 ± 1.729	1.333 ± 8.265

* Variable utilizada como entrada del modelo.

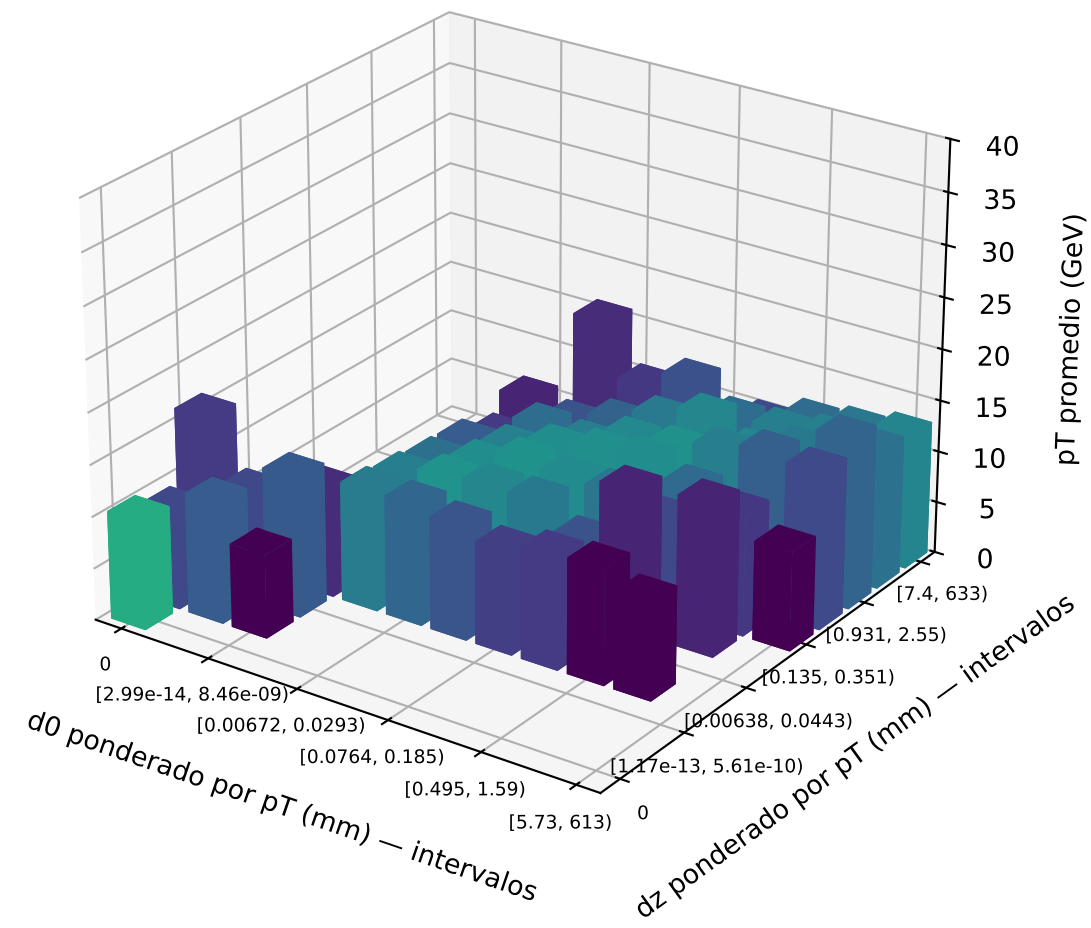
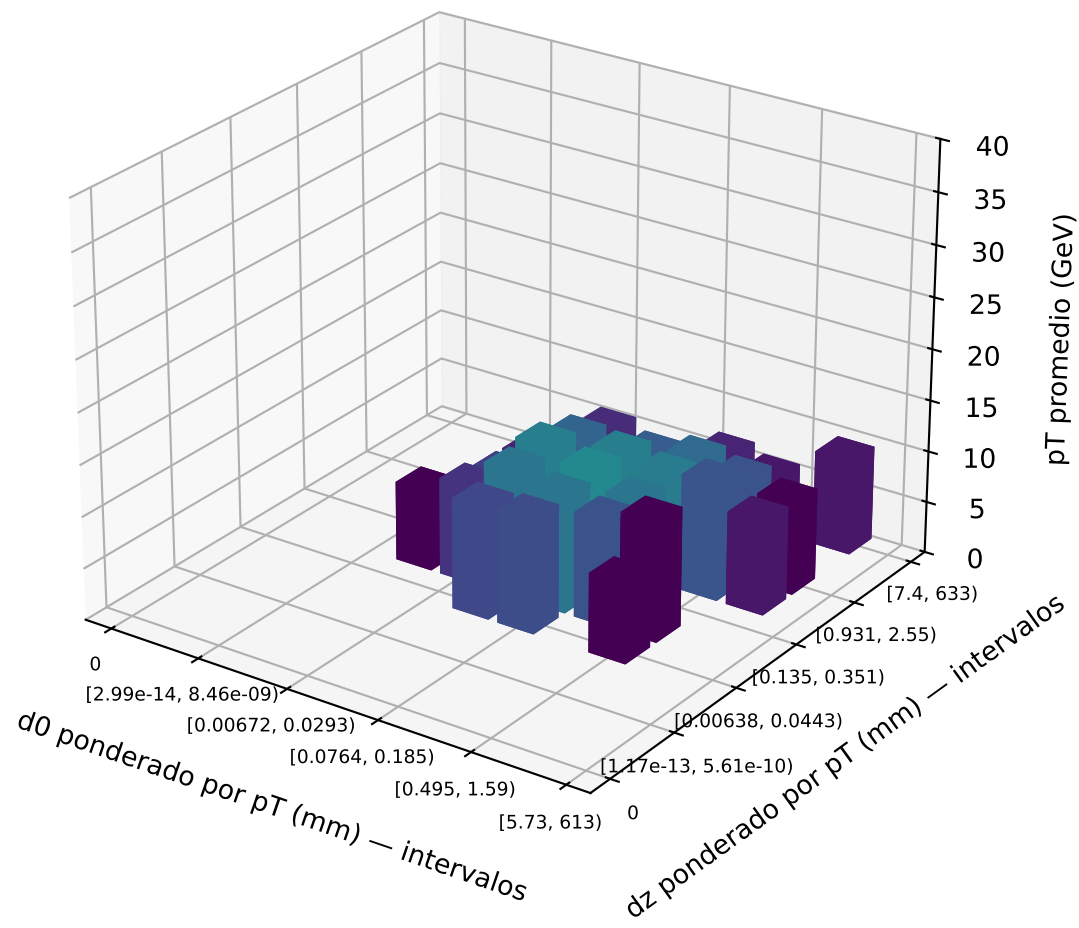
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC16 · semilla 42

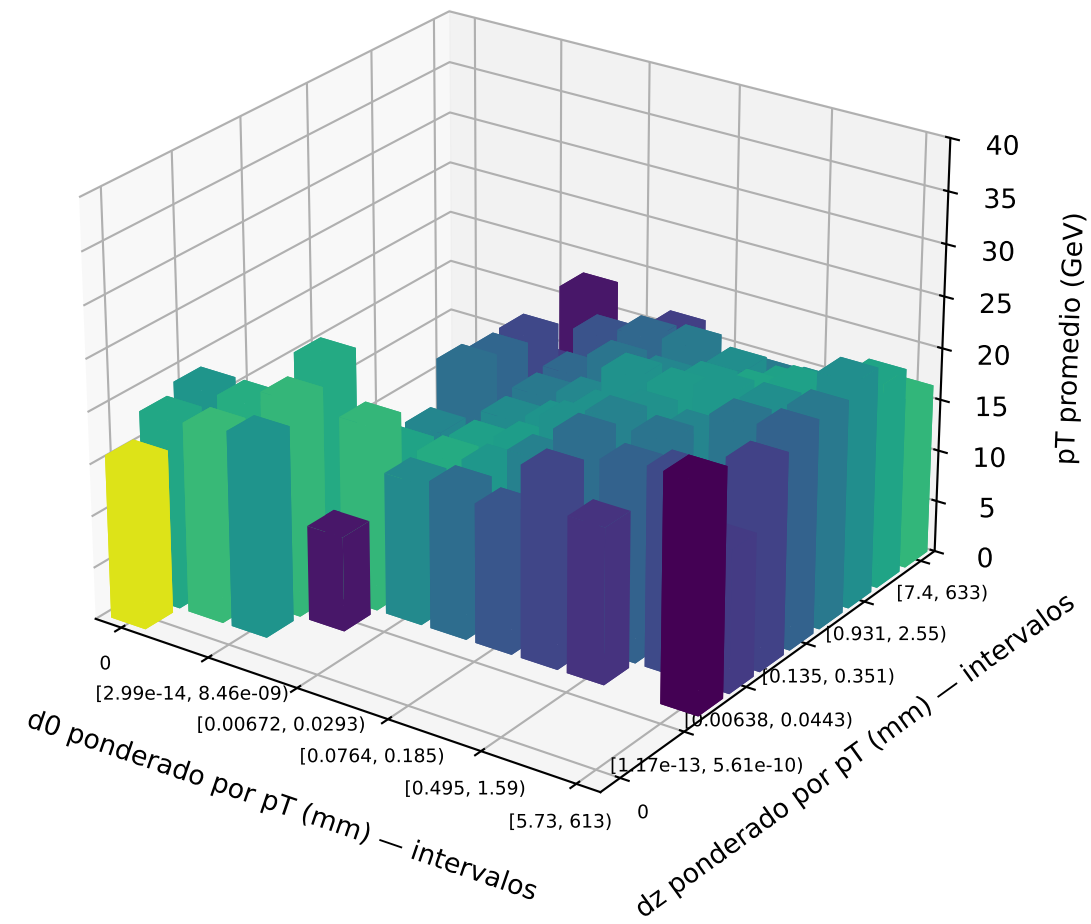
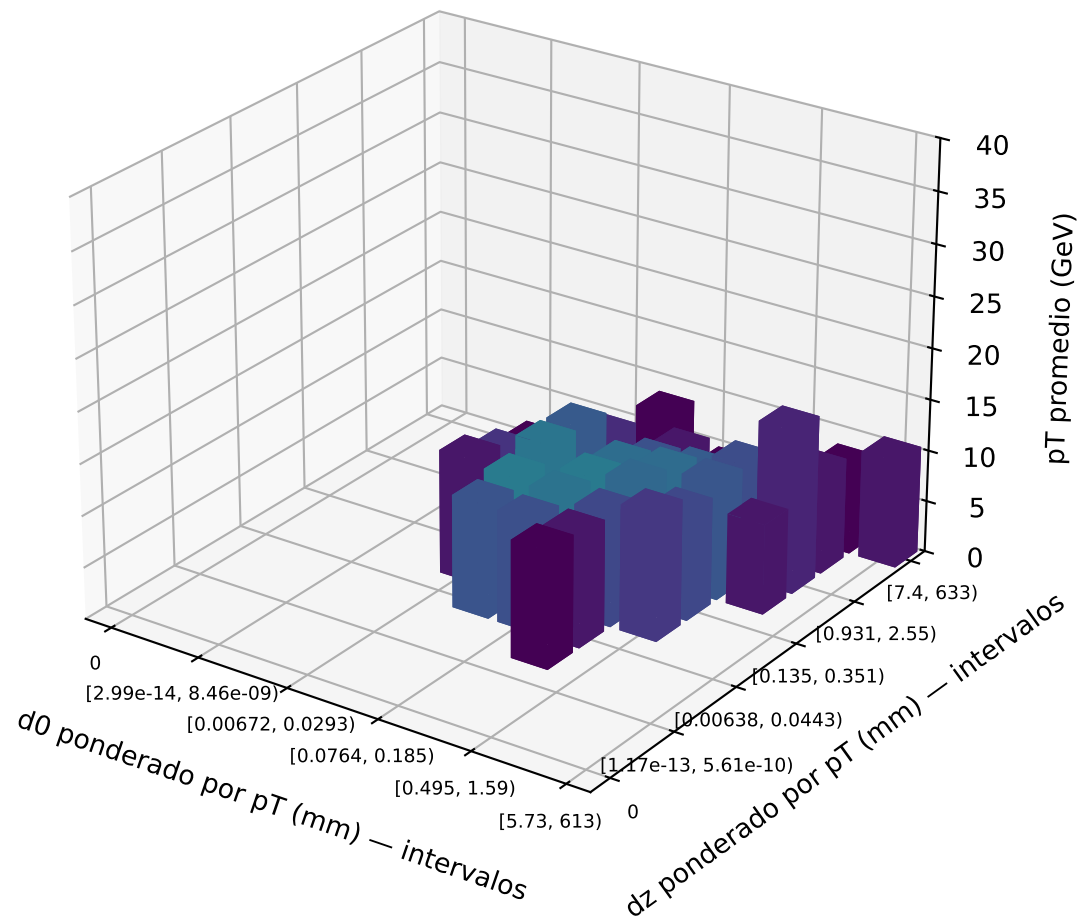
TP — N=952

FN — N=5,723



FP — N=663

TN — N=60,213



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC16 · semilla 42

Variable	Unidad	TP N=952	FN N=5,723	FP N=663	TN N=60,213
jet_pt *	GeV	11.33 ± 3.336	12.42 ± 7.025	11.29 ± 4.017	16.47 ± 11.7
jet_eta *	1	0.07876 ± 1.386	-0.0102 ± 1.419	0.3035 ± 1.333	-0.000619 ± 1.497
jet_abs_eta *	1	1.241 ± 0.6226	1.211 ± 0.7399	1.203 ± 0.6496	1.276 ± 0.7826
jet_mass *	GeV	2.603 ± 1.032	2.518 ± 1.605	2.602 ± 1.05	2.854 ± 1.874
jet_ncharged	count	3.38 ± 1.316	2.611 ± 1.824	3.192 ± 1.254	2.581 ± 1.935
jet_nneutral	count	1.616 ± 1.287	2.002 ± 1.663	1.558 ± 1.154	2.378 ± 1.79
n_tracks_cone	count	3.424 ± 1.276	2.645 ± 1.822	3.232 ± 1.213	2.61 ± 1.935
track_pt_sum *	GeV	6.157 ± 2.561	5.297 ± 5.159	5.892 ± 3.217	6.764 ± 8.093
track_pt_sum_over_jet_pt *	1	0.5463 ± 0.1698	0.4157 ± 0.2565	0.521 ± 0.1756	0.3729 ± 0.267
avg_track_pt *	GeV	1.926 ± 0.9102	1.947 ± 1.436	1.887 ± 0.8376	2.169 ± 2.064
std_track_pt	GeV	0.8882 ± 0.6678	0.7114 ± 0.8939	0.7906 ± 0.6609	1.006 ± 1.603
n_tracks_below_avg_pt	count	1.897 ± 0.9957	1.362 ± 1.281	1.738 ± 0.9144	1.374 ± 1.356
n_tracks_above_avg_pt	count	1.527 ± 0.6808	1.283 ± 0.7663	1.495 ± 0.6588	1.235 ± 0.8027
max_pt_ratio	1	0.2823 ± 0.1248	0.2453 ± 0.1699	0.2644 ± 0.1001	0.2116 ± 0.1556
min_pt_ratio	1	0.09482 ± 0.08899	0.1153 ± 0.1377	0.09961 ± 0.06776	0.08638 ± 0.09937
max_dr_ratio	1	0.128 ± 0.1016	0.1365 ± 0.1389	0.1317 ± 0.0819	0.1051 ± 0.1083
min_dr_ratio	1	0.2244 ± 0.1473	0.2123 ± 0.1786	0.213 ± 0.1203	0.1809 ± 0.155
dr_max_pt	1	0.1386 ± 0.07229	0.1372 ± 0.09522	0.143 ± 0.07752	0.1269 ± 0.09887
dr_min_pt	1	0.2176 ± 0.09638	0.1933 ± 0.1117	0.2159 ± 0.09243	0.1905 ± 0.1169
dr_max_dr	1	0.2679 ± 0.07736	0.2288 ± 0.1137	0.2624 ± 0.07616	0.2211 ± 0.1202
dr_min_dr	1	0.09988 ± 0.05565	0.1096 ± 0.08254	0.1052 ± 0.06076	0.1043 ± 0.08823
pt_difference *	GeV	2.121 ± 1.528	1.7 ± 2.273	1.891 ± 1.665	2.467 ± 4.048
r50	1	0.1485 ± 0.06911	0.1437 ± 0.09287	0.1561 ± 0.06985	0.1335 ± 0.09723
r95	1	0.264 ± 0.07773	0.2264 ± 0.113	0.26 ± 0.07701	0.215 ± 0.1188
mean_abs_d0 *	mm	0.172 ± 0.2732	1.439 ± 8.58	0.2516 ± 0.6609	1.209 ± 7.761
max_abs_d0 *	mm	0.3553 ± 0.5395	3.848 ± 16.63	0.6767 ± 1.705	3.198 ± 16.33
std_abs_d0 *	mm	0.1317 ± 0.2087	1.474 ± 6.439	0.2844 ± 0.7657	1.23 ± 6.563
mean_abs_dz *	mm	0.4265 ± 1.029	2.197 ± 12.83	0.5224 ± 2.217	1.616 ± 9.517
max_abs_dz *	mm	0.8321 ± 1.233	5.561 ± 21.35	1.222 ± 3.94	4.222 ± 22.66
std_abs_dz *	mm	0.2968 ± 0.3263	2.09 ± 7.691	0.5077 ± 1.747	1.635 ± 9.346
pt_weighted_abs_d0 *	mm	0.1815 ± 0.2694	1.359 ± 8.491	0.2329 ± 0.5743	1.152 ± 7.772
pt_weighted_abs_dz *	mm	0.4542 ± 1.039	1.911 ± 12.62	0.5307 ± 2.128	1.329 ± 8.239

* Variable utilizada como entrada del modelo.

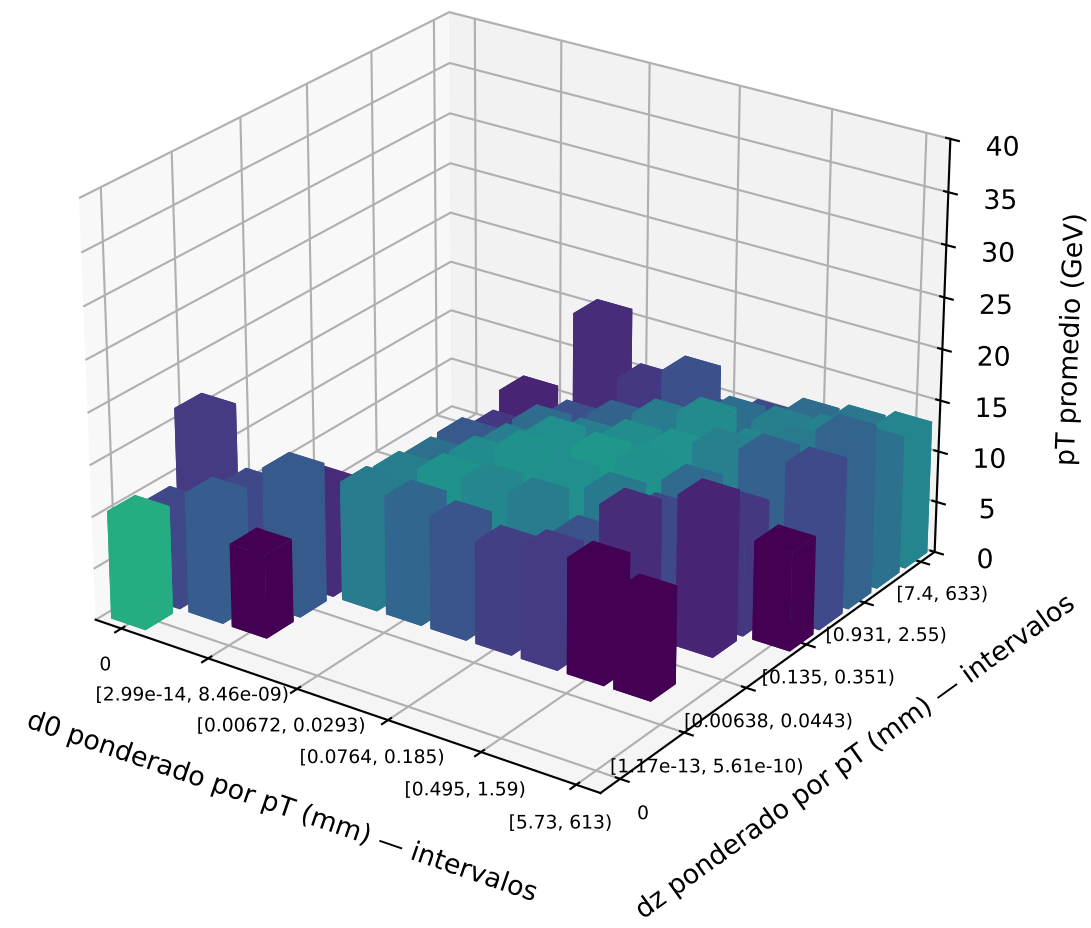
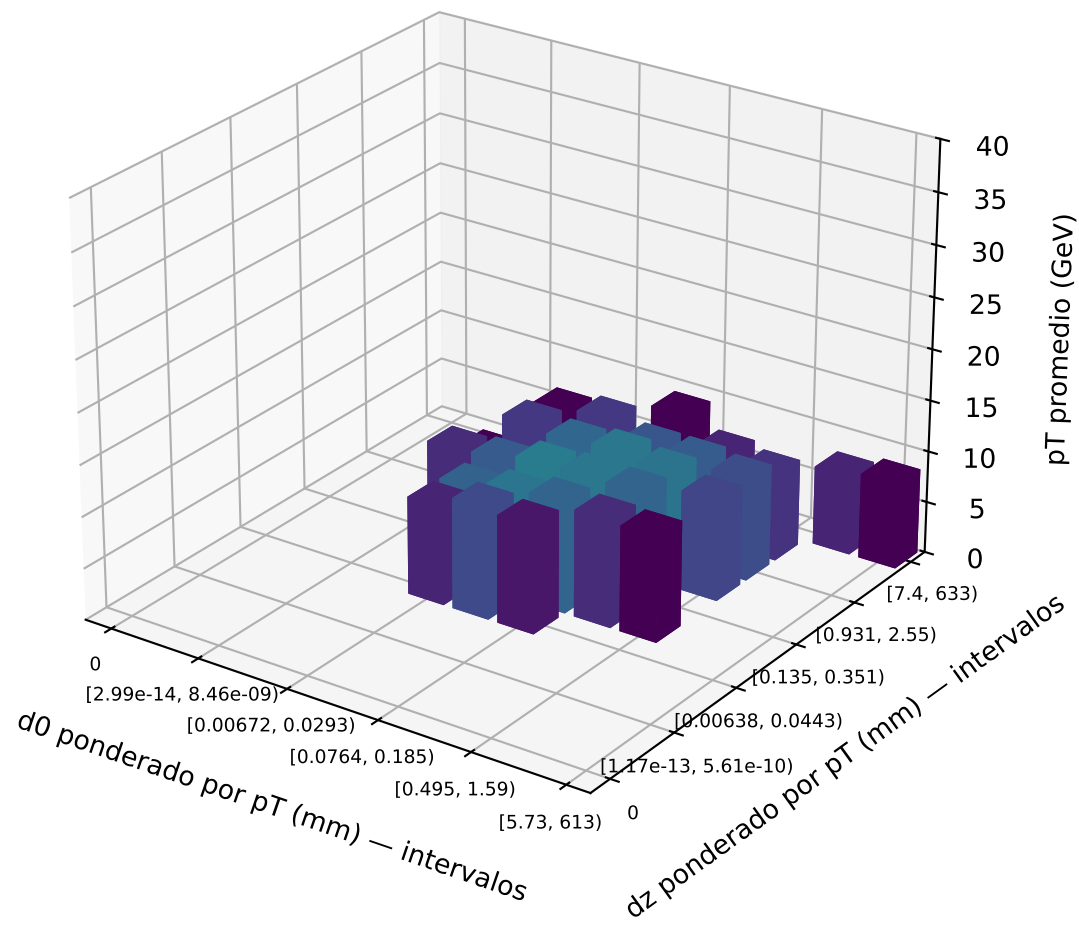
Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.

Perfil de pT medio en intervalos de d0/dz — VQC16 · semilla 73

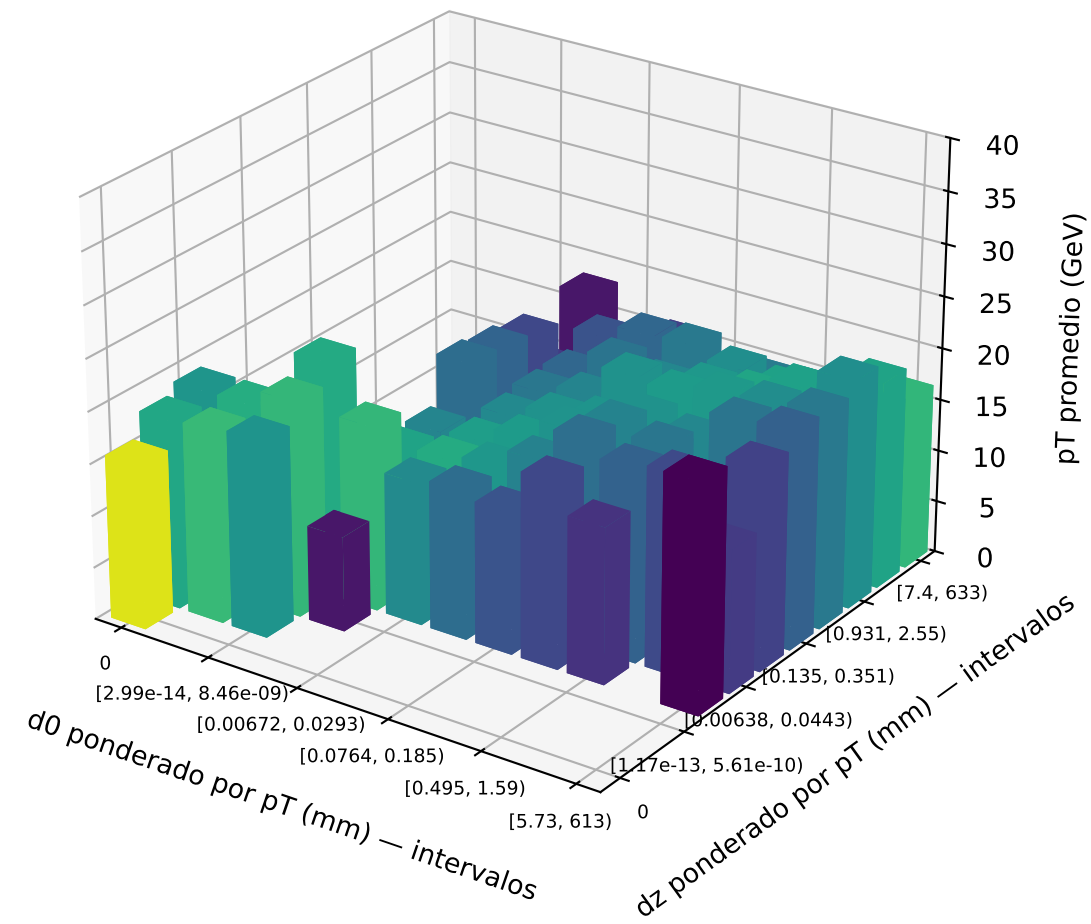
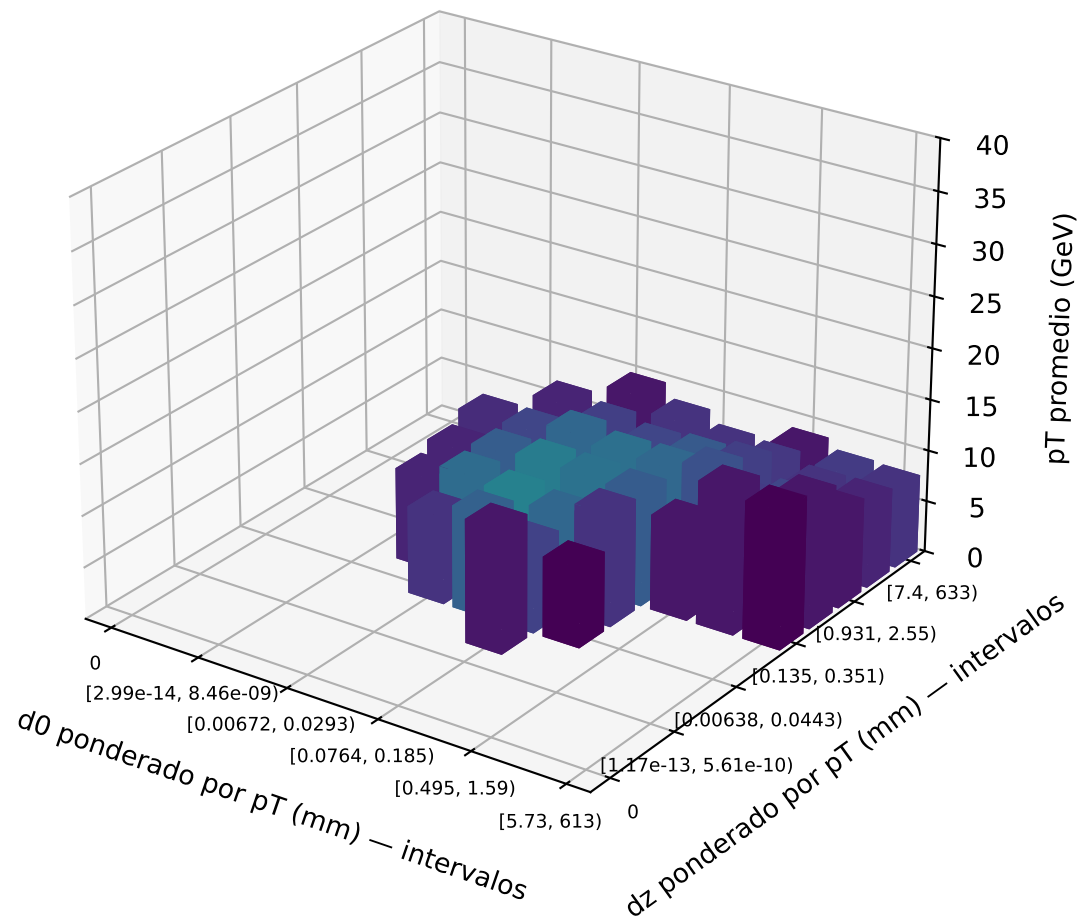
TP — N=692

FN — N=5,983



FP — N=774

TN — N=60,102



Conteo absoluto por celda N (escala logarítmica)

Intervalos de x/y expresados en mm (ancho visual uniforme; no representa distancia física). Altura = pT promedio (GeV), no frecuencia; color = N absoluto, escala logarítmica común. Las distribuciones usan todos los jets de validación (no una muestra aleatoria).

Estadísticas de variables — VQC16 · semilla 73

Variable	Unidad	TP N=692	FN N=5,983	FP N=774	TN N=60,102
jet_pt *	GeV	10.79 ± 3.263	12.43 ± 6.901	10.62 ± 3.488	16.49 ± 11.7
jet_eta *	1	0.1162 ± 1.541	-0.01066 ± 1.399	0.05706 ± 1.53	0.001993 ± 1.495
jet_abs_eta *	1	1.396 ± 0.6637	1.194 ± 0.7282	1.37 ± 0.6819	1.274 ± 0.7825
jet_mass *	GeV	2.15 ± 0.9066	2.574 ± 1.587	2.123 ± 0.8569	2.861 ± 1.875
jet_ncharged	count	2.679 ± 1.24	2.726 ± 1.833	2.545 ± 1.105	2.588 ± 1.938
jet_nneutral	count	1.825 ± 1.404	1.961 ± 1.643	1.658 ± 1.272	2.378 ± 1.79
n_tracks_cone	count	2.759 ± 1.21	2.756 ± 1.83	2.628 ± 1.106	2.616 ± 1.938
track_pt_sum *	GeV	5.207 ± 2.68	5.444 ± 5.076	5.004 ± 2.494	6.777 ± 8.101
track_pt_sum_over_jet_pt *	1	0.483 ± 0.1934	0.4287 ± 0.2554	0.4771 ± 0.1917	0.3732 ± 0.2672
avg_track_pt *	GeV	2.011 ± 0.9817	1.936 ± 1.412	2.044 ± 1.071	2.167 ± 2.064
std_track_pt	GeV	0.7691 ± 0.6177	0.7329 ± 0.8918	0.7228 ± 0.6689	1.008 ± 1.604
n_tracks_below_avg_pt	count	1.447 ± 0.9504	1.437 ± 1.29	1.348 ± 0.8727	1.379 ± 1.358
n_tracks_above_avg_pt	count	1.312 ± 0.5646	1.318 ± 0.779	1.28 ± 0.5309	1.238 ± 0.8046
max_pt_ratio	1	0.2872 ± 0.1353	0.2464 ± 0.1673	0.283 ± 0.1223	0.2112 ± 0.1554
min_pt_ratio	1	0.1242 ± 0.1246	0.111 ± 0.1328	0.1295 ± 0.111	0.08597 ± 0.0988
max_dr_ratio	1	0.1514 ± 0.1272	0.1334 ± 0.135	0.1599 ± 0.1148	0.1047 ± 0.1078
min_dr_ratio	1	0.2413 ± 0.1572	0.2108 ± 0.1761	0.2413 ± 0.1424	0.1805 ± 0.1547
dr_max_pt	1	0.1358 ± 0.07874	0.1376 ± 0.09374	0.1375 ± 0.07668	0.127 ± 0.09892
dr_min_pt	1	0.2084 ± 0.0991	0.1954 ± 0.1111	0.2035 ± 0.09534	0.1906 ± 0.117
dr_max_dr	1	0.2428 ± 0.08936	0.2334 ± 0.1122	0.2365 ± 0.08847	0.2214 ± 0.1202
dr_min_dr	1	0.1076 ± 0.06853	0.1083 ± 0.08049	0.1093 ± 0.0647	0.1043 ± 0.08823
pt_difference *	GeV	1.772 ± 1.432	1.759 ± 2.259	1.636 ± 1.48	2.471 ± 4.052
r50	1	0.1423 ± 0.07629	0.1446 ± 0.09132	0.1439 ± 0.07379	0.1336 ± 0.09725
r95	1	0.2405 ± 0.08835	0.2308 ± 0.1116	0.2351 ± 0.08835	0.2152 ± 0.1188
mean_abs_d0 *	mm	0.1847 ± 0.5566	1.383 ± 8.394	0.4968 ± 1.726	1.208 ± 7.766
max_abs_d0 *	mm	0.3738 ± 1.506	3.694 ± 16.27	1.138 ± 3.912	3.197 ± 16.34
std_abs_d0 *	mm	0.1389 ± 0.6357	1.415 ± 6.301	0.5054 ± 1.843	1.229 ± 6.567
mean_abs_dz *	mm	0.5727 ± 2.092	2.103 ± 12.54	1.538 ± 8.939	1.605 ± 9.476
max_abs_dz *	mm	1.251 ± 6.287	5.307 ± 20.81	3.85 ± 21.6	4.194 ± 22.55
std_abs_dz *	mm	0.482 ± 2.901	1.991 ± 7.471	1.745 ± 10.39	1.621 ± 9.282
pt_weighted_abs_d0 *	mm	0.1715 ± 0.3701	1.309 ± 8.307	0.4044 ± 1.485	1.152 ± 7.778
pt_weighted_abs_dz *	mm	0.52 ± 1.189	1.84 ± 12.35	0.9099 ± 4.471	1.326 ± 8.234

* Variable utilizada como entrada del modelo.

Desviación estándar poblacional observada (ddof=0); no es error estándar ni intervalo de confianza.

Estadísticas descriptivas condicionales; no implican causalidad ni secciones eficaces físicas.